National Comprehensive Cancer Network®

NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®)

Adolescent and Young Adult (AYA) Oncology

Version 2.2026 — February 27, 2026

NCCN.org

NCCN recognizes the importance of clinical trials and encourages participation when applicable and available. Trials should be designed to maximize inclusiveness and broad representative enrollment.

NCCN Guidelines for Patients® available at www.nccn.org/patients

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NCCN Guidelines Version 2.2026
Adolescent and Young Adult (AYA) Oncology

NCCN Guidelines Index
Table of Contents
Discussion

***Smita Bhatia, MD, MPH/Chair € ‡**
O'Neal Comprehensive Cancer Center at UAB

***Alberto S. Pappo, MD/Vice Chair € ‡**
St. Jude Children's Research Hospital/ The University of Tennessee Health Science Center

Melissa Acquazzino, MD, MS € ‡
Fred & Pamela Buffett Cancer Center

Wendy A. Allen-Rhoades, MD, PhD € †
Mayo Clinic Comprehensive Cancer Center

Marie Barnett, PhD θ
Memorial Sloan Kettering Cancer Center

Scott C. Borinstein, MD, PhD € ‡
Vanderbilt-Ingram Cancer Center

Christopher Cann, MD ‡
Fox Chase Cancer Center

Sun Choo, MD € ‡
UC San Diego Moores Cancer Center

Rashmi Chugh, MD ‡ †
University of Michigan Rogel Cancer Center

Shira Dinner, MD † ‡ ξ
Robert H. Lurie Comprehensive Cancer Center of Northwestern University

Adam DuVall, MD, MPH † ‡ €
The UChicago Medicine Comprehensive Cancer Network

Ralph Ermoian, MD §
Fred Hutchinson Cancer Center

Douglas Fair, MD, MS € †
Huntsman Cancer Institute at the University of Utah

Jeanelle Folbrecht, PhD θ £
City of Hope National Medical Center

Julie Germann, PhD θ €
UT Southwestern Simmons Comprehensive Cancer Center

Robert Goldsby, MD €
UCSF Helen Diller Family Comprehensive Cancer Center

Juneko Grilley-Olson, MD †
Duke Cancer Institute

Robert Hayashi, MD € ‡ ξ
Siteman Cancer Center at Barnes-Jewish Hospital and Washington University School of Medicine

Alex Y. Huang, MD, PhD € ‡
Case Comprehensive Cancer Center/ University Hospitals Seidman Cancer Center and Cleveland Clinic Taussig Cancer Institute/ University Hospitals Rainbow Babies & Children's Hospital/

Mary S. Huang, MD € ‡
Dana-Farber/Brigham and Women's Cancer Center | Mass General Cancer Center

Leidy Isenalumhe, MD, MS † € ‡
Moffitt Cancer Center

Linda A. Jacobs, PhD, CRNP € †
Abramson Cancer Center at the University of Pennsylvania

Cathy Lee-Miller, MD € ‡
University of Wisconsin Carbone Cancer Center

Michael P. Link, MD €
Stanford Cancer Institute

John A. Livingston, MD, MS † ‡ €
The University of Texas MD Anderson Cancer Center

Maryam Lustberg, MD, MPH †
Yale Cancer Center/Smilow Cancer Hospital

Marcio Malogolowkin, MD € †
UC Davis Comprehensive Cancer Center

Stephanie Samolovitch, MSW ¥
Patient Advocate

Jodi Skiles, MD, MS € ξ
Indiana University Melvin and Bren Simon Comprehensive Cancer Center

Heather Symons, MD, MHS ‡ €
Johns Hopkins Kimmel Cancer Center

Deepak Vadehra, DO †
Roswell Park Comprehensive Cancer Center

Nicholas Yeager, MD €
The Ohio State University Comprehensive Cancer Center - James Cancer Hospital and Solove Research Institute

Patricia A. Young, MD, MS ξ ‡
UCLA Jonsson Comprehensive Cancer Center

NCCN

Sara Espinosa, PhD
Sarah Montgomery, BA

[NCCN Guidelines Panel Disclosures]

ξ Bone marrow transplantation
* Discussion Section Writing Committee Member
‡ Hematology/Hematology oncology
† Medical oncology
¥ Patient Advocacy
€ Pediatric oncology
θ Psychiatry and psychology, including health behavior
§ Radiation oncology
£ Supportive care including palliative, pain management, pastoral care, and oncology social work

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National Comprehensive Cancer Network®

NCCN Guidelines Version 2.2026
Adolescent and Young Adult (AYA) Oncology

NCCN Guidelines Index
Table of Contents
Discussion

NCCN AYA Oncology Panel Members
Summary of the Guidelines Updates

Definition of the AYA Oncology Population and Purpose of the NCCN Guidelines for AYA Oncology (AYAO-1)
Screening, Assessment, and Evaluation (AYAO-2)
Comprehensive Initial Assessment (AYAO-3)
Treatment-Related Issues (AYAO-4)
Fertility, Reproductive Endocrine, and Sexual Health Considerations (AYAO-5)
Comprehensive Psychosocial/Behavioral Considerations Throughout Treatment:
- Individual (AYAO-7)
- Relationships (AYAO-9)
- Socioeconomic Issues (AYAO-11)

Survivorship (AYAO-12)
Disease-Specific Issues Related to Age (AYAO-A)
Screening Recommendations for AYA Survivors (AYAO-B)
Palliative Care Across the Disease Continuum and End-of-Life Considerations (AYAO-C)

Abbreviations (ABBR-1)

Find an NCCN Member Institution: https://www.nccn.org/home/member-institutions.

NCCN Categories of Evidence and Consensus: All recommendations are category 2A unless otherwise indicated.

See NCCN Categories of Evidence and Consensus.

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Updates in Version 2.2026 of the NCCN Guidelines for Adolescent and Young Adult (AYA) Oncology from Version 1.2026 include:

MS-1
• The discussion section has been updated to reflect the changes in the algorithm.

Updates in Version 1.2026 of the NCCN Guidelines for Adolescent and Young Adult (AYA) Oncology from Version 2.2025 include:

Global Changes
• "Renal" changed to "kidney" throughout the guidelines.
• References updated throughout the guidelines.

AYAO-1
• Purpose of the NCCN Guidelines for AYA Oncology, sentence 1 modified: AYA patients with cancer have a number of unique medical and psychosocial concerns that have been identified by panels of experts, including (although not limited to) *interpersonal relationships,* fertility preservation, parenting, education, and employment attainment and retention.

AYAO-2
• Footnote c modified: ~~Howlader N, et al (eds). SEER Cancer Statistics Review, 1975-2017, National Cancer Institute. Bethesda, MD, https://seer.cancer.gov/csr/1975_2017, based on November 2019 SEER data submission, posted to the SEER website, April 2020.~~ *NCCR*Explorer: An interactive website for NCCR cancer statistics. National Cancer Institute website. Available at: https://nccrexplorer.ccdi.cancer.gov.*
• Footnote e modified: These centers provide a multidisciplinary approach involving a team of providers with expertise in AYA cancer treatment *with access to clinical trials for the AYA population,* and management of specific *issues including* mental health ~~and developmental issues such as fertility~~, education *including genetic counseling, fertility*, career development, employment *management with disease demands, survivorship, and wellness promotion.*~~, family planning, pregnancy, sexually transmitted diseases, smoking, and substance use disorders.~~

AYAO-3
• Page has been significantly revised.

AYAO-4
• Page has been significantly revised.

AYAO-5
• First column, bullet 3 added: Assess for routine reproductive/gynecologic health care when transitioning from pediatric to adult care.
• Second column, bullet 1 modified: Initiate referral for ~~fertility preservation clinics~~ *reproductive health specialist* and/or provide resources for off-site/remote sperm banking as soon as possible for all patients who are interested in ~~discussing~~ *pursuing* fertility preservation.
• Second column, bullet 3 modified: ~~Provide~~ *Assist* AYA patients ~~with~~ *in identifying* financial resources for fertility preservation.

AYAO-6
• General, bullet 1 modified: After assessing the patient's risk for impaired fertility, and the patient's preferences for fertility preservation, recommend a form of appropriate fertility preservation and/or make a referral to a fertility preservation specialist *or sperm bank*.
• Individuals with ovaries, bullet 2 modified: For patients who cannot delay treatment for oocyte or embryo cryopreservation *and are at high risk for impaired fertility*, discuss or refer the patient for consideration of ovarian tissue cryopreservation.
• Footnote j modified: Creating safe spaces to ~~disclose~~ *discuss* gender identity and sexual orientation can mitigate and provide community partnership/resources for fertility and psychosocial experts in sexual and gender minorities (GLMA directory): https://www.glma.org/find_a_provider.php

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NCCN Guidelines Version 2.2026
Adolescent and Young Adult (AYA) Oncology

NCCN Guidelines Index
Table of Contents
Discussion

Updates in Version 1.2026 of the NCCN Guidelines for Adolescent and Young Adult (AYA) Oncology from Version 2.2025 include:

AYAO-7

- Evaluation
  - Psychosocial factors
    - ◊ Sub-bullet 5 modified: ~~Adjustment to illness~~ *Evaluate distress and adjustment through screening for depression and anxiety*
    - ◊ Sub-bullet 7 modified: Evaluate for current and past psychiatric symptoms *and treatments*, including anxiety, depression, suicidal thoughts, eating disorders, and self injurious behavior
    - ◊ Sub-bullet 9, sub-sub-bullet 6 modified: ~~Homeless~~ *Unhoused* or nonstable living environment
    - ◊ Sub-bullet removed: Patient or family/caregiver living in the United States without legal status/documentation
- Supportive Care Services/Interventions
  - Bullet 1 modified: "*Discuss with patient/guardian regarding* refer*ral* for neuropsychological assessment..."
  - Bullet 2 modified: If *psychosocial support specialists are present (including: psychology,* child life specialists*, or integrative medicine)* ~~or appropriate psychosocial support specialists are present~~, have them meet with the patient soon after diagnosis to address the patient's concerns regarding treatment or procedures, *provide psychoeducation,* and assist with coping mechanisms to reduce anxiety related to procedures.
  - Bullet 5 modified: *Facilitate adherence to treatment by* consider*ing* flexible treatment dates, consultation times, and procedures (evenings/weekends).
  - Bullet 8 added: For patients with minoritized identities:
    - ◊ Sub-bullet 1 added: Practice cultural humility. Remain open to the impact of cultural and identity factors in treatment decision-making (See Cultural Humility in Cancer Care)
    - ◊ Sub-bullet 2 modified: ~~For LGBTQIA2S+ AYAs,~~ Offer additional psychosocial support and referrals surrounding stressors, stigma, or rejection related to their sexuality or gender identity. *(See Cancer and Gender Diversity and Young Adult Survivors United)*

AYAO-8

- Evaluation
  - Bullet 1, behavioral factors
    - ◊ Sub-bullet 2 added: *Ongoing assessment for* tobacco, vaping, alcohol, cannabis, or other substance use *as habits may change throughout care or related to treatment and symptom management*
  - Bullet 3 modified: Existential/spiritual ~~issues~~ *beliefs or exploration*
- Supportive Care Services/Interventions
  - Bullet 1, sub-bullet 2 modified: ~~Evaluate for any past history or potential barriers to adherence with medical treatment.~~ *Refer to health or pediatric psychologist for adherence intervention if needed; include a list of barriers and facilitators.*
  - Bullet 3 added: Provide education and/or guidance on cannabinoid use (NCCN Guidelines for Antiemesis).
  - Bullet 5 modified: Provide education about the impact of treatment on sexual health *and safe sexual practices (including oral, anal, and vaginal sex)*, ~~including safe sexual practices~~ in light of risk of infection, risk for bleeding, prevention of pregnancy, and sexually transmitted ~~diseases~~ *infections*.
  - Bullet added: Recommendations for diet and exercise
  - Bullet removed: Provide education about the impact of early cannabis use on cognitive development and mental health.
    - ◊ Sub-bullet removed: If AYA chooses to continue use, provide education on risks and benefits of varying methods of ingestion and dosing.
  - Bullet 8 added: Refer patients with sleep disturbance to supportive medicine or behavioral health for interventions (refer to NCCN Guidelines for Palliative Care)
  - Bullet 9 modified: ~~Consider evidence-based~~ *Assess patient's use of or desire for* integrative therapies~~/~~ *and* interventions.
    - ◊ Sub-bullet added: Refer to integrative oncologist when patient is interested in evidence-based integrative therapies to manage side effects of treatment and overall well-being.

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Updates in Version 1.2026 of the NCCN Guidelines for Adolescent and Young Adult (AYA) Oncology from Version 2.2025 include:

- ◊ Sub-bullet added: Monitor use of dietary supplements that might impact effectiveness of treatment or treatment medications.
- Bullet 11 modified: ~~If appropriate, consider referral to palliative care.~~ *If cancer and its therapy are expected to be associated with significant morbidity, referral to palliative care is warranted.*

• Footnotes
- Footnote q added: Stout NL, et al. Cancer 2020;126:2750-2758.
- Footnote r added: https://integrativeonc.org/practice-guidelines
- Footnote s added: https://www.cancer.gov/about-cancer/treatment/cam/hp/dietary-interactions-pdq

AYAO-9

• Evaluation, bullet 1 modified: Family/caregiver/*primary supports* status
- Sub-bullet 4 added: AYA patients who are parents
- Sub-bullet removed: Patient with young children

AYAO-10

• Page has been significantly revised.

AYAO-11

• Supportive Care Services/Interventions
- Bullet 1
  - ◊ Sub-bullet 1 modified: Provide information regarding ~~drug~~ *medication* assistance programs for patients with limited or no insurance.
  - ◊ Sub-bullet 3 modified: Direct AYA patients to legal resources/advocates for assistance with understanding health insurance coverage *and transition to independent health insurance coverage.*
- Bullet 4 modified: Provide school support~~, and~~ education services*, and work reintegration* for patients ~~in high school or , college~~.
- Bullet removed: Identify resources for respite care for AYA patients with young children.

AYAO-12

• Bullet 4 modified: Encourage relationship with primary health care provider for routine health issues *and support end-of-treatment transition to primary care physician (PCP) with clear communication about PCP role and oncology team role.*
• Bullet 6 modified: Ensure patient is provided continuous access to mental health support (eg, support groups, individual therapy, AYA-specific organizations) *and is aware of risks for mental health following treatment.*
• Bullet 11 modified: ~~Young~~ *AYA* cancer survivors should be screened for risky behavior (ie, substance use, tobacco/nicotine use, binge drinking/excessive alcohol use*, cannabis use*) on a regular basis and counseled on cessation.
• Bullet 12 modified: ~~Young~~ *AYA* cancer survivors should be educated on safe sex practices/contraceptive options if they are not trying to actively conceive regardless of their infertility risk post treatment.

AYAO-13

• Column 1, title modified: *RADIATION* EXPOSURE
• Row 1, column 1 modified: Any ~~exposure~~ *radiation therapy (RT)*
- Recommendation modified: Recommend a dental exam and cleaning every 6 mo for patients who received ~~chemotherapy and/or radiation therapy (~~RT~~)~~ to head or upper neck regions

• Abdominal or pelvic radiation
- Recommendation, bullet 4 added: Consider high-risk obstetrics referral if pregnant

• Limb radiation
- Recommendation, bullet 1 modified: Physical therapy to address *post-radiation* ~~muscle~~ fibrosis*-related deficits*

• Row moved to AYAO-13 A: Intrathecal chemotherapy and high CNS-penetrating systemic chemotherapy

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UPDATES

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Updates in Version 1.2026 of the NCCN Guidelines for Adolescent and Young Adult (AYA) Oncology from Version 2.2025 include:

AYAO-13 A
- Column 1, title modified: *SYSTEMIC THERAPY* EXPOSURE
- Row 1, column 1 modified: Any *chemotherapy*
  - Recommendation added: Recommend a dental exam and cleaning every 6 mo for patients who received chemotherapy to head or upper neck regions, especially for head and neck cancer survivors

AYAO-A 1 of 8
- CNS tumors, bullet 2 added: NCCN Guidelines for Pediatric CNS Cancers
- Bone and soft tissue sarcomas, bullet 4 added: Osteosarcoma
- Colorectal cancer, bullet 5 modified: Greater *germline and somatic* mutation frequency*, including mismatch repair deficiency* ~~of BRCA2, ATM, MSH2, and ATR~~
- Colorectal cancer, bullet removed: Consider mismatch repair gene deficiency in these patients
- Disease type added: Germ cell tumors
  - Bullet 1 added: NCCN Guidelines for Testicular Cancer
  - Bullet 2 added: NCCN Guidelines for Ovarian Cancer Including Fallopian Tube Cancer and Primary Peritoneal Cancer
- Disease type added: Breast Cancer
  - Bullet 1 added: NCCN Guidelines for Breast Cancer
- Disease type added: Thyroid Cancer
  - Bullet 1 added: NCCN Guidelines for Thyroid Carcinoma
- Footnote a added: BRCA2, ATM, MSH2, and ATR have been noted more in the AYA than in the adult population (Tricoli JV, et al. Cancer 2018;124:1070-1082.)

AYAO-A 2 of 8
- Management of Cancer During Pregnancy, bullet 11 modified: Most cancers can be adequately treated with medication that ~~is safe~~ *has been deemed not to cause harm* for both the pregnant patient and the fetus. However, full counseling about all options regarding the pregnancy, including termination per patient ~~preference~~ *decision* or to expedite use of necessary teratogenic medication should be incorporated in all cancer treatment discussions for pregnant AYA patients.

AYAO-A 3 of 8
- New section added: Reported Outcomes and Recommendations for Select Teratogenic Agents During Pregnancy

AYAO-B 1 of 8
- Neuroendocrine Axis Screening
  - Bullet 2, central hypothyroidism
    - ◊ Sub-bullet 1 modified: High-risk population: total radiation dose to HPA axis ≥~~30~~*15* Gy
  - Bullet 5 added: Routine endocrine, ophthalmology, and dental evaluations are recommended for patients with selected radiation exposure and/or total body irradiation (TBI) for hematopoietic cell transplant (HCT).
  - Bullet 6 added: Radiation-related effects - Consider multidisciplinary consultation, which includes a radiation oncologist for the optimal method to reduce radiation-induced late effects.
- Footnote a added: Radiation doses refer to dose when treatment is delivered in conventional fractionation.

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Updates in Version 1.2026 of the NCCN Guidelines for Adolescent and Young Adult (AYA) Oncology from Version 2.2025 include:

AYAO-B 2 of 8

- Dermatology Evaluation
  - Bullet added: Provide recommendation for management of treatment-related hair loss.
- Neuropsychological Evaluation
  - Bullet 2, sub-bullet 2 added: Neurotoxicity: Conduct routine evaluation for symptoms of peripheral neuropathy. See NCCN Guidelines for Adult Cancer Pain.
- Cardiovascular Risk Assessment and Screening
  - Bullet 5 added: Cardiac toxicity: Provide periodic echocardiograms for select patients. A post-treatment baseline electrocardiogram (ECG) is recommended after completion of treatment. When cumulative dose of anthracycline is anticipated to exceed 250 mg/m^2, consider adding a cardioprotectant (eg, dexrazoxane); however, there may be variability depending on institutional guidelines.

AYAO-B 3 of 8

- Pulmonary Function Screening
  - Bullet 1 modified: High-risk population: chest radiation ≥15 Gy (or radiation to large volume of lung), TBI (≥6 Gy in single fraction or ≥12 Gy fractionated), *partial thoracic RT with mean lung dose ≥14 Gy,* bleomycin, combination of chest radiation and bleomycin, and selected alkylating agents (ie, busulfan, carmustine)
  - Bullet 3 added: PFT is the desired method for optimally evaluating pulmonary function in patients who can perform this test
- Thyroid Disease Screening, bullet 1 modified: Thyroid disorders: hypothyroidism ~~(very common)~~, thyroid cancer ~~(common)~~, and hyperthyroidism ~~(uncommon)~~
- Colorectal Cancer Screening, bullet 2 modified: Screening recommendation: colorectal cancer screening, ie, colonoscopy every 5 years~~, or~~ multitarget stool DNA*/RNA-based* test every 3 years, *or blood-based cell-free DNA test every 3 years,* based on informed decision-making between survivor and clinician starting at age 30 or 5 years after radiation, whichever occurs last
- Screening for Kidney and/or Bladder Disease
  - Bullet 2 added: Kidney toxicity: Provide periodic monitoring of kidney function and electrolytes in patients treated with platinum-based therapies and ifosfamide-based chemotherapy.

AYAO-B 4 of 8

- Assessment for Gonadal Function
  - Bullet 2, Individuals with testes
    - ◊ Sub-bullet 2 modified: Impaired fertility: *Permanent or t*emporary azoospermia or oligospermia can occur for variable periods of time post-therapy.
      - – Sub-sub-bullet 1 modified: High level of ~~increased~~ risk:
        - ▪ *Dose modified:* Radiation to testes: ≥6~~.0~~ Gy
      - – Sub-sub-bullet 2, dose modified: Radiation to testes: ~~0.7~~ *>1*–5.9 Gy
      - – Sub-sub-bullet 3, dose modified: Radiation to testes: 0.2–~~0.6~~ *<1* Gy

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NCCN Guidelines Version 2.2026
Adolescent and Young Adult (AYA) Oncology

NCCN Guidelines Index
Table of Contents
Discussion

Updates in Version 1.2026 of the NCCN Guidelines for Adolescent and Young Adult (AYA) Oncology from Version 2.2025 include:

AYAO-B 5 of 8

- Individuals with ovaries
  - Bullet 2
    - ◊ Sub-bullet added: For patients who desire more information regarding potential fertility and/or interventions to preserve future fertility, referral to gynecology, reproductive medicine, or endocrinology is recommended for antral follicle count, ovarian reserve evaluation, consultation regarding assisted reproductive technologies, and post-pubertal survivors treated with potentially gonadotoxic chemotherapy and/or radiotherapy.
    - ◊ Sub-bullet removed: Referral to reproductive endocrinology is recommended for antral follicle count, ovarian reserve evaluation, and consultation regarding assisted reproductive technologies in patients at risk who desire information about potential fertility and interventions to preserve future fertility.
    - ◊ Sub-bullet removed: Referral to gynecology, reproductive medicine, or endocrinology is recommended for post-pubertal survivors treated with potentially gonadotoxic chemotherapy and/or radiotherapy who desire information regarding potential fertility and/or interventions to preserve future fertility.
- Screening for t-AML or Myelodysplasia
  - Bullet added: The risk decreases significantly after 10 years from exposure to the therapeutic agent
- Audiologic Evaluation
  - Bullet added: Ototoxicity: Conduct routine evaluations for tinnitus and periodic audiogram to monitor hearing loss associated with platinum-based chemotherapy. Consider sodium thiosulfate (STS) to reduce the risk of ototoxicity associated with cisplatin in patients with localized, non-metastatic, solid tumors. There are concerns about the use of STS in the metastatic setting.

AYAO-B 6 of 8

- Section added: Cancer Predisposition
  - Bullet added: Assessment for cancer predisposition is strongly recommended in the following settings:
    - ◊ Sub-bullet added: A tumor that has a high likelihood of being part of a cancer predisposition syndrome (adrenocortical carcinoma, desmoid tumors, endolymphatic sac tumors, gastrointestinal stromal tumor, malignant peripheral nerve sheath tumor, medullary thyroid cancer, osteosarcoma, ovarian Sertoli cell or Sertoli-Leydig cell tumors, paraganglioma, pheochromocytoma, pleuropulmonary blastoma, renal cell carcinoma, rhabdoid tumor, schwannoma, atypical teratoid rhabdoid tumor, choroid plexus carcinoma, ciliary body medulloepithelioma, hemangioblastoma, optic pathway glioma, pineoblastoma, pituitary blastoma, retinoblastoma, subependymal giant cell astrocytoma [SEGA], cystic nephroma, juvenile myelomonocytic leukemia, meningioma, myelodysplastic syndrome)
    - ◊ Sub-bullet added: Any bilateral tumor
    - ◊ Sub-bullet added: More than one primary cancer
    - ◊ Sub-bullet added: One or more first-degree relatives with cancer
    - ◊ Sub-bullet added: Consanguinity
    - ◊ Sub-bullet added: Diagnosis of early-onset cancers (breast, colon, gastrointestinal, ovarian)
    - ◊ Sub-bullet added: Diagnosis of a cancer predisposition syndrome in a relative
  - Bullet added: For patients who may be at risk for cancer predisposition based on history, consider referral to genetic counseling or clinical genetics and referral for preconception or prenatal counseling.
- Section added: Screening for Other Treatment-Related Toxicities
  - Bullet added: Immunotherapy-induced toxicity - There may be long-term toxicity, but more data on the AYA population are needed.
  - Bullet added: Osteoporosis, osteonecrosis and osteopenia in selected populations
  - Bullet added: Quality of life
- New section added: Core Outcome Measures (Also for AYAO-7 of 8)

Continued

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Updates in Version 1.2026 of the NCCN Guidelines for Adolescent and Young Adult (AYA) Oncology from Version 2.2025 include:

AYAO-C 1 of 2

- Palliative Care Across the Disease Continuum, sentence 1 modified: Palliative care focuses on symptom control, reduction of physical suffering or discomfort, and optimizing quality of life at any stage of a life-threatening disease (NCCN Guidelines for Palliative Care)*. A palliative approach to care is indicated across the cancer trajectory from diagnosis to end-of-life care.*
- End-of-life considerations
  - Bullet removed: Palliation of symptoms is an important aspect of care.
  - Bullet added: Alleviating Suffering
    - Sub-bullet added: A palliative approach to care, alleviating suffering, and care consistent with patient goals are vitally important in the end-of-life phase of therapy.
    - Sub-bullet added: Work to address physical, psychosocial, social, and spiritual suffering through systematic screening and partnership with interdisciplinary care team members. Consider referral to specialty palliative care when there is suffering that is refractory to primary team management
  - Bullet added: Communication (eg, with advanced care planning)
    - Sub-bullet added: Determine who the AYA wants to be involved in decision-making. Identify individuals who will require psychosocial support during the AYAs cancer journey.
    - Sub-bullet added: Provide AYAs with clear cancer prognostic information
    - Sub-bullet added: Provide care that is aligned with AYA's goals of care
  - Bullet added: Supportive Care Services (if needed)
    - Sub-bullet added: Determine who the AYA wants involved for decision-making.
    - Sub-bullet added: Identify individuals who will require psychosocial support during the AYA's cancer journey.
    - Sub-bullet added: When AYAs have children or young siblings, conduct an assessment of their psychosocial needs and/or refer to programs that have the capacity to assess these psychosocial needs.
    - Sub-bullet added: Clinical teams should be aware of local availability and regulations surrounding 1) home-based palliative care, and 2) concurrent care—the ability to enroll on hospice while also receiving disease-directed therapy
    - Sub-bullet added: Programs should have a bereavement program, which, at a minimum, maintains lists of bereavement resources and makes, at least, one contact with bereaved family members after an AYA's death.

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DEFINITION OF THE ADOLESCENT AND YOUNG ADULT ONCOLOGY POPULATION

The adolescent and young adult (AYA) oncology patient is defined as an individual aged 15–39 years at the time of initial cancer diagnosis. This definition is based on the National Cancer Institute (NCI) Progress Review Group recommendations for a national agenda to advance AYA oncology.

PURPOSE OF THE NCCN GUIDELINES FOR AYA ONCOLOGY

AYA patients with cancer have a number of unique medical and psychosocial concerns that have been identified by panels of experts, including (although not limited to) interpersonal relationships, fertility preservation, parenting, education, and employment attainment and retention. The relative importance of these issues understandably varies markedly across the broad age range defined. These issues should be considered as part of the overall therapeutic plan for the patient. Specific recommendations are highlighted in the NCCN Guidelines for Adolescent and Young Adult (AYA) Oncology. Many centers have established AYA centers to accommodate the specific needs of patients with cancer in this age group. Consideration should be given to referring such patients to one of these AYA centers of excellence if feasible. Available literature demonstrates improved outcomes in patients receiving AYA-focused care through a comprehensive and multidisciplinary approach. AYA patients are encouraged to seek care at centers with established social, clinical, therapeutic, and psychosocial support programs.

The NCCN Guidelines for Adolescent and Young Adult (AYA) Oncology have been developed as supportive care guidelines and not as treatment guidelines. The purpose of the guidelines is to identify and increase awareness of unique issues in AYA oncology. In addition, these guidelines will identify resources available to the AYA population, include appropriate tabular materials, and make recommendations for patient care.

- **AYA patients diagnosed with cancer should be recognized as a distinct age group that has unique medical and psychosocial needs. The distribution of cancer types is dramatically different across the age spectrum of the AYA population.[a]**
- **The distinct biology of disease as well as other age-related issues in the AYA population (fertility, long-term side effects, insurance/financial issues, transportation to clinic appointments, child care, psychosocial support, and adherence to therapy) should be considered in the treatment decision-making process and during the transition of care from pediatric to adult medical teams.[b]**
- **Participation in clinical trials as well as enrollment on tumor banking and biologic protocols should be strongly encouraged in the AYA population.**

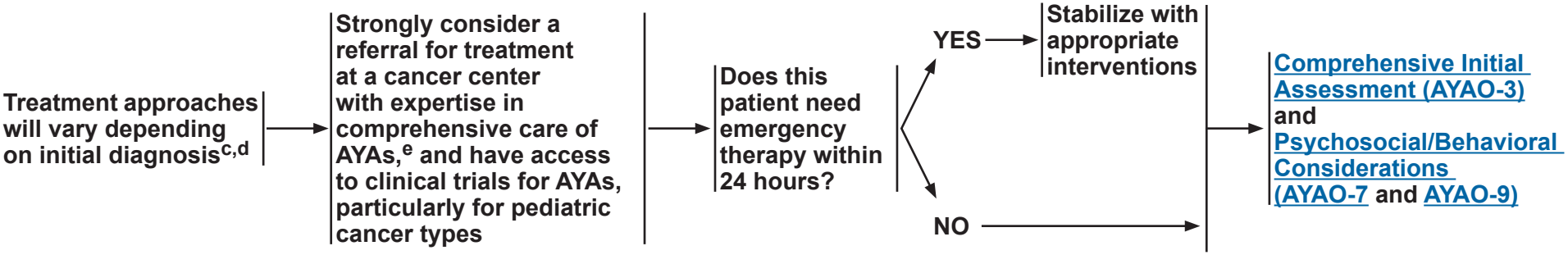
[a] Howlader N, et al (eds). SEER Cancer Statistics Review, 1975-2017, National Cancer Institute. Bethesda, MD, https://seer.cancer.gov/csr/1975_2017, based on November 2019 SEER data submission, posted to the SEER website, April 2020.
[b] Perez GK, et al. Am Soc Clin Oncol Educ Book 2020;40:1-15.

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SCREENING, ASSESSMENT, AND EVALUATION



[c] NCCR*Explorer: An interactive website for NCCR cancer statistics. National Cancer Institute website. Available at: https://nccrexplorer.ccdi.cancer.gov.
[d] Definition of AYA Population (AYAO-1).
[e] These centers provide a multidisciplinary approach involving a team of providers with expertise in AYA cancer treatment with access to clinical trials for the AYA population, and management of specific issues including mental health, education including genetic counseling, fertility, career development, employment management with disease demands, survivorship, and wellness promotion.

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COMPREHENSIVE INITIAL ASSESSMENT

- Communication and education preferences and needs
  - Speak with the patient individually to solicit questions/concerns without a caregiver/partner present so the patient can speak freely.
  - Provide cancer information and education at an age- and developmentally appropriate level. See Online Resources for AYA Patients and Survivors
- Fertility and sexual health/identity
  - Discuss risks of impaired fertility due to cancer and its therapy, as well as fertility preservation options, and consider referral to a fertility preservation/reproductive health program. See Fertility, Reproductive Endocrine, and Sexual Health Considerations (AYAO-5)
  - Discuss contraception prior to initiating therapy.
    - Consult with OB/GYN for patients with ovaries/uterus and consult the CDC Summary Chart of U.S. Medical Eligibility Criteria to assist with the safety and efficacy of selection of appropriate contraception for individuals with childbearing potential.
  - Consider a pregnancy test prior to each cycle of therapy in accordance with institutional requirements.
  - Assess gender identity, pronouns, sexual activity, and concerns.
- Psychosocial assessment
  - See Comprehensive Psychosocial/Behavioral Considerations
    - Individual (AYAO-7 and AYAO-8)
    - Relationships (AYAO-9)
    - Socioeconomic Issues (AYAO-11)
- Take a complete family history and refer for subsequent evaluation by a genetic counselor for risk assessment and counseling as needed. Approximately 8%–9% of all patients with cancer will have a germline mutation in a cancer-predisposing gene.[f]
  - Risk factors for breast cancer
    - See NCCN Guidelines for Genetic/Familial High-Risk Assessment: Breast, Ovarian, Pancreatic, and Prostate
    - Chest irradiation
  - Risk factors for colon cancer
    - See NCCN Guidelines for Genetic/Familial High-Risk Assessment: Colorectal, Endometrial, and Gastric
  - Risk factors for sarcomas
    - See NCCN Guidelines for Soft Tissue Sarcoma
    - See NCCN Guidelines for Bone Cancer
  - Risk factors for multiple endocrine neoplasia (MEN)
    - See NCCN Guidelines for Neuroendocrine and Adrenal Tumors
  - Risk factors for pediatric diffuse high-grade gliomas
    - See NCCN Guidelines for Pediatric Central Nervous System Cancers

- Treatment-related issues → Treatment-Related Issues (AYAO-4)
- Fertility/endocrine → Fertility, Reproductive Endocrine, and Sexual Health Considerations (AYAO-5)
- Distress and psychosocial factors impacting treatment → Psychosocial/Behavioral Considerations (AYAO-7 and AYAO-9)

[f] Parsons DW, et al. JAMA Oncol 2016;2:616-624; Zhang J, et al. N Engl J Med 2015;373:2336-2346; Gröbner SN, et al. Nature 2018;555:321-327.

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TREATMENT-RELATED ISSUES

Dose schedules

- AYA patients should be offered enrollment in open clinical trials for their specific disease when available and appropriate and supportive care should follow well-established guidelines such as those available at www.NCCN.org.
- Dose reductions may be appropriate and are often based on avoiding severe, irreversible organ damage.
  - Unless clear clinical data indicate otherwise, assume that the patient population has a significant long-term survival and that significant end-organ damage may compromise long-term function and quality of life.
- Establish maximum cumulative dosing parameters and monitor cumulative dosing and schedule for certain medications associated with irreversible organ damage and fertility issues when certain lifetime exposure is encountered. See AYAO-10 for specific agents.

Toxicities

- Reversible toxicities do not necessarily warrant dose reductions.
  See NCCN Guidelines for Supportive Care for the management of treatment-related toxicities, including:
  - NCCN Guidelines for Adult Cancer Pain
  - NCCN Guidelines for Antiemesis
  - NCCN Guidelines for Cancer-Related Fatigue
  - NCCN Guidelines for Palliative Care
  - NCCN Guidelines for Prevention and Treatment of Cancer-Related Infections
  - See NCCN Guidelines for Hematopoietic Growth Factors for growth factor support and management of cancer- and chemotherapy-induced anemia for those who refuse blood transfusions.
- Patients should be given a summary of potential therapy-related side effects. Documentation of consent is recommended and ongoing monitoring for treatment-specific acute toxicities is essential.
- Screening is recommended for treatment-related toxicities, including specific core outcomes that should be evaluated accordingly to the initial diagnosis.[g] See Screening Recommendations for AYA Survivors (AYAO-B).

[g] van Kalsbeek RJ, et al. Nat Med 2023;29:1340-1348.

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FERTILITY, REPRODUCTIVE ENDOCRINE, AND SEXUAL HEALTH CONSIDERATIONS

- **Addressing fertility and sexual health and function should be an essential part of the care of AYAs with cancer who are at any level of risk for impaired fertility or sexual dysfunction due to cancer treatments, regardless of gender identity, sexual orientation, or financial status.[h]**
- **Perform an assessment of the risk for gonadotoxicity and impaired fertility due to cancer and its treatment and discuss that assessment and options for fertility preservation with the patient. Do this as soon as possible prior to the start of therapy and throughout the course of treatment.[i]**
- **Assess for routine reproductive/gynecologic health care when transitioning from pediatric to adult care.**

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- **Initiate referral for reproductive health specialist and/or provide resources for off-site/remote sperm banking as soon as possible for all patients who are interested in pursuing fertility preservation.**
- **Consider the emotional impact of conversations surrounding fertility preservation, especially for the younger AYA and sexual gender minority (SGM) patient.**
  - **See Comprehensive Psychosocial/Behavioral Considerations**
    - ◊ **Individual (AYAO-7 and AYAO-8)**
    - ◊ **Relationships (AYAO-9)**
    - ◊ **Socioeconomic Issues (AYAO-11)**
- **Assist AYA patients in identifying financial resources for fertility preservation.**

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Fertility, Reproductive Endocrine, and Sexual Health Considerations (AYAO-6)

[h] Meacham LR, et al. J Adolesc Young Adult Oncol 2020;9:662-666; Mulder RL, et al. Lancet Oncol 2021;22:e68-e80; Mulder RL, et al. Lancet Oncol 2021;22:e57-e67; Mulder R, et al. Lancet Oncol 2021;22:e45-e56.
[i] Green DM, et al. Pediatr Blood Cancer 2014;61:53-67.

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FERTILITY, REPRODUCTIVE ENDOCRINE, AND SEXUAL HEALTH CONSIDERATIONS

General[j]

- After assessing the patient's risk for impaired fertility, and the patient's preferences for fertility preservation, recommend a form of appropriate fertility preservation and/or make a referral to a fertility preservation specialist or sperm bank.[k]
- Discuss effects of treatment on sexual function during and after treatment. Consider referral to a specialist as appropriate. See NCCN Guidelines for Survivorship.
- Discuss contraception before, during, and after treatment.
  - Consult with OB/GYN for patients with ovaries/uterus and consult the CDC Summary Chart of U.S. Medical Eligibility Criteria to assist with the safety and efficacy of selection of appropriate contraception for individuals at risk of pregnancy.

Individuals with Testes

- Sperm banking is the preferred choice for patients without erection or ejaculation issues. For patients who can delay cancer treatment, consider more than one collection of ejaculate prior to initiating treatment.
  - Suggest a local sperm bank or available online sperm banking kit.[l]
  - A semen analysis should be performed by the sperm bank, which should be assessed to ensure viable sperm has been frozen before starting cancer treatment, if time allows.
- For patients with erection or ejaculation issues, there are other options available,[m] including:
  - PDE5 inhibitors
  - Vibratory stimulation
  - Electro-ejaculation
  - Collection of retrograde ejaculate
- Testicular transposition out of the field of radiation can be considered for patients in whom the radiation field will include the testes.
- For those who cannot ejaculate, or who now have azoospermia or insufficient sperm in the ejaculate to freeze, discuss surgical sperm extraction as an alternative strategy, such as the testicular sperm extraction (TESE) procedure.[m]
- Discuss effects of treatment on gonadal hormone function. After completion of treatment, screen or refer the patient to a specialist as appropriate.[n]

Individuals with Ovaries

- For patients who can delay cancer treatment for approximately 3 weeks, discuss oocyte or embryo cryopreservation via immediate (or random start) controlled ovarian stimulation (COS).
- For patients who cannot delay treatment for oocyte or embryo cryopreservation and are at high risk for impaired fertility, discuss or refer the patient for consideration of ovarian tissue cryopreservation.
- For patients in whom the radiation field will include the ovaries, discuss oophoropexy or transposition of the ovaries out of the field of radiation.
- Discuss effects of treatment on gonadal hormone function during and after treatment. Some individuals face primary ovarian insufficiency and should be screened and treated by a specialist. Additionally, for individuals who did not undergo fertility preservation prior to treatment, some may still be eligible for fertility preservation after treatment is completed. After completion of treatment, screen or refer the patient to a specialist as appropriate.[o]
- Menstrual suppression
  - Progestin-only methods, combined hormonal contraceptives, or gonadotropin-releasing hormone (GnRH) agonists may be used in protocols that are predicted to cause prolonged thrombocytopenia and present a risk for menorrhagia. It is controversial whether menstrual suppression provides adequate protection for the ovaries. GnRH agonists may protect ovarian function[p]; however, other fertility preservation modalities should still be considered and, if possible, pursued.

[j] Creating safe spaces to discuss gender identity and sexual orientation can mitigate challenges and foster community partnerships with/provide community resources for fertility and psychosocial experts in sexual and gender minorities (GLMA directory): https://www.glma.org/find_a_provider.php

[k] Practice Committee of the American Society for Reproductive Medicine. Fertil Steril 2019;112:1022-1033; Oktay K, et al. J Clin Oncol 2018;36:1994-2001; Mulder RL, et al. Lancet Oncol 2021;22:e57-e67; Mulder RL, et al. Lancet Oncol 2021;22:e45-e56.

[l] Lee JS, et al. J Clin Oncol 2006;24:2917-2931; Loren AW, et al. J Clin Oncol 2013;31:2500-2510; Oktay K, et al. J Clin Oncol 2018;36:1994-2001.

[m] Practice Committee of the American Society for Reproductive Medicine. Fertil Steril 2019;112:1022-1033.

[n] Skinner R, et al. Lancet Oncol 2017;18:e75-e90.

[o] van Dorp W, et al. J Clin Oncol 2016;34:3440-3450.

[p] Lambertini M, et al. J Natl Cancer Inst 2022;114:400-408.

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COMPREHENSIVE PSYCHOSOCIAL CONSIDERATIONS THROUGHOUT TREATMENT: INDIVIDUAL

EVALUATION

- Psychosocial factors:
  - Evaluate for current and past trauma history (eg, adverse childhood experiences [ACEs], medical-related trauma, abusive relationships)
  - Developmental and cognitive function
  - Communication and information delivery preferences
    - ◊ Assess health literacy
  - Preferred coping style of patient/family/caregiver
  - Evaluate distress and adjustment through screening for depression and anxiety
    - ◊ If interest is expressed, provide opportunity for patient to share their cancer story
    - ◊ See NCCN Guidelines for Distress Management
  - Evaluate for current and past psychiatric symptoms and treatments, including anxiety, depression, suicidal thoughts, eating disorders, and self injurious behavior
  - Involvement/interruption of school/work
  - Living status
    - ◊ Alone
    - ◊ Spouse/partner
    - ◊ Friends/roommates
    - ◊ Parents
    - ◊ Children
    - ◊ Unhoused or nonstable living environment
  - Impact of cancer on identity
    - ◊ Personal values
    - ◊ Self-esteem
    - ◊ Relational identity
    - ◊ Body image and physical changes
    - ◊ Strengths/resilience
    - ◊ Future goals
  - Impact of cancer on support network
    - ◊ Social isolation
    - ◊ Lack of will to discuss diagnosis due to feelings of shame or embarrassment
    - ◊ Loss of family, caregiver, or friends
  - Assess for gender expression, gender identity, chosen name, pronouns, and sexual orientation
  - Patient or family/caregiver involvement with criminal justice system or family court system
  - Patient preference for information sharing and involvement of certain biological or chosen family members in care (and exclusion of others)

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SUPPORTIVE CARE SERVICES/INTERVENTIONS

- Discuss with patient/guardian regarding referral for neuropsychological assessment if there are concerns regarding the patient's cognitive function (eg, attention, memory, executive function) and/or prior to educational and career transitions, including returning to school/work after treatment.
- If psychosocial support specialists are present (including psychology, child life specialists, or integrative medicine), have them meet with the patient soon after diagnosis to address the patient's concerns regarding treatment or procedures, provide psychoeducation, and assist with coping mechanisms to reduce anxiety related to procedures.
- Refer AYA patients to a social worker, mental health provider, and community-based resources serving AYA patients to screen for symptoms of depression, anxiety, suicidal ideation/behaviors, and self-injurious behavior.
- Offer psychosocial support and counseling to help alleviate distress (NCCN Guidelines for Distress Management).
- Facilitate adherence to treatment by considering flexible treatment dates, consultation times, and procedures (evenings/weekends).
- Refer for educational and career services to address training/education, employment, disability disclosure, vocational adjustment training, and transition services (ie, social services, vocational counseling, occupational therapy, financial counselors).
- For all AYA patients, provide counseling around sexual health conversations and decision-making regarding the risks of treatment-related fertility impairment and discuss options for fertility preservation prior to the start of therapy. See Fertility, Reproductive Endocrine, and Sexual Health Considerations (AYAO-5).
- For patients with minoritized identities:
  - Practice cultural humility. Remain open to the impact of cultural and identity factors in treatment decision-making (see Cultural Humility in Cancer Care).
  - Offer additional psychosocial support and referrals surrounding stressors, stigma, or rejection related to their sexuality or gender identity (see Cancer and Gender Diversity and Young Adult Survivors United).
  - Ensure record system accurately states patient's pronouns and preferred name.
  - For transgender youth, consider referral to a specialized gender clinic for psychosocial support and coordination of gender-affirming medical care at the patient's discretion.
- Refer patients to legal services (if applicable) for estate or legacy planning, child custody concerns, or other legal issues.

Behavioral factors → AYAO-8
Relationships → AYAO-9
Socioeconomic issues → AYAO-11

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COMPREHENSIVE BEHAVIORAL CONSIDERATIONS THROUGHOUT TREATMENT: INDIVIDUAL

EVALUATION

- Behavioral factors:
  - Adherence to therapy
  - Ongoing assessment for tobacco, vaping, alcohol, cannabis, or other substance use as habits may change throughout care or related to treatment and symptom management
  - Assess for access to firearms or other dangerous objects in the home
  - Sexual behavior/risks/concerns
  - Assess nutritional requirement and potential deficits based on age
  - Exercise needs, hobbies, and recreational activities
  - Sleep patterns
- Use of both integrative therapies and complementary and alternative medicine (CAM)
- Existential/spiritual beliefs or exploration

SUPPORTIVE CARE SERVICES/INTERVENTIONS

- Adherence to therapy and safety plans
  - Educate about the expectations of treatment and explain the patient's responsibility to adhere to therapy.
  - Refer to health or pediatric psychologist for adherence intervention if needed; include a list of barriers and facilitators.
  - Provide education and/or guidance about all medications and their dose, purpose, and adverse effects prior to the start of treatment and every time there is a change in treatment.
  - Simplify and modify dosing schedule, when medically possible, to fit into AYA's lifestyle and normal activities.
  - Consider the use of technology (eg, smartphone applications).
  - Provide access to systematic and standardized symptom management for side effects related to cancer treatment (NCCN Guidelines for Supportive Care).
- Refer to smoking cessation program if needed (NCCN Guidelines for Smoking Cessation).
- Provide education and/or guidance on cannabinoid use (NCCN Guidelines for Antiemesis).
- Refer patients with signs, symptoms, or a history of a substance use disorder to a risk reduction or substance use counseling program.
- Provide education about the impact of treatment on sexual health and safe sexual practices (including oral, anal, and vaginal sex), in light of risk of infection, risk for bleeding, prevention of pregnancy, and sexually transmitted infections.
- Recommendations for diet and exercise
  - Provide education about potential diet/nutritional changes associated with cancer treatment and possible interventions. Refer to a registered dietitian-certified specialist in oncology (RD-CSO).
  - Provide education on physical conditioning and related health benefits during and following cancer treatment.[q] Refer to a rehabilitation specialist (ie, physiatrist, physical therapist, occupational therapist) to address physical impairments and initiate physical activity interventions. Medical evaluation and clearance by a physician (such as an oncologist or a physiatrist) are recommended prior to initiating exercise in patients in whom exercise modifications or precautions might be needed.
- Refer patients with sleep disturbance to supportive medicine or behavioral health for interventions (refer to NCCN Guidelines for Palliative Care).
- Assess patient's use of or desire for integrative therapies and interventions.
  - Refer to integrative oncologist when patient is interested in evidence-based integrative therapies to manage side effects of treatment and overall well-being.[r]
  - Monitor use of dietary supplements that might impact effectiveness of treatment or treatment medications.[s]
- Refer patients experiencing challenges with their faith or belief in a just or fair world to faith-based resources or activities (eg, church youth groups, mentors). Refer to a chaplain or spiritual counselor.
- If cancer and its therapy are expected to be associated with significant morbidity, referral to palliative care is warranted.

[q] Stout NL, et al. Cancer 2020;126:2750-2758.
[r] https://integrativeonc.org/practice-guidelines
[s] https://www.cancer.gov/about-cancer/treatment/cam/hp/dietary-interactions-pdq

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Relationships (AYAO-9)
Socioeconomic issues (AYAO-11)

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COMPREHENSIVE PSYCHOSOCIAL/BEHAVIORAL CONSIDERATIONS THROUGHOUT TREATMENT: RELATIONSHIPS

EVALUATION

- Family/caregiver/primary supports status
  - Interaction and relationship with parents/caregivers
  - Interaction and relationship with spouse/partner
    - ◊ Dating and intimacy
  - Interaction and relationship with sibling(s) and other family members/caregivers
  - AYA patients who are parents
- Peer relationships
- Gender identification and sexual orientation
- Participation in community and social activities (eg, religious organizations, clubs, athletics/recreation, music, youth groups)
- Communications with health care professionals
  - Assistance with preparation of advance directive
  - Assistance with end-of-life planning and goal setting (eg, what the patient would like to achieve prior to death)
  - Addressing issues related to privacy and confidentiality:
    - ◊ Encouragement of completion of HIPAA release form when the patient is of age of majority
  - Ongoing consideration for the level of information the AYA patient wishes to have and share regarding disease and/or treatment
  - Consideration of patient's heritage, culture, and familial structure when addressing psychosocial concerns and needs, including decision-making and end-of-life care

→ Supportive Cancer Services/ Interventions (AYAO-10)

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COMPREHENSIVE PSYCHOSOCIAL/BEHAVIORAL CONSIDERATIONS THROUGHOUT TREATMENT: RELATIONSHIPS

SUPPORTIVE CARE SERVICES/INTERVENTIONS

- Promote collaborative communication between AYA patients:
  - Parents/caregivers
  - Children
  - Spouse/partner
  - Siblings and other family members/caregivers
  - Friends
  - Social network
- Early in the treatment process encourage education and the completion of a medical power of attorney and a living will at age of majority.
  - Provide access to AYA-specific advanced care planning guides to determine a health care proxy.
- Provide identified family members/caregivers and partners with information about psychosocial support and behavioral services.
  - Increase awareness of and normalize the possible psychosocial issues associated with cancer diagnosis in AYAs, so that family members/caregivers and partners may continue to support the patient.
- Refer patients to resources that particularly offer support for AYA patients who are parents:
  - Identify resources for respite care for AYA patients with young children.
  - See Bright Spot Network
  - See Pickles Group
  - See Young Adult Survivors United
- Provide education and access to AYA-specific activities (see Online Resources for AYA Patients and Survivors)
  - Inpatient: Recreational activities to reduce boredom and foster social connections with peers
  - Outpatient
    - ◊ Social networking events and opportunities
    - ◊ Camp and retreat programs
  - Education on peer support: Establishing relationships with AYA patients and maintaining previous relationships
- Provide education and access to AYA-specific psychosocial support (see Online Resources for AYA Patients and Survivors)
  - Parent/caregiver support groups
  - AYA support groups (see Cactus Cancer Society)
  - Psychoeducational programs
- Provide education and access to supportive/palliative care and integrative oncology for stress and symptom management
  - Palliative/supportive care for symptom management
  - Integrative oncology including mind-body therapies, acupuncture, tai chi, yoga, music therapy, etc.
- Create flexible visiting hours and an environment that will encourage peers to visit AYA patients.
  - Ensure staff are trained to recognize normative expectations for AYA interactions with peers.
- Communicate directly with individual patients.
  - Provide psychoeducation and assistance exploring and documenting advance directive preferences.
  - Ask for permission to share information with identified family members/caregivers or supports.
  - Provide developmentally appropriate information about their cancer, treatment options, and potential side effects (see Online Resources for AYA Patients and Survivors). See NCCN Patient Guidelines for Adolescent and Young Adult Cancer.
  - If AYA patient is >18 years of age, provide information on and necessary forms that legally allow medical information to be shared with caregivers of patient's choice.
  - Always conduct medical and psychosocial care in the language preferred by the patient/family/caregiver. Use certified interpreters and do not rely on family members/caregivers, friends, or non-certified medical staff for interpretation.

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COMPREHENSIVE PSYCHOSOCIAL/BEHAVIORAL CONSIDERATIONS THROUGHOUT TREATMENT: SOCIOECONOMIC ISSUES

EVALUATION

- Insurance availability and security
  - Employer-provided
  - Parent's insurance
  - Medicaid
- Health insurance marketplace
- Assessment of risk for losing insurance
  - Loss of employment
  - Age out of parent's insurance
- Assessment of financial toxicity
- Risk for financial loss or bankruptcy
- Child care
- Transportation
- Accommodation if traveling to receive treatment
- Stability of housing and basic household socioeconomic needs

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SUPPORTIVE CARE SERVICES/INTERVENTIONS

- Link qualified AYA patients to Medicaid, Social Security, and/or disability insurance.
  - Provide information regarding medication assistance programs for patients with limited or no insurance.
  - Provide information regarding hospital pharmacy vouchers or low-cost medication programs.
  - Direct AYA patients to legal resources/advocates for assistance with understanding health insurance coverage and transition to independent health insurance coverage.
- Educate AYA patients about benefits for which they may qualify, such as short- or long-term disability, state disability benefits, and public assistance.
- Provide information on obtaining financial assistance for fertility-based services. Local and institutional grants may be available.
- Provide school support, education services, and work reintegration for patients.
- Refer for career counseling and/or education support as indicated.
- Encourage discussion with guidance counselors and educators about impact of cancer care on education.
- Refer to mental health expert as needed to assess the psychosocial impact of financial toxicity (eg, loss of employment, withdrawing from school, not being able to socialize with friends due to decreased income).
- Refer to transportation assistance programs (eg, van ride programs, voucher programs).
- Provide AYAs with a list of recommended and reliable online sources and financial support programs to access information related to their cancer.
  See Online Resources for AYA Patients and Survivors.
- Integrate financial assistance into AYA survivorship care plans.
- Consider need for long-term follow-up care for monitoring and treatment of late effects long after completion of treatment.

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SURVIVORSHIP

AYA cancer survivor[t] →

- **Develop a "Cancer Treatment Summary and Survivorship Care Plan." See NCCN Guidelines for Survivorship.**
- **Always conduct medical and psychosocial care in the language preferred by the patient/family/caregiver. Use certified interpreters and do not rely on family members/caregivers, friends, or non-certified medical staff for interpretation.**
- **Provide a periodic evaluation focusing on history, physical examination, and screening based on treatment exposures and risk for treatment-related late effects.**
- **Encourage relationship with primary health care provider for routine health issues and support end-of-treatment transition to primary care physician (PCP) with clear communication about PCP role and oncology team role.**
- **Provide assistance in establishing care with primary health care provider.**
- **Ensure patient is provided continuous access to mental health support (eg, support groups, individual therapy, AYA-specific organizations) and is aware of risks for mental health following treatment.**
- **Recommend vaccinations:**
  - **Human papillomavirus (HPV) vaccine is recommended for all survivors aged 9–26 years and may be given through age 45 (however, best protection is achieved at earlier ages). Clinicians should discuss HPV vaccination with survivors aged 27–45 years, and determine on an individual basis whether or not vaccination may be of benefit for individual survivors within this age group.[u]**
  - **Please see the CDC website for Interim Clinical Considerations for Use of COVID-19 Vaccines in the United States.**
  - **Annual influenza vaccine**
  - **See NCCN Guidelines for Survivorship for additional guidance on vaccines.**
- **Counsel regarding striving to meet physical activity guidelines and methods to reduce risk of long-term chronic health problems, recurrence, or new tumors (eg, avoiding smoking, increasing level of physical activity).**
- **Advocate for appropriate health care coverage.**
- **Due to the transient lifestyle of the AYA population (ie, attending college far from home, traveling to different locations to establish careers), it is imperative for the initial care team to communicate with the new care team (once established).**
- **AYAs should be screened for risky behavior (ie, substance use, tobacco/nicotine use, binge drinking/excessive alcohol use, cannabis use) on a regular basis and counseled on cessation.**
- **AYAs should be educated on safe sex practices/contraceptive options if they are not trying to actively conceive regardless of their infertility risk post treatment.**
- **AYAs should also be asked about sexual dysfunction on a routine basis and how it may be impacting interpersonal relationships/self-image.**

→ **Radiation Exposure and Recommendations Table (AYAO-13)**

[t] An individual is considered a cancer survivor from the time of diagnosis, through the balance of their life. Family members, friends, and caregivers are also impacted. Adapted with permission from the National Coalition for Cancer Survivorship as shown in the National Cancer Institute's About Cancer Survivorship Research: https://cancercontrol.cancer.gov/ocs/about/staff.html.

[u] Further details are here: http://www.cdc.gov/vaccines/hcp/acip-recs/vacc-specific/hpv.html.

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SURVIVORSHIP

RADIATION EXPOSURE	RECOMMENDATION Screening Recommendations (AYAO-B)	
Any radiation therapy (RT)	• Recommend a dental exam and cleaning every 6 mo for patients who received RT to head or upper neck regions	
Total body irradiation (TBI)	• Thyroid disease screening • Cardiovascular risk factor screening (dyslipidemia) when TBI >10 Gy • Ocular screening (cataracts)	• Screening for secondary malignant neoplasms (SMNs) • Impaired glucose metabolism screening when TBI >10 Gy • Gonadal dysfunction screening
Cranial or craniospinal radiation	• Neuroendocrine axis screening • Neuropsychological evaluation • Ocular screening • Gonadal dysfunction screening	• Audiologic evaluation for cranial radiation doses ≥30 Gy • Colorectal cancer screening (for lumbar, sacral, or whole spine)
Chest radiation	• Individual assigned female at birth: see NCCN Guidelines for Breast Cancer Screening and Diagnosis • Cardiovascular risk assessment and screening	• Screening for cardiomyopathy • Screening for valvular heart disease • Pulmonary function screening
Neck radiation	• Thyroid disease screening	• Consider carotid artery screening
Abdominal or pelvic radiation	• Colorectal cancer screening • Assessment of gonadal and sexual function/ dyspareunia • Screening for dyslipidemia (for abdominal radiation) • Consider high-risk obstetrics referral if pregnant	• Assessment for bowel dysfunction • Screening for kidney or bladder disease • Screening for impaired glucose metabolism (for abdominal radiation) • Consider screening for cardiac disease (cardiomyopathy and vascular disease) if RT to upper abdomen
Limb radiation	• Physical therapy to address post-radiation fibrosis-related deficits	• Orthopedic screening to assess fragility fractures • Avascular necrosis

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SURVIVORSHIP

SYSTEMIC THERAPY EXPOSURE	RECOMMENDATION Screening Recommendations (AYAO-B)	
Any chemotherapy	• Recommend a dental exam and cleaning every 6 mo for patients who received chemotherapy to head or upper neck regions, especially for head and neck cancer survivors	
Intrathecal chemotherapy and high CNS-penetrating systemic chemotherapy (high-dose methotrexate, Ara-C)	• Neuropsychological evaluation	
Corticosteroid	• Osteoporosis screening	
Alkylating agents	• Screening for kidney or bladder disease • Assessment of gonadal function	• Screening for treatment-related acute myeloid leukemia (t-AML) or myelodysplasia • Pulmonary function screening (for selected agents)
Anthracyclines	• Screening for cardiomyopathy • Screening for t-AML or myelodysplasia	
Bleomycin	• Pulmonary function screening	
Cisplatin/ carboplatin	• Cardiovascular risk assessment • Screening for kidney and/or bladder disease • Audiologic evaluation	• Screening for t-AML or myelodysplasia • Screening for peripheral neuropathy • Screening for gonadal function
Epipodophyllotoxins	• Screening for t-AML or myelodysplasia	
Vinca alkaloid	• Screening for peripheral neuropathy	

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DISEASE-SPECIFIC ISSUES RELATED TO AGE

Consider referral to an institution with multidisciplinary expertise in the management of the disease types specified below.

Acute Lymphoblastic Leukemia (ALL)
- NCCN Guidelines for ALL
- NCCN Guidelines for Pediatric ALL

Lymphomas
- NCCN Guidelines for Hodgkin Lymphoma
- NCCN Guidelines for Pediatric Hodgkin Lymphoma
- NCCN Guidelines for B-Cell Lymphomas
- NCCN Guidelines for Pediatric Aggressive Mature B-Cell Lymphomas

Central Nervous System (CNS) Tumors
- NCCN Guidelines for CNS Cancers
- NCCN Guidelines for Pediatric CNS Cancers

Bone and Soft Tissue Sarcomas
- NCCN Guidelines for Bone Cancer and NCCN Guidelines for Soft Tissue Sarcoma
- Rhabdomyosarcoma
  - Uncommon outside of the pediatric population
  - Rhabdomyosarcoma in AYA patients often has high-risk features and may be associated with inferior outcomes. Consider referral to a center familiar with rhabdomyosarcoma therapy depending on the clinical situation[1,2]
- Ewing sarcoma
  - Inferior outcomes for patients >18 years of age[3]
- Osteosarcoma

Colorectal Cancer
- NCCN Guidelines for Colon Cancer and NCCN Guidelines for Rectal Cancer
- Higher incidence of mucinous histology
- Higher incidence of signet ring cells and microsatellite instability (MSI)
- More likely advanced stage at diagnosis
- Greater germline and somatic mutation frequency, including mismatch repair deficiency[a,4]
- Decreased incidence of chromosomal instability
- Increased risk for additional malignancies

Melanoma
- Melanocytic tumors of uncertain malignant potential (MELTUMP) are more frequently seen in younger patients, and when suspected, consultation with a pathologist with expertise in atypical melanocytic lesions is recommended.
- Conventional melanomas in AYAs have a similar behavior and genomic signature when compared to melanomas in older patients. These patients should be offered similar treatment options. See NCCN Guidelines for Melanoma: Cutaneous.

Germ Cell Tumors
- NCCN Guidelines for Testicular Cancer
- NCCN Guidelines for Ovarian Cancer Including Fallopian Tube Cancer and Primary Peritoneal Cancer

Breast Cancer
- NCCN Guidelines for Breast Cancer

Thyroid Cancer
- NCCN Guidelines for Thyroid Carcinoma

[a] *BRCA2*, *ATM*, *MSH2*, and *ATR* have been noted more in the AYA than in the adult population (Tricoli JV, et al. Cancer 2018;124:1070-1082).

[1] Ferrari A, et al. Lancet Child Adolesc Health 2022;6:545-554.
[2] Harrison DJ, et al. Pediatr Blood Cancer 2024;71:e30847.
[3] Leavey PJ, et al. J Clin Oncol 2021;39:4029-4038.
[4] Stoffel EM, et al. Gastroenterology. 2018;154:897-905.e1.

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DISEASE-SPECIFIC ISSUES RELATED TO AGE

Management of Cancer During Pregnancy (See Discussion)

- **While cancer in pregnancy is relatively uncommon, it complicates about 1 out of 1000 pregnancies.**
- **Cervical and breast cancer account for about 50% of cancers diagnosed during pregnancy and hematologic cancers account for about 25% of cancers diagnosed during pregnancy. Other cancers less commonly seen include melanoma, ovarian, thyroid, and colon. Other cancers are exceedingly rare.**
- **Individuals diagnosed with cancer during pregnancy should be treated by a multidisciplinary team involving medical, surgical, and radiation oncologists; gynecologic oncologists; obstetricians; and perinatologists as appropriate.**
- **Selection of an appropriate treatment plan is dependent on the cancer type, tumor biology, tumor stage, and gestational age of the fetus.**
- **Referral to tertiary cancer centers with expertise in diagnosis and treatment of cancer during pregnancy, maternal-fetal medicine, and knowledge of the physiologic changes that occur during pregnancy should be strongly encouraged.**
- **Cancer therapy during the first trimester is associated with the highest risk.**
- **If feasible, chemotherapy should be avoided during the first trimester because of greater risk of teratogenic effects and intrauterine fetal death.**
- **Certain chemotherapeutic agents are safer than others depending on the agent's mutagenic effect, ability to cross the placental barrier, and side effect profile.**
- **RT is contraindicated during pregnancy. In very rare instances when RT is necessary, it should be delivered in the lowest effective therapeutic doses using techniques to minimize fetal exposure.[4]**
- **Risk of transplacental transfer of cancer is diagnosis dependent.**
- **Most cancers can be adequately treated with medication that has been deemed not to cause harm for both the pregnant patient and the fetus. However, full counseling about all options regarding the pregnancy, including termination per patient decision or to expedite use of necessary teratogenic medication should be incorporated in all cancer treatment discussions for pregnant AYA patients.**
- **For disease-specific recommendations on management of cancer during pregnancy, see:**
  - **NCCN Guidelines for Breast Cancer**
  - **NCCN Guidelines for Cervical Cancer**
  - **NCCN Guidelines for Chronic Myeloid Leukemia**
  - **NCCN Guidelines for Hodgkin Lymphoma**
  - **NCCN Guidelines for Myeloproliferative Neoplasms**

[4] Silverstein J, et al. JCO Oncol Pract 2020;16:545-557.

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DISEASE-SPECIFIC ISSUES RELATED TO AGE

Reported Outcomes and Recommendations for Select Teratogenic Agents During Pregnancy[b,1-6]

Antineoplastic Agent	Reported Pregnancy Outcomes	Recommendations
Antimetabolites		
Capecitabine	• First trimester (T1): No malformations or adverse events	Use with caution
Cytarabine	• T1–third trimester (T3): Intrauterine fetal demise (IUFD), neonatal death (ND), intrauterine growth restriction (IUGR), dilated cardiomyopathy, atrial septal defect, ventricular septal defect, limb malformations, bilateral microtia, bilateral atresia of external auditory canals, absent digits, absent radii, brachycephaly, hypoplasia of anterior cranial base and mid face, micrognathia	Avoid use if possible
5-FU	• T1–T3: IUFD, ND, IUGR, syndactyly, cleft hands, absent phalanges, high arched palate, microcephaly, flat nasal bridge, congenital bilateral ureteral reflux, down syndrome, talipes, inguinal hernia	Avoid use if possible
Gemcitabine	• T1–T3: No malformation or adverse events	Use with caution
6-Mercaptopurine	• T1–T3: IUFD, ND, IUGR	Avoid use if possible
Methotrexate	• T1: Spontaneous miscarriage, cranial dysostosis with delayed ossification, hypertelorism, wide nasal bridge, micrognathia, ear abnormalities, skeletal abnormalities, ambiguous genitalia • Second trimester (T2)–T3: No malformations or adverse events	• T1: Avoid use if possible • T2–T3: Use with caution
Pemetrexed	• Limited to animal models: Cleft palate, protruding tongue, fused lumbar vertebrae, incomplete ossification of talus and skull, decreased fetal weight	Avoid use if possible
Trifluridine and tipiracil	• Limited to animal data: Decreased fetal weight, cleft palate, ectrodactyly, alterations of great vessels, skeletal anomalies	Avoid use if possible
Platinum Agents		
Carboplatin	• T2–T3: 1 reported spontaneous miscarriage, fetus with gastroschisis • Additional reports without malformations or adverse events	Avoid use if possible
Cisplatin	• T2–T3: 1 reported cerebral ventriculomegaly and cerebral atrophy • Additional reports without malformations or adverse events	Avoid use if possible

[b] This is not an exhaustive list of all antineoplastic, immunotherapy, and targeted agents, but is based upon available literature review, case reports, and animal studies. It is difficult to completely attribute all adverse outcomes to an individual agent as the majority of agents are used in combination with other cytotoxic agents. All decisions regarding chemotherapy administration should be based upon explicit discussion with the patient regarding the potential risks of the antineoplastic agent to the patient and fetus in conjunction with the risks of not providing cancer-directed therapy. (https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm)

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DISEASE-SPECIFIC ISSUES RELATED TO AGE

Reported Outcomes and Recommendations for Select Teratogenic Agents During Pregnancy (continued)[b,1-6]

Antineoplastic Agent	Reported Pregnancy Outcomes	Recommendations
Antimitotic Agents		
Docetaxel	• T1–T3: 1 reported pyloric stenosis • Additional reports without malformations or adverse events	Use with caution
Paclitaxel	• T1–T3: 1 pyloric stenosis • Additional reports without malformations or adverse events	Use with caution
Vinblastine	• T1–T3: 1 reported plagiocephaly, 1 syndactyly • Additional reports without malformations or adverse events	Use with caution
Vincristine	• T1–T3: IUFD, ND, spontaneous miscarriage, hydrocephalus, atrial septal defect, absent digits, absent radii	Avoid use if possible
Topoisomerase Inhibitors		
Etoposide	• T2–T3: No malformations or adverse events	Use with caution
Irinotecan	• T2–T3: No malformations or adverse events	Use with caution
Mitoxantrone	• T2–T3: No malformations or adverse events	Use with caution
Anthracyclines		
Bleomycin	• T1–T3: 1 reported plagiocephaly, 1 reported syndactyly • Additional reports without malformations or adverse events	Use with caution
Daunorubicin	• T1–T3: IUFD, IUGR, congenital adherence of iris to the cornea, syndactyly, rectal atresia	Avoid use if possible
Doxorubicin	• T1–T3: IUFD, holoprosencephaly, plagiocephaly, IgA deficiency, pyloric stenosis, syndactyly, hip subluxation, double cartilage in both ears, congenital bilateral ureteral reflux, talipes, Down syndrome	Avoid use if possible
Epirubicin	• T1–T3: IUFD, ND	Avoid use if possible
Idarubicin	• T1–T3: IUFD, IUGR, dilated cardiomyopathy, ventricular septal defect, short digits and limbs, macrognathia	Avoid use if possible

[b] This is not an exhaustive list of all antineoplastic, immunotherapy, and targeted agents, but is based upon available literature review, case reports, and animal studies. It is difficult to completely attribute all adverse outcomes to an individual agent as the majority of agents are used in combination with other cytotoxic agents. All decisions regarding chemotherapy administration should be based upon explicit discussion with the patient regarding the potential risks of the antineoplastic agent to the patient and fetus in conjunction with the risks of not providing cancer-directed therapy. (https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm)

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DISEASE-SPECIFIC ISSUES RELATED TO AGE

Reported Outcomes and Recommendations for Select Teratogenic Agents During Pregnancy (continued)[b,1-6]

Antineoplastic Agent	Reported Pregnancy Outcomes	Recommendations
Alkylating Agents		
Busulphan	• T1–T3: 1 reported pyloric stenosis, 1 unilateral kidney agenesis (both in T2) • Additional reports without malformations or adverse events	Use with caution
Cyclophosphamide	• T1–T3: IUFD, ND, IUGR, IgA deficiency, holoprosencephaly, pyloric stenosis, single coronary artery, inferior vena cava abnormality, imperforate anus, esophageal atresia, umbilical hernia, absent digits, cleft palate, hip subluxation, Down syndrome, talipes, congenital bilateral ureteral reflux, eye abnormalities • One patient with development of thyroid cancer and neuroblastoma in adolescence	Avoid use if possible
Dacarbazine	• T1–T3: 1 reported plagiocephaly, 1 reported syndactyly • Additional reports without malformations or adverse events	Use with caution
Ifosfamide	• T2: No malformations or adverse events	Use with caution
Procarbazine/ Mechlorethamine	• T1–T3: 1 reported hydrocephaly with ND (T1), 1 reported syndactyly (T2)	Use with caution
Multikinase/Tyrosine Kinase Inhibitors		
Encorafenib	• Limited to animal data: Spontaneous abortion	Avoid use if possible
Fruquintinib	• Limited to animal data: Visceral abnormalities, skeletal abnormalities, edema	Avoid use if possible
Imatinib	• T1: Communicating hydrocephalus, cerebellar hypoplasia, craniosynostosis, meningocele, cardiac defects, omphalocele, kidney abnormalities, skeletal abnormalities, cleft palate, polydactyly, pyloric stenosis, hypospadias • T2–T3: No malformations or adverse events	• T1: Avoid use if possible • T2–T3: Use with caution
Regorafenib	• Limited to animal data: Cardiovascular, genitourinary, skeletal malformations	Avoid use if possible

[b] This is not an exhaustive list of all antineoplastic, immunotherapy, and targeted agents, but is based upon available literature review, case reports, and animal studies. It is difficult to completely attribute all adverse outcomes to an individual agent as the majority of agents are used in combination with other cytotoxic agents. All decisions regarding chemotherapy administration should be based upon explicit discussion with the patient regarding the potential risks of the antineoplastic agent to the patient and fetus in conjunction with the risks of not providing cancer-directed therapy. (https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm)

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DISEASE-SPECIFIC ISSUES RELATED TO AGE

Reported Outcomes and Recommendations for Select Teratogenic Agents During Pregnancy (continued)[b,1-6]

Antineoplastic Agent	Reported Pregnancy Outcomes	Recommendations
Monoclonal Antibodies		
Bevacizumab	• With systemic use, evidence limited to animal studies: Gross and skeletal fetal abnormalities, decreased fetal weight	Avoid use if possible
Cetuximab	• Limited to animal studies: Spontaneous abortion, impairment of placental, lung, cardiac, skin, and neural development	Avoid use if possible
Panitumumab	• Limited to animal studies: Spontaneous abortion; no additional evidence of teratogenesis	Avoid use if possible
Ramucirumab	• No animal studies have been conducted to evaluate effects on fetal development; theoretical significant effect on embryofetal development due to VEGF/VEGF2 role in angiogenesis	Avoid use if possible
Rituximab	• T1–T3: ND, spontaneous miscarriage (T1), cardiac malformations, talipes	Avoid use if possible
Sacituzumab govitecan	*No animal studies have been conducted to evaluate effects on fetal development; theoretical significant effect on embryofetal development due to SN-38 genotoxicity	Avoid use if possible
HER2/Anti-ERBB2 Agents		
• Lapatinib • Pertuzumab • Trastuzumab • Trastuzumab-emtansine	• T1–T3: Oligohydramnios, congenital respiratory tract disorders, neonatal kidney failure	Avoid use if possible

[b] This is not an exhaustive list of all antineoplastic, immunotherapy, and targeted agents, but is based upon available literature review, case reports, and animal studies. It is difficult to completely attribute all adverse outcomes to an individual agent as the majority of agents are used in combination with other cytotoxic agents. All decisions regarding chemotherapy administration should be based upon explicit discussion with the patient regarding the potential risks of the antineoplastic agent to the patient and fetus in conjunction with the risks of not providing cancer-directed therapy. (https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm)

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DISEASE-SPECIFIC ISSUES RELATED TO AGE

Reported Outcomes and Recommendations for Select Teratogenic Agents During Pregnancy (continued)[b,1-6]

Antineoplastic Agent	Reported Pregnancy Outcomes	Recommendations
Immune Checkpoint Inhibitors		
• Atezolizumab • Avelumab • Cemiplimab • Dostarlimab • Durvalumab • Ipilimumab • Nivolumab • Pembrolizumab • Relatlimab • Tremelimumab	• T1–T3: 1 spontaneous abortion secondary to maternal antiphospholipid syndrome, 1 ND secondary to pneumonitis, 1 case of transient congenital hypothyroidism	Use with caution
Other Targeted Therapies		
Abemaciclib	• Limited to animal studies: Teratogenicity, decreased fetal weight	Avoid use if possible
Alectinib	• Animal studies: Spontaneous abortion, decreased fetal weight, retroesophageal subclavian artery, dilated ureter, thymic cord, small ventricle, thin ventricle, reduced number of sacral and caudal vertebrae • T1–T3: Case reports without malformations or adverse events	Use with caution
Everolimus	• Limited to animal studies: Spontaneous abortion, IUFD, skeletal abnormalities	Avoid if possible
Olaparib	• Limited to animal studies: Spontaneous abortion, anophthalmia, microphthalmia, skeletal abnormalities, fused exoccipital, diaphragmatic hernia, abnormalities within the lung, thymus, liver, ureter, or umbilical artery	Avoid use if possible
Palbociclib	• Limited to animal studies: Reduced fetal weight, skeletal variations	Use with caution
Ribociclib	• Limited to animal studies: Spontaneous abortion, reduced fetal weight, other fetal abnormalities	Avoid use if possible

[b] This is not an exhaustive list of all antineoplastic, immunotherapy, and targeted agents, but is based upon available literature review, case reports, and animal studies. It is difficult to completely attribute all adverse outcomes to an individual agent as the majority of agents are used in combination with other cytotoxic agents. All decisions regarding chemotherapy administration should be based upon explicit discussion with the patient regarding the potential risks of the antineoplastic agent to the patient and fetus in conjunction with the risks of not providing cancer-directed therapy. (https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm)

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SCREENING RECOMMENDATIONS FOR AYA SURVIVORS

Long-Term Follow-Up Guidelines for Survivors of Childhood, Adolescent, and Young Adult Cancers

- The recommendations represent only key aspects; for more information, refer to the Children's Oncology Group (COG) website, www.survivorshipguidelines.org.
- See also the NCCN Guidelines for Survivorship.
- The COG Guidelines are based on exposures used in the treatment for pediatric cancer. As such, the recommendations are applicable to many survivors of cancers that span across adolescence and young adulthood, such as acute leukemias, Hodgkin and non-Hodgkin lymphomas, medulloblastomas, and sarcomas. In addition, since the treatment exposures for some young adult cancers, such as germ cell tumors in individuals with testes, are similar to pediatric cancer treatments (eg, cisplatin, bleomycin, abdominal irradiation), the recommendations may be applicable. In contrast, the COG recommendations are generally not applicable to survivors of typical adult cancers occurring during young adulthood, such as breast, colorectal, and ovarian cancers.
- The risk for many late effects may be influenced by family history, lifestyle behaviors, and comorbid health conditions. The following recommendations are based on the treatment exposure; timing and intensity of screening may be adapted based on additional risk factors.
- Most survivors will have multiple treatment exposures, and therefore may have multiple screening needs.

Neuroendocrine Axis Screening (selected outcomes)[a]

- Growth hormone deficiency
  - High-risk population: Radiation dose to hypothalamic-pituitary-adrenal (HPA) axis ≥18 Gy
  - Screening recommendation: Height, weight, and body mass index (BMI) every 6 months until growth is completed, then yearly. Note: Most AYA patients will have attained (or nearly attained) final height; the significance and management of growth hormone status among survivors who attained their final height is controversial.
  - Consider endocrine consultation for height below the third percentile on the growth curve.
- Central hypothyroidism
  - High-risk population: Total radiation dose to HPA axis ≥15 Gy
  - Screening recommendation: Thyroid-stimulating hormone (TSH) and free T4, yearly
- Gonadotropin deficiency
  - High-risk population: Total radiation dose to HPA axis ≥30 Gy
  - Screening recommendation: Follicle-stimulating hormone (FSH), luteinizing hormone (LH), and testosterone for individuals with testes and FSH, LH, anti-Müllerian hormone (AMH), and estradiol for individuals with ovaries, as clinically indicated. Referral to a reproductive endocrinologist is recommended.
- Central adrenal insufficiency
  - High-risk population: Total radiation dose to HPA axis ≥30 Gy
  - Screening recommendation: 8:00 AM serum cortisol, yearly after treatment. Refer to endocrinology for further testing if level is <13 mcg/dL or <365 nmol/L.
- Routine endocrine, ophthalmologic, and dental evaluations are recommended for patients with selected radiation exposure and/or TBI for hematopoietic cell transplant (HCT).
- Radiation-related effects - Consider multidisciplinary consultation, which includes a radiation oncologist for the optimal method to reduce radiation-induced late effects.

[a] Radiation doses refer to dose when treatment is delivered in conventional fractionation.

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SCREENING RECOMMENDATIONS FOR AYA SURVIVORS

Dermatology Evaluation
- High-risk population: Patients who previously received RT
- Screening recommendation: Annual skin examinations performed by a dermatologist
- Provide recommendation for management of treatment-related hair loss.

Neuropsychological Evaluation
- The risk of neuropsychological deficits is related to tumor location and specific treatment interventions (radiation dose, radiation treatment volume, and surgical intervention). However, subtle deficits in executive function, sustained attention, memory, and processing speed may occur with cranial radiation, particularly in higher doses given to treat brain tumors and head/neck tumors.
  - Include information/education regarding neuropsychological evaluations, and/or cognitive screening for AYA survivors. Neuropsychological evaluations and/or cognitive screening may be warranted regardless of treatment history and should be offered if patient raises concerns.
- Screening recommendation: All AYA survivors of cancer should undergo routine screening for difficulties in school or work during follow-up so that providers can identify the need for referral early in the survivorship phase.[1] In particular, patients with brain tumors and those treated with cranial/craniospinal radiation, intrathecal chemotherapy, and/or high-dose methotrexate (eg, osteosarcoma therapy) should be referred for neuropsychological evaluation.
  - The goal is to reinforce screening for subtle neurocognitive deficits in AYA survivors including those treated for non-CNS tumors in long-term follow-up, so that those in need of neurocognitive/neuropsychological interventions can be referred for treatment.
  - Neurotoxicity: Conduct routine evaluation for symptoms of peripheral neuropathy. See NCCN Guidelines for Adult Cancer Pain.

Breast Cancer Screening
- High-risk population: See NCCN Guidelines for Genetic/Familial High-Risk Assessment: Breast, Ovarian, Pancreatic, and Prostate
- Screening recommendation: See NCCN Guidelines for Breast Cancer Screening and Diagnosis
- Survivors <30 years who receive thoracic RT should be considered for enhanced screening.

Cardiovascular Risk Assessment and Screening
- High-risk population: ≥15 Gy combined with anthracycline or mediastinal/chest radiation ≥35 Gy alone, or abdominal radiation, or TBI
- Screening recommendation: Optimize cardiovascular risk factors, including blood pressure, lipid profile, and blood glucose
- Baseline ECG post treatment: Not for treatment decision, but definitely for when there is a subsequent issue for comparison
- Screening for ischemic coronary artery disease remains controversial; consider cardiology consultation (5–10 years after radiation) in patients who received ≥35 Gy chest radiation
- Cardiac toxicity: Provide periodic echocardiograms for select patients.[2] A post-treatment baseline electrocardiogram (ECG) is recommended after completion of treatment. When cumulative dose of anthracycline is anticipated to exceed 250 mg/m^2, consider adding a cardioprotectant (eg, dexrazoxane)[3]; however, there may be variability depending on institutional guidelines.

Screening for Cardiomyopathy/Asymptomatic Heart Failure[4]
- High-risk population: Cumulative anthracycline dose ≥250 mg/m^2, doxorubicin equivalent dose; chest radiation ≥30 Gy; combination of anthracycline ≥100 mg/m^2 and chest irradiation ≥15 Gy
- Moderate-risk population: Cumulative anthracycline dose ≥100 mg/m^2 and <250 mg/m^2 with <15 Gy or no chest radiation; or chest radiation alone at dose of 15–30 Gy
- Screening recommendation: Echocardiogram every 2 years for patients who are high risk and every 5 years for patients who are moderate risk. (Note: Frequency of testing is dependent on dose of exposure. The frequency of testing has not been established for breast cancer survivors treated with lower cumulative doses of anthracyclines.) A baseline ECG is recommended at entry into long-term follow-up if anthracycline exposure or chest radiation ≥15 Gy; repeat as clinically indicated.

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SCREENING RECOMMENDATIONS FOR AYA SURVIVORS

Screening for Valvular Heart Disease
- High-risk population: Chest radiation ≥35 Gy
- Low-risk population: Chest radiation 15–35 Gy
- Screening recommendation: Echocardiogram every 2 years for patients who are high risk and every 5 years for patients who are lower risk

Pulmonary Function Screening
- High-risk population: Chest radiation ≥15 Gy (or radiation to large volume of lung), TBI (≥6 Gy in single fraction or ≥12 Gy fractionated), partial thoracic RT with mean lung dose ≥14 Gy, bleomycin, combination of chest radiation and bleomycin, and selected alkylating agents (ie, busulfan, carmustine)
- Screening recommendation: Pulmonary function tests (PFTs) (including diffusing capacity of the lung for carbon monoxide [DLCO] and spirometry) as a post-therapy baseline and then as clinically indicated in patients with abnormal results or progressive pulmonary dysfunction
- PFT is the desired method for optimally evaluating pulmonary function in patients who can perform this test.

Thyroid Disease Screening
- Thyroid disorders: Hypothyroidism, thyroid cancer, and hyperthyroidism
- High-risk population: Radiation field includes or is in close proximity to the thyroid gland (Neuroendocrine Axis Screening [AYAO-B 1 of 8] for high-dose cranial radiation)
- Use of immune checkpoint inhibitors
- Screening recommendation: TSH, free T4, and thyroid/neck exam yearly

Colorectal Cancer Screening
- High-risk population: All patients who received any abdominal or pelvic radiation or TBI or spinal radiation (lumbar, sacral, or whole spine)
- Screening recommendation: Colorectal cancer screening, ie, colonoscopy every 5 years, multitarget stool DNA/RNA-based test every 3 years, or blood-based cell-free DNA test every 3 years, based on informed decision-making between survivor and clinician starting at age 30 or 5 years after radiation, whichever occurs last
- For patients with known colorectal cancer predisposition syndromes, see NCCN Guidelines for Genetic/Familial High-Risk Assessment: Colorectal, Endometrial, and Gastric

Screening for Kidney and/or Bladder Disease
- Kidney insufficiency and secondary kidney/renovascular hypertension
  - High-risk population: Radiation ≥10 Gy, combination of radiation with nephrotoxic agents (eg, cisplatin, ifosfamide, aminoglycosides, amphotericin, immunosuppressants, history of HCT), and history of nephrectomy
  - Screening recommendation: Post-therapy baseline blood urea nitrogen (BUN), creatinine, Na, K, Cl, CO_2, Ca, Mg, and PO_4; repeat as clinically indicated; measure blood pressure yearly
- Kidney toxicity: Provide periodic monitoring of kidney function and electrolytes in patients treated with platinum-based therapies and ifosfamide-based chemotherapy.
- Hemorrhagic cystitis/bladder fibrosis
  - High-risk population: Cyclophosphamide >3 gm/m^2, ifosfamide, and pelvic radiation ≥30 Gy
  - Screening of pertinent urinary tract symptoms is recommended.
- Bladder cancer
  - High-risk population: Cyclophosphamide combined with pelvic radiation
  - Screening of pertinent urinary tract symptoms is recommended.

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SCREENING RECOMMENDATIONS FOR AYA SURVIVORS

Assessment for Gonadal Function

- See also the NCCN Guidelines for Survivorship
- Individuals with testes
  - Individuals with bilateral orchiectomy will have both impaired fertility and Leydig cell dysfunction and will need referral for hormonal replacement therapy.
  - Impaired fertility: Permanent or temporary azoospermia or oligospermia can occur for variable periods of time post-therapy. Below are risk assessments according to the Pediatric Initiative Network (PIN) Risk Stratification System for Treatment-Related Gonadal Insufficiency and Infertility in Childhood Adolescent and Young Adult Cancer[5]:
    - ◊ High level of risk:
      - Alkylator: ≥4 g/m² cyclophosphamide equivalent dose (CED)[6]
      - HCT: Including alkylator ± TBI; myeloablative and reduced-intensity regimens
      - Radiation to testes: ≥6 Gy[b,7,8,9,10]
      - Radiation to hypothalamus: >40 Gy
    - ◊ Significantly increased risk:
      - Cisplatin: >500 mg/m²
      - Radiation to testes: >1–5.9 Gy
      - Radiation to hypothalamus: 30–39.9 Gy
      - Retroperitoneal lymph node dissection
    - ◊ Minimally increased risk:
      - Alkylator: <4 g/m² CED
      - Carboplatin or cisplatin: Any dose <500 mg/m²
      - Radiation to testes: 0.2–<1 Gy
      - Radiation to hypothalamus: 26–29.9 Gy
    - ◊ Screening recommendation: Semen analysis as requested by patient or evaluation of impaired fertility; periodic evaluation over time is recommended as resumption of spermatogenesis can occur post therapy[11,12]
  - Leydig cell dysfunction
    - ◊ High-risk population: Testicular irradiation combined with head/brain irradiation, testicular radiation dose ≥12 Gy in combination with alkylating agents, cyclophosphamide conditioning for HCT, or unilateral orchiectomy[7,8,10,12-15]
    - ◊ Screening recommendation: Early morning testosterone as clinically indicated in patients with clinical signs and symptoms of testosterone deficiency
- Individuals with ovaries
  - Individuals with bilateral oophorectomy will have both impaired fertility and ovarian hormone deficiency and will need referral for hormonal replacement therapy.
  - Impaired fertility (acute ovarian insufficiency or primary ovarian insufficiency): When marked depletion of the ovarian reserve occurs secondary to cancer therapy, ovarian insufficiency can develop soon after the completion of therapy. However, there may be a reproductive window before the onset of primary ovarian insufficiency. Below are risk assessments according to the PIN Risk Stratification System for Treatment-Related Gonadal Insufficiency and Infertility in Childhood Adolescent and Young Adult Cancer[5]:
    - ◊ High level of risk
      - Alkylator: Prepubertal (>12 g/m² CED); pubertal (>8 g/m² CED)
      - HCT: Including alkylator ± TBI; myeloablative and reduced-intensity regimens
      - Radiation to ovary: Prepubertal (≥15 Gy); pubertal (≥10 Gy)
      - Radiation to hypothalamus: >40 Gy
    - ◊ Significantly increased risk
      - Alkylator: Prepubertal (8–12 g/m² CED); pubertal (4–8 g/m² CED)
      - Radiation to ovary: Prepubertal (<15 Gy); pubertal (<10 Gy)
      - Radiation to hypothalamus: 30–39.9 Gy
    - ◊ Minimally increased risk:
      - Alkylator: Prepubertal (<8 g/m² CED); pubertal (<4 g/m² CED)
      - Carboplatin or cisplatin: Any dose

[b] Radiation dose to testes (≤6 Gy azoospermia may be transient, ≥6 Gy azoospermia likely permanent and especially testicular dose ≥20 Gy), combination with alkylating agents, and/or genitourinary surgery.

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SCREENING RECOMMENDATIONS FOR AYA SURVIVORS

- – Radiation to hypothalamus: 22–29.9 Gy
  - ◊ Screening recommendation: Annual history and physical (H&P) (menstrual and pregnancy history, history of hormonal therapy, Tanner staging) only. Laboratory screening (ie, FSH, estradiol, AMH) for those with menstrual cycle dysfunction or those who desire information about future fertility. Bone density evaluation (dual-energy x-ray absorptiometry [DEXA]) at baseline (entry to long-term follow-up) for those who had corticosteroids and/or HCT only.[18]
  - ◊ For patients who desire more information regarding potential fertility and/or interventions to preserve future fertility, referral to gynecology, reproductive medicine, or endocrinology is recommended for antral follicle count, ovarian reserve evaluation, consultation regarding assisted reproductive technologies, and post-pubertal survivors treated with potentially gonadotoxic chemotherapy and/or radiotherapy.
- ▸ Pregnancy in cancer survivors: Referral is recommended for patients with cardiomyopathy, cardiovascular risk factors, and history of pelvic radiation to maternal-fetal medicine.
- ▸ Patients with uterine radiation exposure are at increased risk for uterine vascular insufficiency, resulting in adverse pregnancy outcomes, such as spontaneous abortion, ND, low-birth-weight infant, fetal malposition, and premature labor. Factors associated with increased risk: Wilms tumor and associated Müllerian anomalies (ie, agenesis, hypoplasia), TBI, higher radiation dose to pelvis, and radiation dose ≥30 Gy.[19-21]
- ▸ Monitor for (intervening if necessary) vaginal stenosis that can develop after radiation.[22]

Screening for t-AML or Myelodysplasia

- High-risk populations: Epipodophyllotoxins, alkylating agents, heavy metals, and/or anthracyclines; autologous HCT
- Screening recommendation: Complete blood count (CBC) and bone marrow exam as clinically indicated based on symptoms
- The risk decreases significantly after 10 years from exposure to the therapeutic agent.

Audiologic Evaluation

- High-risk population: Cisplatin ≥360 mg/m^2, carboplatin conditioning for HCT, radiation involving the ear for doses ≥30 Gy, and combination of cisplatin and cranial/ear radiation
- Screening recommendation: Audiology testing as a post-therapy baseline and then every 5 years
- Cancer survivors with hearing impairment should be provided with psychoeducational evaluation and support related to educational, psychological, and social function.
- Ototoxicity: Conduct routine evaluations for tinnitus and periodic audiogram to monitor hearing loss associated with platinum-based chemotherapy. Consider sodium thiosulfate (STS) to reduce the risk of ototoxicity associated with cisplatin in patients with localized, non-metastatic, solid tumors. There are concerns about the use of STS in the metastatic setting.[23]

Cancer Predisposition

- Assessment for cancer predisposition is strongly recommended in the following settings:
  - ▸ A tumor that has a high likelihood of being part of a cancer predisposition syndrome (adrenocortical carcinoma, desmoid tumor, endolymphatic sac tumor, gastrointestinal stromal tumor, malignant peripheral nerve sheath tumor, medullary thyroid cancer, osteosarcoma, ovarian Sertoli cell or Sertoli-Leydig cell tumor, paraganglioma, pheochromocytoma, pleuropulmonary blastoma, renal cell carcinoma, rhabdoid tumor, schwannoma, atypical teratoid rhabdoid tumor, choroid plexus carcinoma, ciliary body medulloepithelioma, hemangioblastoma, optic pathway glioma, pineoblastoma, pituitary blastoma, retinoblastoma, subependymal giant cell astrocytoma [SEGA], cystic nephroma, juvenile myelomonocytic leukemia, meningioma, myelodysplastic syndrome)
  - ▸ Any bilateral tumor
  - ▸ More than one primary cancer
  - ▸ One or more first-degree relatives with cancer
  - ▸ Consanguinity
  - ▸ Diagnosis of early-onset cancers (ie, breast, colon, gastrointestinal, ovarian)
  - ▸ Diagnosis of a cancer predisposition syndrome in a relative
- For patients who may be at risk for cancer predisposition based on history, consider referral to genetic counseling or clinical genetics and referral for preconception or prenatal counseling.

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SCREENING RECOMMENDATIONS FOR AYA SURVIVORS

Screening for Other Treatment-Related Toxicities

- Immunotherapy-induced toxicity - There may be long-term toxicity, but more data on the AYA population are needed.
- Osteoporosis, osteonecrosis,[16] and osteopenia in selected populations[17]
- Quality of life (QOL)

Core Outcome Measures[c]

- All cancer diagnoses
  - Cause-specific mortality: Data extraction of cause of death
  - Subsequent neoplasm: Measurement by occurrence and type of neoplasm
  - Subfertility: Clinical diagnosis after ≥12 months of regular unprotected sexual intercourse
  - Organ dysfunction (site, chemotherapy, radiation, and/or surgery dependent):
    - ◊ Heart failure: Measurement by left ventricular (LV) systolic function on echocardiogram
    - ◊ Pulmonary dysfunction: Clinical diagnosis requiring supplemental oxygen or support
    - ◊ Kidney insufficiency: Clinical diagnosis of chronic kidney disease requiring medication or dialysis/transplant
    - ◊ Hepatoxicity: Clinical diagnosis of liver dysfunction
  - Measurement by height and weight data extraction
  - Psychosocial aspects of QOL: Measured by formal QOL evaluation
  - Physical aspects of QOL: Measured by formal QOL evaluation
- CNS tumor-specific outcome measures
  - Any CNS tumor:
    - ◊ Hypothalamic-pituitary dysfunction: Includes ACTH, GH, TSH, LH/FSH, and ADH deficiencies
    - ◊ Motor problems: Clinical diagnosis of paralysis, motor neuropathy, or movement disorder requiring support
    - ◊ Neurocognitive aspects of QOL: Measured by formal QOL evaluation
  - Craniopharyngioma:
    - ◊ Stroke: Clinical diagnosis requiring intervention or hospitalization
    - ◊ Temperature dysregulation: Clinical diagnosis with core temperature <35°C or requiring intervention
    - ◊ Visual problems: Clinical diagnosis of visual impairments or double vision requiring support
- CNS tumor-specific outcome measures (continued)
  - Embryonal tumor:
    - ◊ Posterior fossa syndrome (site specific): Clinical diagnosis requiring support in daily living activities
    - ◊ Hearing problems (site and therapy [cisplatin and/or radiation] related): Clinical diagnosis of hearing loss/deafness or severe tinnitus requiring support
  - Gliomas:
    - ◊ Visual problems (site specific): Clinical diagnosis of visual impairments or double vision requiring support

[c] The provided core outcome measures are modified from the international consensus statement for childhood cancer (van Kalsbeek RJ, et al. Nat Med 2023; 29:1340-1348) and tailored to focus on AYA patients with cancer, including those with one of the more common AYA CNS tumors, hematologic malignancies, or solid tumors.

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SCREENING RECOMMENDATIONS FOR AYA SURVIVORS

Core Outcome Measures (continued)[c]

- ◊ Seizures: Clinical diagnosis requiring medication or intervention

- Hematologic malignancy-specific measures
  - Any leukemia or lymphoma:
    - ◊ Heart failure (anthracycline and radiation exposure dependent): Measurement by LV systolic function on ECG
    - ◊ Neurocognitive aspects of QOL: Measured by formal QOL evaluation
  - Acute lymphoblastic leukemia (ALL):
    - ◊ Osteonecrosis (steroid dependent): Clinical diagnosis requiring medication or surgery
  - Hodgkin lymphoma:
    - ◊ Myocardial infarction (anthracycline and radiation exposure dependent): Clinical diagnosis including abnormal cardiac enzymes and ECG changes.
    - ◊ Pulmonary dysfunction (therapy and radiation exposure dependent): Clinical diagnosis requiring supplemental oxygen or support
  - Any treated with allogenic HCT:
    - ◊ Chronic graft-versus-host disease: Clinical diagnosis with a global severity score of moderate or severe

- Solid tumor-specific outcomes
  - Sarcomas:
    - ◊ Disfigurements: Clinical diagnosis of amputation or physical disfigurement requiring support
    - ◊ Reduced joint mobility (site dependent): Clinical diagnosis of reduced mobility in large joints requiring support
  - Germ cell tumor:
    - ◊ Male sexual dysfunction: Clinical diagnosis requiring medication or intervention
  - Breast cancer:
    - ◊ Breast reconstruction: Issues related to surgical outcomes, potential complications, and psychosocial impacts of breast reconstruction following mastectomy
  - Colon cancer:
    - ◊ Bowel function: Issues related to bowel function post-surgery, including potential complications such as bowel obstruction, incontinence, and changes in bowel habits
    - ◊ Ostomy: Issues related to living with an ostomy, including the impact on daily living, management of the ostomy, and psychosocial implications

[c] The provided core outcome measures are modified from the international consensus statement for childhood cancer (van Kalsbeek RJ, et al. Nat Med 2023; 29:1340-1348) and tailored to focus on AYA patients with cancer, including those with one of the more common AYA CNS tumors, hematologic malignancies, or solid tumors.

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PALLIATIVE CARE ACROSS THE DISEASE CONTINUUM

Palliative care focuses on symptom control, reduction of physical suffering or discomfort, and optimizing quality of life at any stage of a life-threatening disease (NCCN Guidelines for Palliative Care). A palliative approach to care is indicated across the cancer trajectory from diagnosis to end-of-life care. Referral to palliative care is appropriate when patients are being treated with curative intent and can be initiated at the time of initial diagnosis.[1,2,3] A palliative care team is one with resources and expertise to address the psychosocial, emotional, and physical challenges relevant to the patient.[4,5] Strategies to support a patient, particularly in the AYA population, must be individualized in the context of the family dynamic, including maturity of the patient and level of independence (both desired and actual).

END-OF-LIFE CONSIDERATIONS

- **Alleviating suffering**
  - **A palliative approach to care, alleviating suffering, and care consistent with patient goals are vitally important in the end-of-life phase of therapy.**
  - **It is imperative for health care professionals not to assume that AYA patients may be less inclined to discuss death and other end-of-life issues.[6,7,8,9]**
  - **Work to address physical, psychosocial, social, and spiritual suffering through systematic screening and partnership with interdisciplinary care team members. Consider referral to specialty palliative care when there is suffering that is refractory to primary team management.**
- **Communication (eg, with advanced care planning)**
  - **Determine who the AYA wants to be involved in decision-making. Identify individuals who will require psychosocial support during the AYAs cancer journey.**
  - **Consideration to include palliative care team or hospice services should be discussed early in treatment if prognosis is poor in order to provide the continuity of care and support for both patient and family/caregiver throughout their cancer experience.[1,2]**
  - **Discussion about end-of-life preferences and the formulation of an advance care planning document should begin early in treatment, but details should be individualized according to the preferences of the AYA patient and family/caregiver.[6,10,11]**
  - **Clinical teams should be aware of and work with their palliative care teams regarding local guidelines for concurrent palliative and cancer-directed care.**
  - **Provide AYAs with clear cancer prognostic information.**
  - **Provide care that is aligned with AYA's goals of care.**
- **Supportive care services (if needed)**
  - **Determine who the AYA wants to be involved in decision-making.**
  - **Identify individuals who will require psychosocial support during the AYA's cancer journey.**
  - **Involve child life specialists and/or psychosocial team member to discuss legacy projects and memory work with patient and family/caregiver.**
  - **Individual, family/caregiver, and cultural differences influence the preferred location of death. While many adolescents indicate a preference for dying at home, 80% die in hospitals.[12,13,14,15] Other AYAs and families choose to die in the hospital due to regional scarcities of home hospice, caregiver demand at the end of life, or personal preference. Every effort should be made to query and support AYA's preferred location of death. Clinicians with experience in end-of-life care should facilitate discussion about medical interventions such as nutrition/hydration, sedation, treatment cessation, and place of death.[5]**
  - **Ongoing psychosocial support is extremely important during the transition to end-of-life care. For family/caregiver and friends, grief from loss may begin before death.**
  - **When AYAs have children or young siblings, conduct an assessment of their psychosocial needs and/or refer to programs that have the capacity to assess these psychosocial needs.**
  - **Clinical teams should be aware of local availability and regulations surrounding 1) home-based palliative care, and 2) concurrent care—the ability to enroll on hospice while also receiving disease-directed therapy.**
  - **Programs should have a bereavement program, which, at a minimum, maintains lists of bereavement resources and makes, at least, one contact with bereaved family members after an AYA's death.**

References

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PALLIATIVE CARE ACROSS THE DISEASE CONTINUUM
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ABBREVIATIONS

ACE	adverse childhood experience
ALL	acute lymphoblastic leukemia
AMH	anti-Müllerian hormone
AYA	adolescent and young adult
BMI	body mass index
BUN	blood urea nitrogen
CAM	complementary and alternative medicine
CBC	complete blood count
CED	cyclophosphamide equivalent dose
CNS	central nervous system
COG	Children's Oncology Group
COS	controlled ovarian stimulation
DEXA	dual-energy x-ray absorptiometry
DLCO	diffusing capacity of the lung for carbon monoxide
ECG	electrocardiogram
FSH	follicle-stimulating hormone
GnRH	gonadotropin-releasing hormone
H&P	history and physical
HCT	hematopoietic cell transplant
HIPAA	Health Insurance Portability and Accountability Act
HPA	hypothalamic-pituitary-adrenal
HPV	human papillomavirus
IUFD	Intrauterine fetal demise
IUGR	intrauterine growth restriction
LH	luteinizing hormone
LV	left ventricular
MELTUMP	melanocytic tumors of uncertain malignant potential
MEN	multiple endocrine neoplasia
MSI	microsatellite instability
ND	neonatal death
PCP	primary care physician
PIN	Pediatric Initiative Network
QOL	quality of life
RD-CSO	registered dietitian-certified specialist in oncology
SEGA	subependymal giant cell astrocytoma
SGM	sexual gender minority
SMN	secondary malignant neoplasm
STS	sodium thiosulfate
t-AML	treatment-related acute myeloid leukemia
TBI	total body irradiation
TESE	testicular sperm extraction
TSH	thyroid-stimulating hormone

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NCCN Categories of Evidence and Consensus	
Category 1	Based upon high-level evidence (≥1 randomized phase 3 trials or high-quality, robust meta-analyses), there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2A	Based upon lower-level evidence, there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2B	Based upon lower-level evidence, there is NCCN consensus (≥50%, but <85% support of the Panel) that the intervention is appropriate.
Category 3	Based upon any level of evidence, there is major NCCN disagreement that the intervention is appropriate.

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NCCN Guidelines Version 2.2026
Adolescent and Young Adult (AYA) Oncology

Discussion

This discussion corresponds to the NCCN Guidelines for Adolescent and Young Adult Oncology. Last updated on February 27, 2026.

Table of Contents

Overview MS-2
Guidelines Update Methodology MS-3
Literature Search Criteria MS-3
Sensitive/Inclusive Language Usage MS-4
Risk Factors MS-4
Screening MS-6
Diagnosis MS-6
Comprehensive Care for AYA Patients with Cancer: Special Considerations MS-6
- Age-Appropriate Care MS-7
- Treatment Options MS-8
- Distress and Psychosocial Factors Impacting Treatment MS-11
- Impact of Cancer and Its Treatment on Fertility MS-12
- Fertility Preservation MS-13
- Contraception in AYA Patients with Cancer MS-18
- Management of Cancer During Pregnancy MS-18

Comprehensive Psychosocial and Behavioral Issues MS-21
- Individual Function MS-21
- Relationships MS-26
- Socioeconomic Issues MS-28

Survivorship Issues MS-29
- Late Effects in AYA Cancer Survivors MS-29
- Long-Term Follow-up MS-36

Palliative and End-of-Life Considerations MS-37
Summary MS-38
References MS-39

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Overview

Advances in cancer treatment have resulted in declining cancer mortality rates among the overall adolescent and young adult (AYA) patient population over the last decade; however, AYA patients continue to face significant challenges in cancer care due to gaps in knowledge regarding etiology, basic biology, treatment, and survivorship.[1] One of the main reasons for these knowledge gaps is that AYA patients have a low rate of participation in clinical trials, reported to consistently remain below 10%.[2-4] Participation of AYA patients in clinical trials has been decreasing since 2010, with the exception of AYAs with acute lymphoblastic leukemia (ALL).[4] In addition to the low rate of participation in clinical trials, several other factors contribute to poor outcomes in AYA patients, such as: differences in disease biology, lack of consistency in treatment approaches, poor adherence or intolerance to therapy, lack of health insurance, delays in diagnosis, and physician's lack of familiarity with cancer in the AYA population.[5,6] AYA patients also have unique developmental, medical, and psychosocial issues, which make adjustment to their disease, health maintenance, and financial hardships more challenging.[2,7-10]

The biology, epidemiology, and clinical outcomes affecting AYA patients are usually different from those affecting younger (<15 years) and older patients (>39 years) with cancer.[11,12] In addition, the genetic, physiologic, and pharmacologic changes associated with AYA patients may impact their ability to tolerate cancer therapy and their response to treatment. Moreover, short- and long-term toxicities impacting a young, independent patient—including the impact of treatment on fertility and sexual function—may disincentivize treatment, leading to gaps in adherence and poor outcomes. Addressing these issues and providing options that empower the patient at the time of initial cancer treatment may result in more successful implementation of the planned therapy. Unlike comprehensive geriatric assessment, which is helpful to physicians in developing a coordinated treatment plan and understanding the functional needs of older patients, no similar assessment has been reported for AYA patients. There continues to remain fewer evidence-based data to guide the treatment of AYA patients. AYA patients diagnosed with cancer should be recognized as a distinct age group that has unique medical and psychosocial needs.[13,14] The distinct biology of disease as well as age-related issues in AYA patients (such as fertility, long-term side effects, insurance/financial issues, transportation to clinic appointments, child care, interpersonal relationships, psychosocial support, and adherence to therapy) should be considered in the treatment decision-making process and during the transition of care from pediatric to adult medical teams.[15,16]

Based on the National Cancer Institute's (NCI's) Progress Review Group recommendations, the AYA patient is generally defined as an individual aged 15 to 39 years at the time of initial cancer diagnosis.[2,17] For 2024, the incidence of cancer in the United States in the AYA population was estimated to be 84,100 with 8890 cancer deaths.[18] Globally, up to 1.19 million cancer cases are estimated to be diagnosed in AYAs annually.[19] Compared to children <15 years, 5-year relative survival in AYA patients is worse for those with ALL, acute myeloid leukemia (AML), Hodgkin lymphoma (HL), non-Hodgkin lymphoma (NHL), astrocytomas, Ewing sarcoma, rhabdomyosarcoma, or osteosarcoma.[20] Additionally, for Ewing sarcoma, outcomes are worse for patients ≥18 years.[21-23] However, 5-year relative survival is better in AYA patients with medulloblastomas and germ cell tumors compared to in children with these tumors, possibly reflecting biologic differences in the tumors of each age group. Compared to adults aged ≥40 years, AYA patients tend to have better survival rates, except for those with breast and colorectal cancers.[20,24] Increasing age is associated with poorer prognosis in AYA patients with AML, NHL, Burkitt and Burkitt-like lymphoma, or rhabdomyosarcoma.[25]

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The spectrum of cancer types that affect the AYA population is unique and different from those that affect the population outside of AYA. Thyroid cancers, lymphomas, melanoma, testicular cancer, cervical cancer, bone and soft tissue sarcomas, leukemias, central nervous system (CNS) cancers, breast cancer, and colorectal cancer account for the majority of cancers in this age group.[1,26]

Quality care for AYA patients with cancer is tied to timely detection and initiation of treatment, adherence to treatment, and access to a multidisciplinary team of health care professionals who are well-versed in the specific age-related/developmental issues relevant to this patient population.[27,28] These issues include, but are not limited to, fertility and sexual function; long-term side effects; behavioral, psychosocial, and socioeconomic issues; transportation to clinic appointments; maintaining school and work obligations; child care; treatment adherence; and the unique biology of the disease. The relative importance of these issues varies considerably across the broad age range defined as AYA. Certain institutions have established centers specialized in accommodating the specific needs of AYA patients. A retrospective analysis in Florida evaluating clinical trial enrollment in a comprehensive care center melanoma program found that AYAs with advanced disease (ie, nodal or metastatic disease) were more likely to enroll in a clinical trial. Additionally, researchers reported a nonstatistically significant trend of improved 3-year overall survival in AYA patients whose disease was advanced.[29] Referral of patients to AYA centers of excellence, or centers with established social, clinical, therapeutic, and psychosocial support programs, should be encouraged when feasible. Patients who receive AYA-focused care through a comprehensive and multidisciplinary approach have improved outcomes compared to those who do not receive AYA-focused care.[30-33]

The goals of the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) for Adolescent and Young Adult Oncology are to identify and increase awareness of issues specific to AYA patients and recommend interventions unique to these patients; educate physicians regarding the prevalence of cancer in the AYA population and its long-term consequences; identify special considerations related to the management of cancer in AYA patients with the aim of improving treatment tolerance, adherence, and clinical outcomes; and promote participation in clinical trials as well as enrollment in tumor banking and biologic protocols.

Guidelines Update Methodology

The complete details of the Development and Update of the NCCN Guidelines® are available at www.NCCN.org.

Literature Search Criteria

Prior to the update of the NCCN Guidelines for Adolescent and Young Adult Oncology, an electronic search of the PubMed database was performed to obtain key literature in adolescent and young adult oncology, using the following search terms: cancer infertility, cancer fertility, cancer AND adolescent, or cancer AND young adult. The PubMed database was chosen because it remains the most widely used resource for medical literature and indexes peer-reviewed biomedical literature.[34]

The search results were narrowed by selecting studies in humans published in English. Results were confined to the following article types: Clinical Trial, Phase II; Clinical Trial, Phase III; Clinical Trial, Phase IV; Guideline; Meta-Analysis; Practice Guideline; Randomized Controlled Trial; Systematic Reviews; and Validation Studies. The data from key PubMed articles as well as articles from additional sources deemed as relevant to these guidelines as discussed by the Panel during the Guidelines update have been included in this version of the Discussion section. Recommendations for which high-level evidence is lacking are based on the Panel's review of lower-level evidence and expert opinion.

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Sensitive/Inclusive Language Usage

NCCN Guidelines strive to use language that advances the goals of equity, inclusion, and representation.[35] NCCN Guidelines endeavor to use language that is person-first; not stigmatizing; anti-racist, anti-classist, anti-misogynist, anti-ageist, anti-ableist, and anti-weight-biased; and inclusive of individuals of all sexual orientations and gender identities. NCCN Guidelines incorporate non-gendered language, instead focusing on organ-specific recommendations. This language is both more accurate and more inclusive and can help fully address the needs of individuals of all sexual orientations and gender identities. NCCN Guidelines will continue to use the terms *men, women, female*, and *male* when citing statistics, recommendations, or data from organizations or sources that do not use inclusive terms. Most studies do not report how sex and gender data are collected and use these terms interchangeably or inconsistently. If sources do not differentiate gender from sex assigned at birth or organs present, the information is presumed to predominantly represent cisgender individuals. NCCN encourages researchers to collect more specific data in future studies and organizations to use more inclusive and accurate language in their future analyses.

Risk Factors

With rare exceptions, cancer appears to arise sporadically in most AYAs with a negative family history of cancer. Toxic and environmental exposures that cause cancer in AYAs include, but are not limited to, chemotherapy and/or radiation therapy (RT) leading to second malignant neoplasms (SMNs) in patients treated for cancer during childhood or young adulthood[36-38]; predisposition to clear cell adenocarcinoma of the vagina or cervix in patients with maternal exposure to diethylstilbestrol; and melanomas induced by ultraviolet light/sun exposure. Infections that predispose AYAs to cancer include cervical carcinoma following exposure to human papillomavirus (HPV), HL and Burkitt lymphoma following Epstein-Barr virus (EBV) infection, and Kaposi sarcoma and NHL in people with HIV.[39]

Familial cancer syndromes, associated with germline mutations in a variety of genes, affect only a minority of AYA patients. However, these syndromes greatly increase the risk of cancer during adolescence and young adulthood. Approximately 8% to 9% of all patients with cancer will have a germline mutation in a cancer-predisposing gene.[40-42]

Young individuals with germline mutations of *BRCA1/2* are predisposed to hereditary breast and ovarian cancer syndrome, while those with mutations in *TP53* (Li-Fraumeni syndrome), *PTEN* (Cowden syndrome), or *ATM* (ataxia-telangiectasia), or those who have received mantle field RT for HL are at an increased risk of developing breast cancer during young adulthood.[2,43,44] Screening for breast cancer may be warranted in AYA patients with inherited or familial risk factors. See the NCCN Guidelines for Genetic/Familial High-Risk Assessment: Breast, Ovarian, Pancreatic, and Prostate, available at www.NCCN.org.

In young adults, hereditary polyposis and nonpolyposis syndromes, inflammatory bowel disease, and radiation exposure are predisposing factors for developing colorectal cancer. Hereditary nonpolyposis colorectal cancer (HNPCC or Lynch syndrome) is an autosomal dominant syndrome caused by mutations in one of the four mismatch repair (MMR) genes (*MSH2, MLH1, MSH6,* or *PMS2*), and is associated with early-onset colorectal cancer.[2,6] Familial adenomatous polyposis (FAP) is an autosomal dominant disease caused by germline mutations in the *APC* gene. This syndrome is associated with the presence of thousands of colonic polyps and with the development of colon cancer in most affected patients by age 40 years. Desmoid tumors are considered to be the most common extracolonic manifestations of FAP and may be the presenting manifestation of FAP in AYA patients.[45] Screening for colorectal cancers may be warranted in AYA patients with inherited or familial risk factors.

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See the NCCN Guidelines for Colorectal Cancer Screening, available at www.NCCN.org.

AYA patients with Li-Fraumeni syndrome (resulting from germline mutations in the *TP53* tumor suppressor gene) or germline mutations in the *RB* gene are at higher risk of developing osteosarcoma and rhabdomyosarcoma.[46] Sarcomas represent 25% of tumors in *TP53* mutation carriers. AYAs with germline mutations in the *RB* gene have often been treated for retinoblastoma during early childhood.[47] AYAs with a family history of Li-Fraumeni syndrome have a higher risk of developing not only sarcomas but a wide variety of malignancies, including leukemia, brain tumors, breast cancer, and adrenocortical carcinoma before age 40 years.[48] Rhabdomyosarcoma in AYA patients has high-risk features and may be associated with inferior outcomes. Referral to a center familiar with rhabdomyosarcoma therapy should be considered depending on the clinical situation.[49,50] See the NCCN Guidelines for Soft Tissue Sarcoma, available at www.NCCN.org.

Patients with mutations in the *SDH* gene are at risk for paraganglioma and pheochromocytoma, gastrointestinal stromal tumors (GIST), renal clear cell carcinoma, and papillary thyroid carcinoma. Testing for germline mutations in the *SDH* subunit should be considered for AYA patients with wild-type GIST lacking *KIT* or *PDGFRA* mutations.[51,52] Patients with germline mutations in *NF1* carry a 10% lifetime risk for malignant peripheral nerve sheath tumors, as well as an increased risk for other malignancies including GIST and early breast cancer.[53]

Multiple neuroendocrine neoplasia (MEN) syndromes (MEN1 and MEN2) are autosomal dominant syndromes characterized by the development of multiple endocrine tumors. MEN1 is caused by a germline mutation or inactivation of the tumor suppressor gene *MEN1*, whereas MEN2 is associated with germline mutations in the *RET* proto-oncogene.[54] MEN1 is associated with the development of pituitary, parathyroid, and pancreatic neuroendocrine tumors.[54] Testing for MEN1 should be considered for patients with two or more MEN-associated tumors or in patients with one MEN-associated tumor and a relative with MEN1. MEN2 is further subdivided into MEN2A and MEN2B. Both of these subtypes are associated with a high risk of developing medullary thyroid carcinoma (MTC).[55] Most patients with MEN2B have mucosal neuromas or intestinal ganglioneuromas and pheochromocytoma in addition to MTC.[55] Testing for MEN2A should be considered for patients with two or more MEN2A-associated neoplasms or in patients with a close relative with MEN2A-associated neoplasms. Testing for MEN2B should be considered for patients with MTC, pheochromocytoma, mucosal neuromas of the lips and tongue, medullated corneal nerve fibers, distinctive facies with enlarged lips, “marfanoid” body habitus, or inability to cry tears. See the NCCN Guidelines for Neuroendocrine and Adrenal Tumors, available at www.NCCN.org, for more information. HPV infection has been associated with cervical cancer and few other non-cervical cancers, including anal and oropharyngeal cancers.[56,57] Randomized clinical trials have demonstrated the efficacy of HPV vaccination against cervical intraepithelial neoplasia, anal intraepithelial neoplasia, and oral HPV infections in patients aged 15 to 25 years.[58-60] In the PATRICIA trial, the efficacy of HPV vaccine against all cervical intraepithelial neoplasia associated with HPV-16/18 was highest in the 15- to 17-year-age group and progressively decreased in the 18- to 20-year and 21- to 25-year-age groups, suggesting that early HPV vaccination could substantially reduce the incidence of HPV-associated cancers in the AYA population.[59] However, studies have shown that survivors of childhood or AYA cancers have lower rates of HPV vaccination compared to their peers.[61,62] Furthermore, recommendation for HPV vaccination from the survivor’s medical team was positively associated with vaccination,[61,62] highlighting the importance of recommending the HPV vaccine to AYA patients and their caregivers. HPV immunization (if not previously administered) is recommended by the U.S. Centers for Disease Control and Prevention

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(CSC) Advisory Committee on Immunization Practices (ACIP) for all adolescents aged 11 or 12 years and can be administered in patients as young as 9 years. Catch-up vaccination is recommended in both sexes until age 26 years, since the vaccine has been shown to prevent cervical carcinoma and anal intraepithelial neoplasia.[63-65] The American Cancer Society recommends HPV vaccination for all adolescents between ages 9 and 12 years to promote on-time vaccination rates. Health care providers are also encouraged to start offering the HPV vaccine at age 9 or 10 years.[66] ACIP also recommends that individuals with specific risk factors, including those who are transgender, or those who have immunocompromising medical conditions (such as cancer) be vaccinated through age 26 years.[63,64]

Screening

AYA patients, especially those with risk factors, should be made aware of the importance of early diagnosis and self-examination of the skin, breasts, and testicles as recommended by the American Cancer Society. They should also be educated regarding the benefits of early detection and treatment.[67] Cancer screening in some circumstances, particularly in cervical, breast, and colorectal cancers, can significantly reduce mortality if directed at the appropriate age group and if the results are interpreted and followed up appropriately.[68] However, there are no age-specific screening tests that have been developed that would increase early detection in AYA patients with cancer, and in some instances screening tests have been associated with false-positive results leading to false diagnosis and unnecessary treatments.[69] Therefore, it is necessary to assess the potential risks and benefits of cancer screening in the AYA population. see *Screening Recommendations for AYA Survivors* within the algorithm for additional specific recommendations and resources.

Diagnosis

The onset of new symptoms in AYA patients may not immediately trigger evaluation for malignancy, due to the relatively low incidence of cancer in this age group and the resulting low index of suspicion on the part of patients, their families, and primary care providers. AYAs are at a higher risk of delayed cancer diagnosis, which may result in a more advanced stage of cancer that requires more therapy and is associated with a worse prognosis.[67] Some studies have reported that adolescents experience longer lag times (interval between symptom onset and diagnosis) than children.[70-72] Lack of health insurance, inexperienced physicians, and workup that is inappropriate for the patient's age are some causes of delayed diagnosis in AYA patients, although more research is needed on these contributing factors.[2] Passage of the Affordable Care Act in 2010, which included provisions extending dependent health care coverage up to age 26 years, has improved insurance rates for young adults through age 25 years, although lack of insurance remains a problem for AYA patients who are >26 years.[73] In a retrospective analysis of 503 patients aged 15 to 29 years with previously untreated cancer, the advanced stage of cancer at diagnosis and lack of health insurance were significantly associated with longer lag times.[74] Those with public or no health insurance had longer lag times than those with private health insurance in most of the cancers evaluated. In addition to health insurance, education and employment status are also likely to influence lag time, although these factors were not evaluated in this study.

Comprehensive Care for AYA Patients with Cancer: Special Considerations

AYA patients should be cared for by a multidisciplinary team of providers with expertise in AYA cancer treatment and with access to clinical trials for the AYA population, and management of specific issues including mental health, education including genetic counseling, fertility, career development, employment management with disease demands,

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survivorship, and wellness promotion. Given the rarity of several tumor types diagnosed in this population, all AYA patients should be encouraged and offered enrollment in tumor banking studies and multicenter clinical trials, when available.

All AYA patients should undergo comprehensive assessment following cancer diagnosis, which should include a comprehensive psychosocial assessment, discussion of the cancer and treatment-related risk on fertility and sexual health or identity, the use of education concerning fertility preservation methods and contraception, evaluation of complete family history, and referral for a subsequent genetic counselor evaluation for risk assessment and counseling as needed. Providers should assess the gender identity, pronouns, sexual activity, and concerns of the patient. A pregnancy test should be considered prior to each cycle of therapy in accordance with institutional requirements. Consult with an OB/GYN for patients with ovaries/uterus and review the CDC Summary Chart of U.S. Medical Eligibility Criteria to assist with the safety and efficacy of selection of appropriate contraception for individuals with childbearing potential. Consider referral to a fertility preservation or reproductive health program. It is important to speak with each patient individually, without a caregiver or partner present, to solicit any questions or concerns and to give the patient an opportunity to speak freely to communicate their preferences or needs. Age and developmentally appropriate information related to cancer should be provided.

Age-Appropriate Care

AYA patients can be treated either at pediatric or adult cancer centers.[27] Retrospective analyses have shown that AYA patients with certain pediatric-type cancers, such as ALL,[25,75-77] rhabdomyosarcoma,[78] and Ewing sarcoma,[79,80] have superior outcomes when treated with pediatric protocols. Alternatively, there is a lack of compelling evidence that pediatric protocols improve outcomes in AYA patients with AML, HL, and NHL.[81-83]

As aforementioned, the low rate of participation in clinical trials is one of the main reasons for the lack of improvement in outcomes in AYA patients with cancer.[3,25,84-86] A review of 30 studies of adolescents with cancer (aged 15–19 years) showed that 5% to 34% of these patients were enrolled in clinical trials.[25] Concerningly, with the exception of AYAs with ALL, participation of AYA patients in clinical trials has been decreasing; the decrease in clinical trial accrual from 2010 to 2015 ranged from -10% (aged 15–19 years) to -54% (aged 35–39 years).[4] Care should be provided at medical centers with broad access to clinical trials (standard-of-care registry trials and trials evaluating novel therapies).[27] Pediatric cancer centers enroll more adolescents into clinical trials (35% vs. 12% at non-pediatric cancer centers), and AYA patients treated at pediatric cancer centers have a higher rate of clinical trial enrollment (26%) compared to those treated at adult cancer centers (4%).[87-89] Parsons et al reported that AYA patients who are treated by non-pediatric oncologists are less likely to be enrolled in clinical trials.[90] Nevertheless, a substantial number of AYA patients with pediatric malignancies are not being treated at pediatric cancer centers.[25,91,92]

The treatment and appropriate location of care vary with the type of cancer as well as with the availability of family, community, and institutional support.[67,93] A study conducted in Pittsburgh evaluated clinical trial enrollment of AYAs following the creation of a dedicated AYA Oncology Program, a joint effort by pediatric and medical oncologists. It was found that AYA clinical trial enrollment increased to 32% in comparison to 4% for historic controls.[94] AYA patients should ideally be evaluated at medical centers with extensive experience in treating cancer in this patient population and at centers that have access to supportive care services (psychosocial/educational support and fertility preservation) specific to the

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AYA population as well as to medical subspecialty services appropriate to the cancer diagnosis, such as orthopedic surgeons with experience in limb-sparing surgery for patients with extremity sarcomas.[27] In a supportive care needs survey that assessed the information and service needs of young adults with cancer at a single institution, the majority of young adults with cancer identified the following information as most important: information on their specific malignancy, effects of treatment on fertility, information on maintaining a healthy diet, and exercise/physical fitness during cancer treatment.[95]

Treatment Options

Select AYA patients may tolerate more intensive therapies than older patients, as they have fewer comorbid conditions that limit the intensity of treatment in some older adults.[67] Dose-intensive and dose-dense treatments are associated with improved outcomes in certain malignancies.[21,96,97] Therefore, more intensive therapy may be considered for some AYA patients if such a regimen exists for that particular disease and there are no contraindications. When possible, AYA patients should be enrolled in clinical trials for their specific disease. Additionally, unless clear clinical data indicate otherwise, assume that the patient population has a significant long-term survival and that significant end-organ damage may compromise long-term function and quality of life.

Treatment-related issues in AYA patients may differ from those of populations outside of AYA due to the distinct biology of the disease.[11] Physical and physiologic changes, such as changes in body composition, size and maturity of organs, and hormones associated with the normal pubertal process, may directly affect the drug disposition, drug efficacy, and toxicity of chemotherapy in AYA patients.[98] Appropriate management of symptoms and side effects to reduce the severity and toxicity of treatment should be an integral part of the comprehensive care of AYA patients.[99] Surgery, RT, chemotherapy, and hematopoietic cell transplant (HCT) are the main treatment options for patients who are able to tolerate curative treatment. All of these options are associated with both acute and late side effects.[67,100] Patients should be given a summary of potential therapy-related side effects. The NCCN Panel recommends the patient be provided documentation of consent and be closely monitored for treatment-specific acute toxicities. Please refer to *Screening Recommendations for AYA Survivors* in the algorithm for guidance on screening for treatment-related toxicities including core outcomes, which should be evaluated based on the initial diagnosis.[101]

Surgery

Surgery plays an important role in the management of cancer in AYA patients, especially in breast cancer, thyroid cancer, melanoma, bone and soft tissue sarcomas, colorectal cancer, and CNS cancers that are more common in this patient population. Adolescent patients, whose bodies are still developing, may be more affected by some surgical procedures than older patients who are already at or near their full body size.[67] The extent of surgery is dependent on the type and location of cancer. In some instances, extensive surgery requiring the removal of part of or an entire organ or limb may be necessary. With advances in surgical techniques and chemotherapy, limb-sparing surgery is feasible for the majority of patients with extremity sarcoma and osteosarcoma.[80] Surgery should be performed at high-volume centers by surgeons with expertise in AYA comprehensive care, with access to rehabilitative services to ensure that function is preserved as much as possible.

Radiation Therapy

RT is associated with an increased risk of delayed morbidity and mortality; development of SMNs; pulmonary, cardiac, and thyroid dysfunction; and chronic health conditions and growth abnormalities.[102] AYA patients receiving RT to the testes or ovaries are at risk of developing fertility impairment and reproductive endocrinopathies later in life.[103] Females with

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HL who receive chest RT between ages 10 and 30 years are at increased risk of developing breast cancer.[104] Cranial RT is associated with short stature, auditory deficits, cognitive processing difficulties, poor physical function, and rarely debilitating migraines, which contribute to lower rates of employment, independent living, and marriage among AYA survivors of childhood cancer.[105-107] RT-induced spinal cord dysfunction is thought to be more prevalent in adolescents, presumably because of elongation of the cord during the growth spurt.[108] A multidisciplinary consultation should be considered to determine the optimal method to reduce radiation-induced effects. For patients with a predicted risk of radiation-induced late effects to tissues surrounding but not in targeted tissues, consider a consultation with a radiation oncologist for proton radiotherapy.[109-113] A dermatology evaluation (ie, annual skin examinations performed by a dermatologist) is also recommended for patients who previously received RT.[114]

Systemic Therapy

Pain, fatigue, nausea, vomiting, mucositis, hair loss, infection, and myelosuppression are some of the acute side effects of chemotherapy. Reversible toxicities do not necessarily warrant dose reductions. See established supportive care guidelines, available at www.NCCN.org, for the management of treatment-related toxicities. Every attempt should be made to maintain dose intensity unless it is contraindicated. Dose reductions are often based on avoiding severe, irreversible organ damage. Significant end-organ damage may compromise long-term function and quality of life in AYA patients. Establish maximum cumulative dosing parameters and monitor cumulative dosing and schedule for certain medications associated with irreversible organ damage and fertility issues when certain lifetime exposure is encountered.

Anticipatory nausea and vomiting (ANV), also known as conditioned, learned, or psychological nausea and vomiting, is reported to occur before chemotherapy in approximately 20% of patients at any one chemotherapy cycle and in 25% to 30% of patients by their fourth chemotherapy cycle.[115] Younger patients (<50 years) may be more susceptible to ANV, because they generally receive more aggressive chemotherapy and have poorer emesis control than older patients.[115] Other risk factors for ANV include, but are not limited to, female sex, previous ANV, history of motion sickness, and morning sickness during pregnancy. Prevention strategies include behavior therapies, acupuncture, and administration of anxiolytic medications.[116] See the NCCN Guidelines for Antiemesis, available at www.NCCN.org. Dental exams and cleanings are recommended every 6 months for patients who received chemotherapy to the head or upper neck region, especially for head and neck cancer survivors.[117]

Although there are limited data on the subject, systematic reviews have reported that AYA patients experience more cancer-related fatigue than older patients[118] and that fatigue is one of the most prevalent, severe, and distressing symptoms in this age group.[119] Furthermore, several studies on the topic have shown benefit of interventions targeting fatigue in AYA patients with cancer or cancer survivors, although further research on effective management strategies is needed.[118-121] See the NCCN Guidelines for Cancer-Related Fatigue, available at www.NCCN.org.

Alkylating agent-based chemotherapy is associated with a higher risk of infertility in patients.[103] See the section on *Impact of Cancer and Its Treatment on Fertility* in this Discussion. Anthracycline-based chemotherapy is associated with irreversible cardiac dysfunction, whereas neurotoxic chemotherapies such as methotrexate and cytarabine can result in CNS dysfunction.[99,122] Bleomycin-induced pulmonary toxicity is well-documented in patients with HL and germ cell tumors treated with bleomycin-containing chemotherapy regimens.[123,124] Higher cumulative doses of cisplatin, ifosfamide, or epipodophyllotoxins are associated with

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hearing loss, peripheral neuropathy, kidney dysfunction, and secondary AML, respectively.[125-128] See also the section on *Late Effects in AYA Cancer Survivors*.

Ototoxicity may occur following treatment with platinum-based chemotherapy agents.[129] Although this side effect is not considered life-threatening, it can have a detrimental effect on an AYA patient's quality of life. In 2022, the FDA approved the use of sodium thiosulfate (STS) for reducing the risk of ototoxicity associated with cisplatin in pediatric patients ≥1 month of age with localized, non-metastatic solid tumors.[130] The approval of this indication was based on data from two open-label, phase 3, randomized controlled trials in pediatric patients with cancer who were treated with cisplatin; the incidence of hearing loss was lower in those who received STS than those who did not receive STS.[131,132] However, concerns remain regarding the use of STS in the metastatic setting. A post-hoc analysis of data from the ACCL0431 trial showed that, among patients with disseminated disease, STS was associated with a significantly lower 3-year overall survival rate compared with those who did not receive STS (45% vs. 84%; P = .009).[133]

Immunotherapies are becoming an important component of care for patients with cancer. It is possible that there may be long-term toxicity associated with these agents; however, more data in the AYA population are needed to fully understand the effects.

AYA cancer survivors are at increased risk for reduced bone mineral density, alopecia, and other dermatologic side effects. Due to the risk of osteoporosis and osteopenia, bone mineral density surveillance is recommended upon entry into long-term follow-up with repeat testing once the patient is 25 years old.[134] Chemotherapy-induced alopecia is common, as the treatment targets rapidly growing cells and therefore damages hair follicles.[135,136] Hair loss can have long-term psychosocial and behavioral effects on patients. A randomized control trial investigating the use of scalp cooling to increase hair thickness and density has found promising results in patients with breast cancer, but the long-term effects are still under investigation.[137] Additionally, various dermatologic side effects from cancer treatments include scarring, irritated skin, hyperpigmentation, and hirsutism, and these conditions can have a significant impact on a patient's quality of life.[138,139] AYAs should be provided recommendations for the management of treatment-related hair loss.[140,141]

Hematopoietic Cell Transplant

HCT is a potentially curative treatment option for an increasing number of AYA patients with leukemias and lymphomas.[142] Graft-versus-host disease (GVHD), chronic immunosuppression, and gonadal dysfunction related to high-dose conditioning chemotherapy and RT are the major post-transplant complications associated with HCT.[99,100]

Chronic GVHD has been identified as the leading cause of non-relapse mortality in HCT survivors.[143] AYA patients are at a higher risk of developing chronic GVHD than younger children.[144] Patients >15 years (children <5 years had a probability of <14% compared to a probability of 44% for patients >15 years) and the use of total body irradiation (TBI) were significantly associated with an increased likelihood of developing chronic GVHD following allogeneic HCT.[145] Patients who receive peripheral stem cells during their transplant procedure have a greater risk of chronic GVHD compared to those who receive bone marrow transplant.[146] A report from the Bone Marrow Transplant Survival Study demonstrated that chronic GVHD had a significant impact on the overall health status of HCT survivors, particularly in the areas of functional impairment, activity limitation, and pain.[147] This study also demonstrated that resolution of chronic GVHD resulted in long-term health outcomes that were comparable to survivors who were never diagnosed with chronic GVHD.

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HCT survivors are also at increased risk for late complications, which include recurrent infections, secondary cancers, cardiac dysfunction, growth failure, weight loss, neurocognitive delay, and other end-organ dysfunction.[99,100,148] In addition, the incidence of severe or life-threatening chronic health conditions, endocrine complications, or secondary cancers is also higher among HCT survivors than in non-cancer populations and patients with cancer who are treated conventionally.[142] Allogeneic HCT survivors irradiated at ≤30 years are at higher risk of developing secondary solid cancers.[149]

These findings highlight the increasingly recognized need for long-term follow-up care that incorporates screening and surveillance of AYA survivors of HCT.

Distress and Psychosocial Factors Impacting Treatment

Adherence is defined as the extent to which a person's behavior corresponds with agreed recommendations from a health care provider. Nonadherence to recommended treatment and follow-up care contributes to poor clinical outcomes in AYA patients with cancer.[150,151] Non-adherence to appointments can lead to delayed identification of side effects, complications, or secondary cancers.[5]

Nonadherence to treatment regimens has been an ongoing problem among patients with cancer, and the prevalence of nonadherence has been consistently higher among adolescents compared to younger or older patients with cancer.[150] Nonadherence to oral chemotherapy contributes to reduced treatment efficacy and increased risk of recurrence. Available evidence from clinical trials that have included AYA patients with leukemia and lymphoma suggests that a substantial portion of AYA patients with cancer (27%–63%) have difficulties adhering to their oral treatment regimens.[150,151] Difficulties with medication adherence also extend to survivorship, with 23.8% of AYA cancer survivors reporting nonadherence compared to 14.3% in the comparison group ($P < .001$). Survivors were also more likely to report that they could not afford their medication, and uninsured survivors were more likely to report nonadherence than those who were privately insured.[152]

Nonadherence to other components of cancer treatment (eg, non-adherence to appointments for treatment or follow-up, refusing medical examinations, not preparing for procedures or therapy) was also identified in AYA patients. Treatment nonadherence in clinical trials can also interfere with adequate evaluation of the efficacy of a given treatment regimen, which in turn can invalidate the results of a clinical trial.

Risk factors for nonadherence among AYA patients include patients' emotional functioning (depression and poor/low self-esteem), personal beliefs (perceived severity of cancer diagnosis and the necessity of intervention), growing independence, competing obligations (school, work, and family), and lack of insurance and appropriate psychosocial support.[153] In a randomized controlled trial, video game intervention significantly improved treatment adherence to prophylactic antibiotic use among AYAs with acute leukemia, lymphoma, and soft tissue sarcoma.[154] A review aimed to identify and characterize the various digital health interventions available for support of AYAs with cancer. Numerous modalities, including mobile/tablet apps, video games, wearables, virtual reality, social media, and other web-based interventions were found to be in existence for the purpose of promoting physical activity, assessing pain management, and improving adherence among AYAs.[155] Additional studies evaluating the efficacy of interventions to improve adherence in AYA patients with cancer are needed.[2]

Risk assessment for non-adherence among AYA patients should include consideration of patient maturity, independence, unmet psychosocial and physical needs, and treatment side effects.[156] For AYAs presumed to be at high risk of nonadherence, implementation of individualized interventions

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such as additional supportive care resources (eg, social work, psychology, palliative care) to promote adherence may improve outcomes in AYA patients with cancer. The patient's personal support system (family, friends, or caregiver) should be mobilized and educated to assist in relieving some of the burdens of care and to positively encourage the patient to maintain adherence to therapy. In the absence of data from studies evaluating the effect of interventions to improve adherence in AYA patients with cancer, findings from studies involving AYA patients with other chronic diseases may be able to be extrapolated to this patient population.[5]

NCCN Recommendations on Adherence to Therapy and Safety Plans

- Educate about the expectations of treatment and explain the patient's responsibility to adhere to therapy.
- Refer to health or pediatric psychologist for adherence intervention if needed; include a list of barriers and facilitators.
- Provide education and/or guidance about each medication prior to the start of treatment and every time there is a change in treatment. Review the list of medications as well as their dose, purpose, and adverse effects.[150,151]
- Simplify and modify dosing schedule when medically possible, to accommodate an AYA patient's lifestyle and normal activities.[150,151]
- Consider the use of technology (eg, smartphone applications).
- Provide access to systematic and standardized symptom management for side effects related to cancer treatment.[150,151] See established guidelines for supportive care, available at www.NCCN.org.

Impact of Cancer and Its Treatment on Fertility

Impaired fertility is a major consequence of cancer and its treatment.[157-160] The impact of cancer treatment on fertility is related to the age of the patient at the time of diagnosis and treatment, and is dependent on the type, duration, and dose intensity of treatment. Alkylating agent-based chemotherapy, high-dose cranial RT that can impair hypothalamic pituitary function, and targeted RT to the uterus, ovaries, or testes are primary risk factors for gonadal dysfunction and decreased fertility.[161-166] Gonadal exposure to low-dose RT can result in oligospermia or azoospermia in individuals with testes. Higher-dose RT is associated with both ovarian and uterine dysfunction.

Young females with HL treated with chemotherapy are at risk of developing primary ovarian insufficiency, irrespective of their age at the time of treatment (38% for those diagnosed between ages 30 and 40 years; 37% for those diagnosed between 9 and 29 years of age).[167] The cumulative risks for primary ovarian insufficiency are much higher after alkylating agent-based chemotherapy.[167,168] In a large cohort of females treated between ages 15 and 40 years for HL, the cumulative risk of premature ovarian insufficiency after alkylating agent-based chemotherapy was 60% compared to only 3% or 6% after non-alkylating agent-based chemotherapy.[168] Independent risk factors for acute ovarian insufficiency include increasing RT doses to the ovaries and exposure to procarbazine and cyclophosphamide at ages 13 to 20 years.[162] An analysis of 590 females who were diagnosed with HL before age 18 years showed that RT to the pelvis was associated with decreased incidence of parenthood (hazard ratio [HR], 0.66; 95% CI, 0.48–0.90; $P = .01$).[169]

Among young females treated with adjuvant chemotherapy for breast cancer, the risk for premature menopause is significantly higher for those >35 years with newly diagnosed breast cancer treated with chemotherapy.[170,171] Similarly, among survivors of HL diagnosed between ages 14 and 40 years, those who were aged 22 to 39 years at first treatment were at a higher risk of developing premature menopause after treatment compared to younger patients (aged 14–21 years).[172] Treatment with MOPP (mechlorethamine, vincristine, procarbazine, and prednisone)/ABV (doxorubicin, bleomycin, and vinblastine) significantly

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increased the risk of ovarian insufficiency. After 10 years of treatment, the actuarial risk of premature menopause was 64% after high cumulative doses (>8.4 g/m^2) and 15% after low doses (≤4.2 g/m^2) of procarbazine.[172]

In those treated with alkylating agent-based chemotherapy and RT to the testes, germ cell dysfunction with resultant infertility is more common than Leydig cell dysfunction and testosterone insufficiency.[173] Leydig cell dysfunction is characterized by increased plasma concentrations of luteinizing hormone (LH) combined with low levels of testosterone. Germ cell dysfunction is associated with reduced testicular volume, increased follicle-stimulating hormone (FSH) concentrations, and reduced plasma concentrations of inhibin B. Leydig cell dysfunction occurs at RT doses higher than those associated with germ cell dysfunction. AYA patients treated with testicular RT ≥20 Gy are at high risk for Leydig cell dysfunction, whereas testicular RT ≥2 Gy can impair spermatogenesis resulting in permanent azoospermia.[173] TBI used as part of high-dose conditioning therapy prior to HCT can also affect the testes, resulting in permanent infertility in the majority of AYA patients with testes.[103]

Azoospermia is associated with chemotherapy and radiation. Whether it is transient or permanent depends on the type of treatment involved, with radiation and alkylating agents posing the greatest risk for long-term damage.[174] Azoospermia has been reported in >90% of males receiving procarbazine-based chemotherapy regimens such as MOPP and may not resolve over time, resulting in permanent infertility.[175] Alternatively, the ABVD (doxorubicin, bleomycin, vinblastine, and dacarbazine) regimen has been shown to be less gonadotoxic, with a vast majority of patients regaining normal fertility after completion of treatment.[176]

Given that gonadal insufficiency/gonadotoxicity is a well-known side effect of cancer therapy,[177] the Pediatric Initiative Network (PIN) proposed a standardized risk stratification model, which subdivides the risk of gonadal insufficiency/infertility into minimally increased risk, significantly increased risk, and high level of increased risk.[178] This classification is based on exposure to alkylating agents or heavy metals, HCT, and radiation exposure.[178] The male model also incorporates the risk of surgery and does not stratify risk according to pubertal status, unlike the female model.[178] Although limitations exist, this model is meant to estimate and standardize infertility risk and promote conversation concerning fertility preservation. Details regarding the risk assessment can be found under the “Assessment for Gonadal Function” section under *Screening Recommendations for AYA Survivors* in the algorithm.

The NCCN Guidelines recommend discussing the risks of impaired fertility due to cancer and its treatment with all patients at the time of diagnosis, prior to initiating treatment. This is especially important for patients who will be starting therapies with a high risk of affecting fertility, as described above.

Fertility Preservation

As the AYA age range includes the primary reproductive years, fertility preservation is an issue of crucial importance and should be an essential part in the management of the cancer.[15,103,179-183] The American Society of Clinical Oncology (ASCO) Clinical Practice Guidelines recommend that providers discuss the options for fertility preservation with all new patients with cancer at the time of diagnosis.[184] Nevertheless, fertility preservation is currently one of the most under-prescribed and least implemented services in AYA patients with cancer.[103,179,185] A study that reviewed 231 records of AYA patients with leukemia/lymphoma, sarcoma, or breast or testicular cancers showed that infertility risk was discussed 26% of the time, and fertility preservation options were discussed 24% of the time.[186] However, it is possible that more discussions about infertility occurred without having been documented. Another study that analyzed the electronic medical records of 454 AYA patients at a single cancer center showed that the risk of infertility was discussed with 83% of patients, with

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females more likely to be informed than males (odds ratio [OR], 3.57; 95% CI, 1.33–9.60; *P* = .01).[187] A study of 146 adolescent males at risk for fertility impairment due to cancer treatment across eight different pediatric oncology centers found that only 53.4% attempted to bank sperm, with 43.8% successfully banking. Parent or medical team recommendation was associated with increased likelihood of sperm banking completion.[188]

A systemic review reported that the following factors can hinder the decision-making process for females with cancer: 1) lack of information about fertility preservation options; 2) fear of perceived risks of fertility treatment (eg, delaying cancer treatment, hormone treatment for a hormone-sensitive cancer); 3) lack of physician referral; 4) emotional distress; 5) relationship/parental status; and 6) financial concerns.[189] The American Society for Reproductive Medicine (ASRM) recommends that conversations concerning fertility be undertaken by an interdisciplinary medical team composed of oncologists, reproductive endocrinologists and urologists, and reproductive surgeons trained in fertility preservation methods. The initial step of discussing intended therapies and associated reproductive risks is entrusted to the oncologist, who can subsequently make referrals as appropriate.[190] Psychosocial providers can assist patients and families in the decision-making process about fertility preservation, particularly when AYA patients are distressed about the potential for impaired fertility associated with cancer treatment.[184] The emotional impact of conversations surrounding fertility, especially for the younger AYA patient, must be taken into consideration. Finally, genetic and financial counselors may also be consulted to discuss the risks and probability of heritable disease transmission and review financial options for fertility preservation, respectively.[190]

The Oncofertility Consortium, a group of clinicians and researchers in the United States, was formed in 2007 by the U.S. National Institutes of Health (NIH) to address reproductive barriers facing AYA patients and to identify research priorities in this area.[191] Future aims that were developed during a 2011 meeting are as follows:

- Determine optimal techniques for cryopreservation of reproductive tissue and gametes.
- Further investigate in vitro follicle maturation in primates.
- Investigate AYA patients' psychosocial needs as part of the fertility preservation plan.
- Improve patient-provider communication regarding fertility preservation.
- Develop and carry out multicenter studies, utilizing the pre-existing infrastructure of the National Physicians Cooperative.

Ideally, fertility preservation should be initiated prior to the start of treatment. However, in some situations, when it is impractical or impossible to pursue fertility preservation prior to initiating therapy, it may be appropriate to readdress later in the course of treatment. ASRM recommends that fertility preservation programs be affiliated with an experienced assisted reproductive technology (ART) program. Alternatives to fertility preservation, such as the use of donor gametes, embryos, and adoption, should be discussed by the team. For individuals with ovaries scheduled to undergo pelvic RT, the option of gestational surrogacy may also be reviewed. In the case of cryopreservation, appropriate arrangements should be discussed and documented in the case of donor death.[190]

Options for Individuals with Ovaries

Oophoropexy

For individuals scheduled to undergo pelvic RT, oophoropexy or ovarian transposition may be a promising option involving surgical displacement of the ovaries out of the RT field to minimize ovarian damage and preserve ovarian function.[192]

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Embryo Cryopreservation

Embryo cryopreservation after in vitro fertilization (IVF) is an option for fertility preservation.[184] This method is an option for post-pubertal individuals with ovaries and requires a committed sperm donor who is available with short notice. It has been highly successful in individuals <40 years.[103,179] According to data published by ASRM on infertile and donor populations, the live birth rate per cycle start for individuals <35 years was 46.8%, while that for those aged 35 to 37 years was 34.4%, and that for those aged 38 to 40 years was 21%.[190] In one study that assessed pregnancy outcomes following embryo cryopreservation, letrozole was used in combination with FSH to protect patients with breast cancer against the harmful effects of increased estrogen.[193] Out of 33 patients, the live birth rate was 45%, with 39% of live births resulting in twins. These rates are not significantly different from those for infertile couples not affected by cancer, except for implantation rate, which was greater in the patients with breast cancer (40.7% vs. 26.1%). A little more than half (55%) of the embryos were transferred to a gestational carrier, with no significant differences in outcomes (ie, implantation, live birth, twinning rates) between self-transfers and gestational carriers.

Mature Oocyte Cryopreservation

Mature oocyte cryopreservation is an alternative for post-pubertal individuals, and, like embryo cryopreservation, requires hormone stimulation and subsequent oocyte retrieval.[103,179] Evidence from randomized trials[194-197] and a meta-analysis[198] suggest that IVF with cryopreserved oocytes results in fertilization and pregnancy rates similar to that of fresh oocytes. Like embryo cryopreservation, pregnancy rates with mature oocyte cryopreservation also decline with advancing age.[190]

Ovarian Tissue Cryopreservation

Cryopreservation of ovarian cortical tissue is a promising strategy for fertility preservation when there is insufficient time for oocyte or embryo cryopreservation and/or the patient is prepubertal. This technique does not require hormonal stimulation, so there is no delay in initiation of treatment.[103] This procedure would not be appropriate for certain patients with cancer if potential exists for reintroduction of malignant cells with grafting. It is also not recommended for carriers of *BRCA* mutations due to the increased risk for ovarian cancer. Orthotopic transplantation has been met with success, with 130 live births reported from cryopreserved and thawed ovarian tissue since 2017.[190] Following transplantation, ovarian tissue is thought to regain function within 60 to 240 days and may last for 7 years.[190] Pregnancy and live birth rates are expected to be higher in donors <40 years due to ovarian follicular reserve. It is recommended that transplantation be performed with the purpose of regaining fertility and not gonadal endocrine function.[190] While ovarian tissue cryopreservation is still considered investigational at some institutions, it may be discussed as an option for fertility preservation, if available.

Some studies, including randomized trials, have evaluated the role of menstrual suppression with gonadotropin hormone-releasing hormone (GnRH) agonists to preserve ovarian function during chemotherapy.[199-207] Some meta-analyses have shown that GnRH agonists may be beneficial for fertility preservation.[208-210] However, the impact of such meta-analyses is limited by factors such as only examining those with breast cancer and only including trials that were not adequately powered and did not use blinding and/or a placebo condition, among other flaws.[211,212] There are also limited data available on the long-term impact of GnRH on preservation of ovarian function,[208] although a 5-year follow-up analysis of a randomized trial showed that administration of a GnRH agonist does not significantly impact primary ovarian insufficiency or future pregnancy rate.[207] Therefore, although data suggest that menstrual suppression with GnRH agonists may protect ovarian function, further investigation is needed.

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Nevertheless, GnRH agonists and agents such as the progestin medroxyprogesterone and oral contraceptives may be used in patients with hematologic malignancies and thrombocytopenia and/or anemia who are at risk for menorrhagia for the purpose of menstrual suppression.[213] However, caution is needed in endometrial cancer survivors where progestin therapy has been associated with high rates of cancer recurrence,[214] which may be prevented by combining metformin with medroxyprogesterone.[215]

Options for Individuals with Testes

Sperm Cryopreservation

Individuals should also be counseled regarding options for fertility preservation and referral to a reproductive urologist should be considered. Semen cryopreservation before the start of treatment is the most reliable and well-established means of preserving fertility in post-pubertal AYA patients with cancer.[103,179] The success of sperm banking may be limited in some patients, such as those with HL and testicular cancer, who may already have azoospermia associated with the disease. Depending on the type of chemotherapy, semen collection may be possible after initiation of chemotherapy; however, the impact of chemotherapy and RT on the risk of genetic defects in the offspring remains unknown.[216]

If the patient has medical, religious, emotional, or developmental issues that preclude traditional ejaculation methods, other options to obtain sperm for cryopreservation may be discussed, including: vibratory stimulation, electroejaculation, surgical testicular sperm extraction (TESE), the administration of phosphodiesterase type 5 (PDE5) inhibitors, and the use of alpha-agonists. Traditionally more effective in those with neurologic injuries, penile vibratory stimulation may be used to trigger the ejaculatory reflex. Electroejaculation, normally performed under anesthesia, involves the transrectal stimulation of pelvic tissues, including the prostate and seminal vesicles, to yield ejaculate. Surgical TESE is reserved for individuals who have azoospermia and/or are unable to ejaculate via other methods. The use of PDE5 inhibitors may be used to obtain ejaculate in those experiencing erectile dysfunction; similarly, alpha agonists may be warranted in cases of retrograde ejaculation.[190]

Testicular Tissue Cryopreservation

Cryopreservation and subsequent transplantation of spermatogonial stem cells may be an option for prepubertal and pubertal individuals in whom semen cryopreservation is not possible.[103,179] For those with hematologic or testicular malignancies, autologous transplantation of cryopreserved testicular tissue may not be appropriate for fear of reintroduction of tumor cells. However, immature testicular tissue cryopreservation is still considered experimental.[190]

Similar to the recommendation for those with ovaries, there is limited evidence regarding the efficacy of hormone suppression in reducing the risk of impaired fertility in individuals with testes during chemotherapy.[184]

Fertility, Reproductive Endocrine, and Sexual Health Considerations

The NCCN Guidelines emphasize that fertility preservation and sexual health and function should be an essential component of the comprehensive care of AYA patients with cancer who are at any level of risk for impaired fertility and possible sexual dysfunction due to cancer treatments, regardless of gender identity, sexual orientation, or financial status.[178,217-219] An assessment of the risk for gonadotoxicity and impaired fertility due to cancer and its therapy should be performed and discussed with patients. Options for fertility preservation should be discussed with all patients as soon as possible prior to the start of therapy and throughout the course of therapy.[220] AYA patients may be transitioning from pediatric to adult care, in which case, they should be assessed for routine reproductive or gynecologic health care. Care should be coordinated by a multidisciplinary team. Providers should initiate referral to a reproductive health specialist and/or provide resources for off-

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site/remote sperm banking as soon as possible for all patients who are interested in pursuing fertility preservation.[221-223] Consultation with a financial counselor and local or institutional grants may be available to provide financial assistance with fertility preservation needs. Follow-up with a fertility specialist post-treatment may also be helpful for some patients. Consider the emotional impact of conversations surrounding fertility preservation, especially for the younger AYA patient and sexual gender minority (SGM) patients. Creating safe spaces to discuss gender identity and sexual orientation can mitigate and foster community partnerships with or provide community resources for fertility and psychosocial experts in sexual and gender minorities (GLMA directory).

All Patients

- After assessing the patient's risk for impaired fertility and the patient's preferences for fertility preservation, recommend a form of appropriate fertility preservation and/or make a referral to a fertility preservation specialist or sperm bank.[190,217,224]
- Discuss the effects of treatment on sexual function during and after treatment and refer the patient to a specialist as appropriate. See NCCN Guidelines for Survivorship, available at www.NCCN.org.
- Discuss contraception before, during, and after treatment. Consult with an OB/GYN for patients with ovaries/uterus and consult the CDC Summary Chart of U.S. Medical Eligibility Criteria to assist with the safety and efficacy of selection of appropriate contraception for individuals at risk of pregnancy.

Individuals with Ovaries

- For individuals who can delay cancer treatment by approximately 3 weeks, oocyte or embryo cryopreservation via immediate (or random start) controlled ovarian stimulation (COS) should be discussed.
- For individuals who are unable to delay treatment for oocyte or embryo cryopreservation and are at high risk for impaired fertility, discuss or refer the patient for consideration of ovarian tissue cryopreservation.
- Oophoropexy or transposition of the ovaries out of the field of radiation should be considered for individuals for whom the radiation field will include the ovaries.
- The effects of treatment on gonadal hormone function during and after treatment should be discussed with patients. Patients who face primary ovarian insufficiency should be screened and treated by a specialist. Furthermore, patients who do not undergo fertility preservation prior to initiating treatment may still be appropriate candidates for fertility preservation post-treatment and should be screened and referred to a specialist.[225]
- Progestin-only methods, combined hormonal contraceptives, or GnRH agonists can be used in protocols that are predicted to cause prolonged thrombocytopenia and thus present a risk for menorrhagia.[213]
- GnRH agonists may protect ovarian function[226]; however, other fertility preservation modalities should still be considered and, if possible, pursued.

Individuals with Testes

- Sperm banking is the preferred choice of fertility preservation for individuals without erection or ejaculation issues. For patients who can delay cancer treatment, consider >1 collection of ejaculate, prior to initiating treatment. AYA patients can use either a local sperm bank or an available online sperm banking kit.[184,224,227] A semen analysis should be performed by the sperm bank, which should be assessed to ensure viable sperm has been frozen before starting cancer treatment, if time allows. Oncology centers that treat AYA patients should develop a system for offering sperm banking to all AYA patients in a systematic and patient-centered manner.

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- Should the patient have any medical, religious, emotional, or developmental issues that preclude traditional ejaculation methods, alternative methods, including PDE5 inhibitors, vibratory stimulation, electro-ejaculation, and collection of retrograde ejaculate can be considered.[190]
- Testicular transposition out of the field of radiation can be considered for patients in whom the radiation field will include the testes.[228]
- Surgical sperm extraction, such as the TESE procedure, may be an alternative strategy for those who cannot ejaculate or who have azoospermia or insufficient sperm in the ejaculate to freeze.[190]
- Discuss the effects of treatment on gonadal hormone function. Screen or refer the patient to a specialist as appropriate after completion of treatment.[229]

Contraception in AYA Patients with Cancer

The NCCN Guidelines recommend discussion about sexual function before, during, and after treatment; this should include a discussion on the use of contraception.[230,231] Condoms may be used safely by AYAs with cancer.[232] AYAs with ovaries have unique contraception needs and the options are dependent on the type of cancer, its treatment, and treatment-related complications.[230]

Long-acting reversible contraception (LARC) with intrauterine devices (IUDs) or implantable contraceptives are more effective than short-term contraceptive methods, which include the use of estrogen and progestin with various delivery systems.[233] LARC has been shown to be superior to short-acting contraceptives.[234,235] In a study of 4167 participants (aged 14–45 years), LARC was associated with higher 12-month adherence rates than oral contraceptive pills (86% vs. 55%).[234] In a large, prospective study involving 7487 participants, the contraceptive failure rate was significantly higher for those using oral contraceptive pills, patch, or ring compared to those using LARC (4.55 vs. 0.27), and the progression rates among participants <21 years were twice as great as in those ≥21 years.[235]

The Society of Family Planning guidelines recommend the use of IUDs or implantable contraceptives for most females who are receiving treatment for cancer.[236] The use of any method of contraception is recommended for those who have been free of cancer for at least 6 months and have no history of hormonally mediated cancers, chest RT, anemia, osteoporosis, or venous thromboembolism (VTE).[236] The use of IUDs is considered the preferred first-line contraceptive option for females with a history of breast cancer, although for those treated with tamoxifen, a levonorgestrel-containing intrauterine system (IUS) may be preferable since it has been shown to reduce tamoxifen-induced endometrial changes without increasing the risk of breast cancer recurrence. A levonorgestrel-containing IUS may also be used to minimize menstrual blood loss in patients with iron-deficiency anemia.[236]

Due to the risk of VTE associated with the use of combined hormonal contraceptive methods, the CDC recommend that the use of these contraceptive methods should be avoided in patients of childbearing potential with active cancer or who have been treated for cancer in the last 6 months.[237,238]

Management of Cancer During Pregnancy

All patients with childbearing potential must receive a pregnancy test prior to initiating therapy. Cancer is diagnosed in about 0.1% of pregnant individuals and is the second most common cause of maternal death during pregnancy.[239] Melanoma, breast cancer, cervical cancer, lymphomas, and leukemias are the most common cancer types diagnosed during pregnancy.[240-244] HL is the most common hematologic malignancy diagnosed during pregnancy, and accounts for 6% of cancers occurring in pregnancy.[242] An analysis of 1963–2007 data from the Swedish Multi-

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Generation Register and the National Cancer Register showed a lower-than-expected number of cancers diagnosed during pregnancy, and a rebound in the number of instances of melanoma, CNS cancers, breast cancer, and thyroid cancer in postpartum individuals.[243] This rebound may be due to changes in the mammary and thyroid glands being overlooked during the postpartum period. Despite some persisting beliefs, there is no evidence of pregnancy-associated relapse in survivors of HL.[245]

While there is limited research on the prognosis of cancer during pregnancy, a few meta-analyses and systematic reviews have suggested that the prognosis of certain cancers (eg, breast, melanoma, vulvar) may be worse when occurring concurrently with pregnancy compared to the same cancers occurring outside pregnancy.[246-249] These results may be confounded by factors related to the patient's pregnancy, delays in diagnosis, and differences in treatment decisions, making a definitive conclusion difficult. Maternofetal transmission of cancer across the placenta is rare, but has been reported for metastatic melanoma,[250] lung cancer, and certain hematologic malignancies.[244,251-254] Most cancers can be adequately treated with medication that has been deemed not to cause harm for both the pregnant patient and the fetus. However, full counseling about all options regarding the pregnancy, including termination per patient decision or to expedite use of necessary teratogenic medication, should be incorporated in all cancer treatment discussions for pregnant AYA patients.

Accurate diagnosis of the type and stage of cancer using appropriate imaging studies (ultrasound, chest x-ray, and mammogram) with abdominal shielding and limiting fetal exposure to ionizing radiation is an essential step in the management of cancer during pregnancy.[255-257] When possible, non-radioactive imaging modalities, including MRI and ultrasound, should be used.[244] There is insufficient evidence regarding the safety of gadolinium-based contrast agents in pregnant individuals; however, these agents have been demonstrated to cross the placenta in animal studies.[244,257] There are currently no available data on the safety of iodinated contrasts during pregnancy. Surgery is possible at any time during pregnancy depending on the anatomical location of the tumor, although it may be beneficial to delay surgery, when possible, until after fetal viability due to the associated risks including miscarriage, low birth weight, and premature delivery.[244,255,257] Selection of an appropriate treatment plan for pregnant individuals is dependent on the cancer type, tumor biology, and tumor stage, similar to the management of cancer in those who are not pregnant. In addition to the disease characteristics in pregnant individuals, the gestational age of the fetus is a significant factor in the selection of treatment.[255]

RT is contraindicated during pregnancy.[258] However, in very rare instances when it is necessary, such as oncologic emergencies like spinal cord compression, superior vena cava compression, and brain metastases,[259,260] RT may be administered in the lowest effective therapeutic dose (using techniques such as uterine shielding to minimize fetal exposure), with the goal of controlling maternal cancer and providing the fetus with the best chance for survival with normal development.[261] The dose to the fetus can be reduced by using modified RT administration techniques or adding additional shielding between the treatment machine and the patient.[261] Early collaboration among the radiation oncologist, medical physicist, medical and/or surgical oncologist, and obstetrician is essential. The American College of Radiology has developed guidelines with an objective to assist practitioners in identifying pregnancy, preventing unnecessary irradiation of pregnant AYAs, tailoring examinations to effectively manage RT dose, and developing strategies to quantify and evaluate the potential effects of RT delivered to patients who are pregnant.[262] In 2014, an international consensus panel made up of researchers and clinicians who are experts in cancer treatment during

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pregnancy developed similar guidelines for RT in individuals who are pregnant.[256]

Chemotherapy should be avoided during the first trimester because of greater risk of teratogenic effects, which include major congenital malformations, impaired organ function, spontaneous abortions, and fetal death. See *Disease-Specific Issues Related to Age: Reported Outcomes and Recommendations for Select Teratogenic Agents During Pregnancy* in the algorithm for additional details.[257,263-266] While the use of chemotherapy during the second and third trimesters has not been associated with significant teratogenic effects, it may be associated with maternal and fetal risks including low birth weight, preterm labor, and intrauterine growth restriction.[242,263,265,267-269] However, a multicenter, prospective case-control study of children born to those with cancer (129 cases, 129 controls) showed no significant impact of chemotherapy treatment on cognitive, cardiac, and general development of the offspring.[270] Potential benefits and risks of chemotherapy for both the pregnant individual and fetus must be carefully evaluated prior to initiation of treatment. Delaying treatment until after fetal maturity, with careful follow-up to rule out disease progression, is a safe option for those diagnosed with early-stage cancers.[271,272] In some individuals diagnosed with advanced-stage disease with an urgent need to start chemotherapy in the first trimester, pregnancy termination may be considered and the potential teratogenic risks should be discussed if the pregnant individual decides to continue with the pregnancy.[242,244] Due to the severe teratogenic effects of methotrexate, it should not be used for the treatment of cancer in individuals at any stage of pregnancy.[265] Older-generation alkylating agents (eg, procarbazine, busulfan), thalidomide, lenalidomide, pomalidomide, and tretinoin are also considered teratogenic and are contraindicated during pregnancy.[256] The safety and efficacy of hormonal agents and targeted therapies have not yet been evaluated in well-controlled studies including those who are pregnant.[255-257,273-275] At the present time, the use of such agents in patients who are pregnant is not recommended.

Supportive care for the management of treatment-related side effects should be integrated into treatment planning based on the trimester of pregnancy. Granulocyte colony-stimulating factors for the management of neutropenia and antiemetics for the management of nausea and vomiting have been used in patients who are pregnant without any significant side effects.[256,274,276,277]

The Panel members acknowledge that the management of cancer during pregnancy poses significant diagnostic and therapeutic challenges for the pregnant individual, fetus, and physician. The guidelines recommend that AYA patients diagnosed with cancer during pregnancy receive individualized treatment from a multidisciplinary team involving medical, surgical, radiation, and gynecologic oncologists; obstetricians; and perinatologists as appropriate.[242,256] Referral to tertiary cancer centers with expertise in the comprehensive care of high-risk obstetric patients and in the diagnosis of cancer during pregnancy and knowledge of the physiologic changes that occur during pregnancy is strongly encouraged.

Several of the NCCN Guidelines for cancers that are diagnosed more commonly during pregnancy include disease-specific recommendations for managing cancer during pregnancy. For more information, see the NCCN Guidelines for Breast Cancer, NCCN Guidelines for Cervical Cancer, NCCN Guidelines for Chronic Myeloid Leukemia, and NCCN Guidelines for Myeloproliferative Neoplasms, available at www.NCCN.org. The American Academy of Dermatology has published recommendations for the diagnosis and management of melanoma during pregnancy.[278] Similarly, the American Cancer Society has published guidelines for the management of HL in pregnancy.[279] Finally, the American Thyroid Association has published guidelines for the diagnosis and management of thyroid disease, including thyroid cancer, during pregnancy.[280]

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Comprehensive Psychosocial and Behavioral Issues

AYA patients diagnosed with and treated for cancer have psychosocial and behavioral issues that are distinct from those of pediatric and adult patients.[2,7-10] AYA patients aged 20 to 29 years are significantly less likely to use professional mental health services than teens and patients aged 30 to 39 years. AYA patients in the 20- to 29-year age group are also significantly more likely to report an unmet need with regard to receiving age-appropriate information about their cancer. Some of the challenges faced by AYA patients and survivors include maintaining an active and independent lifestyle, coping with treatment-related side effects and stress, seeking and understanding information, accepting cancer, and maintaining a positive attitude.[281,282] AYAs go through developmental stages marked by rapid changes in cognitive and emotional growth, and these issues need to be considered while delivering developmentally appropriate psychosocial and supportive care to AYA patients.[283]

Few measurement tools have been developed to better understand health-related quality of life in AYA patients with cancer.[2] Palmer and colleagues developed an AYA Oncology Psychosocial Screening Tool to assist clinicians in supporting psychosocial coping during active treatment and promoting healthy post-treatment survivorship in AYA patients. This screening tool has four main areas: a distress thermometer, a checklist of “areas of concern,” a tick box for information provision, and signatures. Further validation of this tool and its use will help clinicians to improve psychosocial care for AYA patients, regardless of where they receive treatment.[284]

Psychosocial needs for AYA patients should be assessed across the following domains: 1) individual function (psychosocial, emotional, and behavioral issues); 2) relationships (family, caregiver, peers, and health care professionals); and 3) socioeconomic issues. Age and developmentally appropriate supportive care services and interventions should be used to address each of these domains.

Individual Function

Psychosocial Issues

AYA patients have to cope with cancer treatment while attaining key developmental milestones such as identity development, including sexual identity; peer involvement; initiating intimate and emotional relationships; establishing autonomy from parents; maintaining personal values; fostering self-esteem and resilience; and independently making decisions about their future that involve education, career, or employment.[285-288] The impact of diagnosis and treatment of cancer on their physical appearance, sexual development, and sexual function can lead to shame, social isolation, and regressive behaviors if not addressed promptly. Cancer and its often intensive and lengthy treatments put AYA patients at risk for disruptions in their normal activities. Interruptions of school or work due to treatment may have negative consequences for their long-term career opportunities, financial status, and lifetime earnings.[281] During the treatment period, AYA patients should have the opportunity to live as normal a life as possible, continue their education and/or careers, and participate in the many milestones of their lives.[289] Physical and/or occupational therapy may help AYA patients transition back to a lifestyle appropriate for their age group.[290] Additionally, patients should be evaluated for past trauma history, including adverse childhood experiences (ACEs), medical-related trauma, and abusive relationships.

Integral to the comprehensive care of individuals with cancer, including AYAs, is an assessment for gender expression, gender identity, preferred pronouns, and sexual identity. SGMs, or the lesbian, gay, bisexual, transgender/transsexual, and queer/questioning (LGBTQ) community, are a population that is unrepresented in medicine. This group includes a spectrum of identities; a comprehensive list of acronyms, definitions, and

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additional resources may be found on the NIH website (https://www.lgbtqiahealtheducation.org), overseen by the NIH Sexual & Gender Minority Research Coordinating Committee. The terms “lesbian,” “gay,” and “bisexual” denote one’s sexual orientation or attraction.[291] According to the NIH, the term “sex” is meant to refer to one’s chromosomal makeup and reproductive anatomy and is most often delineated as either male or female. Intersex refers to those individuals whose sexual anatomy or chromosomal constitution does not adhere to the norm (eg, Klinefelter syndrome, or the presence of XXY chromosomes).[292] Gender, however, is understood to be the integration of social, environmental, cultural, and behavioral factors and encompasses terminologies including gender identity, gender norms, and gender relations.[293]

It is estimated that approximately 3.4% to 12% of the U.S. population identifies as LGBTQ. SGMs experience significant inequity related to cancer diagnosis and treatment. Proposed reasons for this incongruity include decreased rates of cancer screening, behavioral risk factors (eg, increased rates of smoking and alcohol use), obesity and decreased exercise, nulliparity (among SGM assigned female at birth [AFAB]), and receptive anal sex (among SGM assigned male at birth [AMAB]). The SGM population also faces numerous health care discrepancies, such as significant discrimination, poorer insurance coverage, and inadequate physician knowledge of specific LGBTQ health needs, leading to psychological ramifications and overall dissatisfaction with treatment.[294] In a nationwide online survey of 273 LGBTQ individuals who had been diagnosed with cancer, patients reported to be influenced by provider LGBTQ knowledge and skills, perceived safety of the clinical encounters for the purpose of identity disclosure, and inclusion of members of their support system. SGM patients reported having a variety of clinical experiences and were more likely to be self-advocates of their care.[295]

The prevalence of cancer risk factors, such as tobacco use, differs among LGBTQ individuals when compared to non-LGBTQ persons. A cross-sectional online survey reported that LGBTQ individuals disclosed higher rates of past 30-day tobacco media exposure and had higher odds of past 30-day use of tobacco products when compared to non-LGBTQ individuals.[296] In addition to the increased prevalence of certain risk factors and a greater physical disease burden, SGM individuals are more likely to have mental health concerns. Attributable to a number of factors, when compared to heterosexual persons, SGM individuals are twice as likely to be depressed, 2.5 times as likely to attempt suicide, and are at an increased risk for anxiety and other psychological conditions.[297] Similarly, LGBTQ individuals are at an increased risk for self-harm or non-suicidal self-injury.[298]

A higher incidence of bullying is reported among youth and AYAs. This figure is even higher among LGBTQ youth, who report more accounts of physical and cyberbullying. In fact, it has been reported that 80% of LGBTQ adolescents have experienced victimization and harassment by peers.[299] Inequity also exists within SGM subpopulations. For instance, nonbinary individuals, or those who do not identify as strictly male or female, reported a higher incidence of bullying (86.7%) and polyvictimization than transgender or LGBTQ cisgender individuals.

Seven cancer sites have been reported to disproportionately affect the LGBTQ population for a variety of theorized reasons: anal, breast, cervical, colorectal, endometrial, lung, and prostate cancers.[291] In a survey of 388 oncology providers at a single institution, 91.7% of physicians specializing in the aforementioned seven cancer sites reported that they would be comfortable treating LGBTQ individuals and would encourage education of unique health matters. However, only 49.5% of providers answered LGBTQ health-related knowledge questions correctly.[291] Among 149 respondents to a survey conducted among 450 oncologists from 45

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NCI-Designated Cancer Centers, while 95.3% reported that they were comfortable treating lesbian, gay, bisexual (LGB) individuals, only 53.1% were confident in their assessment of the health care needs of the LGB population. The percentage was even lower for oncologists who were comfortable treating transgender patients (82.5%), with only 36.9% being confident in their ability to understand the health needs of transgender individuals.[294]

It is therefore imperative that providers recognize the gravity of their role when caring for SGM AYAs. Members of the oncology care team must work collaboratively to create an inclusive, safe, and comfortable space to facilitate conversation with LGBTQ individuals surrounding gender, sexual orientation, sexual behaviors, and other relevant experiences.[299] Support persons, as identified by SGM patients, may be involved in cancer care as appropriate. Referrals for psychosocial support should be initiated as appropriate and referrals to a specialized gender clinic should be considered for transgender youth.[300,301] Recognizing that members of the SGM population face significant health and health care inequities and unique psychosocial issues, providers must develop a thorough and comprehensive understanding of such matters and consider cancer screening as appropriate.[302] Research also suggests that LGBTQ individuals have to navigate addressing unique survivorship issues, including but not limited to arranging follow-up care and coming out to multiple providers, contemplating the effects of systemic therapy on LGBTQ sexuality, and coping with delicate family and relationship issues.[303] Awareness and a thorough understanding of such concerns are required for the care of SGMs.

Emotional Issues

Cancer-related issues such as confrontation with mortality and loss of fertility can result in significant emotional distress and psychiatric symptoms such as depression and anxiety in AYA patients. These feelings are related to patients' cognitive capacity to understand the severity of their disease while sometimes lacking fully mature cognitive and emotional coping abilities.[281] Psychological distress is significantly greater among AYAs compared with older adults,[304-308] and prescription rates of anxiolytics and hypnotics are higher in AYA survivors compared to their peers, suggesting an increased emotional burden.[309]

In a longitudinal study that assessed the prevalence of psychological distress in 215 AYA patients with cancer (aged 15–39 years) during the first year following diagnosis, distress symptoms exceeded population norms at the time of diagnosis and at 12-month follow-up.[305] In this study, 12% of AYA patients reported clinically significant chronic distress throughout the first 12 months following diagnosis and an additional 15% reported delayed distress. Distress is also prevalent in AYA cancer survivors; however, most AYA survivors with distress reported that they had not spoken with a mental health professional (74.7% with moderate distress, 52.2% with severe distress).[310]

In addition to distress, depression and anxiety are commonly experienced by AYA cancer survivors. An analysis from the Childhood Cancer Survivor Study (CCSS) showed that survivors of AYA cancer (n = 2589) report higher rates of depression (OR, 1.55; 95% CI, 1.04–2.30) and anxiety (OR, 2.00; 95% CI, 1.17–3.43) compared to their siblings (n = 391).[311] Another study of 5341 cancer survivors diagnosed at ≤25 years found that survivors were more likely to be prescribed antidepressants compared to age- and gender-matched controls (26.9/1000 person-years for survivors vs. 22.5/1000 person-years in controls; HR, 1.19; 95% CI, 1.12–1.28).[312]

The need for information, counseling, and practical support was reported in 57%, 41%, and 39% of AYA patients, respectively, at 12 months after cancer diagnosis.[307] Kazak and colleagues reported that intensive cancer treatments during adolescence are associated with inferior psychosocial outcomes and health beliefs in survivors compared to their age-matched

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peers.[313] Psychological problems are also associated with an increased risk for obesity and poor health behavior, which may increase future risk for chronic health conditions and secondary neoplasms.[314] It is therefore recommended that AYAs diagnosed with cancer meet with child life specialists, if available, soon after diagnosis to address concerns that the patient may have regarding treatment or procedures and assist with coping mechanisms to reduce anxiety.[315]

Behavioral Issues

AYA patients with cancer may engage in risky behaviors (tobacco, alcohol, cannabis, or substance use) that may impair their health. Advanced age at cancer diagnosis, lower household income, less education, no pulmonary-related cancer treatment, and no brain RT were independently associated with a statistically significant relative risk of smoking initiation.[316] The risk factors associated with alcohol use included fair or poor self-assessed health, depression, anxiety, somatization, activity limitations, and cancer-related fears and uncertainty.[317] Low perception of susceptibility to late effects, older adolescence compared to early adolescence, and worry about cancer and its treatment were the strongest predictors of substance use.[318] While AYA patients may be aware of the complications associated with tobacco, alcohol, cannabis, or use of other substances during their treatment, they may not avoid them throughout treatment as these habits may make them feel “normal” and like part of their peer group. Clinicians working with this population should be aware of such matters and address these issues in a sensitive and confidential manner.[289,319]

Studies have shown increased rates of mental illness and cognitive impairment among adolescent individuals who use cannabis compared to adults with similar usage habits.[320] Heavy or regular use of cannabis in adolescents has been associated with impairments in attention, learning, memory, planning, and psychomotor speed. An earlier age of onset of cannabis use exacerbates these adverse effects. If an AYA patient chooses to continue use of cannabis, education on methods for lowering risk of adverse effects is recommended. For example, the patient may be counseled to avoid high tetrahydrocannabinol (THC)-content products, avoid synthetic cannabinoids, choose routes of administration other than inhalation of combusted cannabis, limit frequency of use, and never drive while impaired.[321] Additionally, as there are insufficient data, it is uncertain whether cannabis affects the metabolism and efficacy of chemotherapeutic agents.

In 2006, e-cigarettes, or vaping products, were introduced to the U.S. market.[322] Marketed originally as tools for smoking cessation and safer alternatives to cigarettes, the aerosols of these products are now known to contain active compounds, including nicotine, THC, cannabidiol, vitamin E acetate, and select flavorings and additives. It is reported that among youth, the overall use of nicotine-containing products has increased since the release of e-cigarettes and devices such as the JUUL. Past 30-day vaping among U.S. high school students rose from 1.5% in 2011 to 20.8% in 2018. Alarmingly, past 30-day e-cigarette use among U.S. middle school students increased from 0.6% in 2011 to 4.9% in 2018.[323] The CDC and U.S. Food and Drug Administration (FDA), as well as other health authorities, have reported an extensive number of e-cigarette or vaping use-associated lung injury (EVALI) cases since 2019. Many patients (the majority of whom have been AYAs) have been hospitalized, with several requiring intensive care and respiratory support.[324,325]

AYA patients are also vulnerable to sexual and reproductive health complications that should be addressed prior to, during, and after completion of treatment.[230] Traditional risk-taking behaviors of AYAs coupled with a compromised immune system may put AYA patients with cancer and survivors at a greater risk of sexually transmitted infections and, in certain cases, malignancy. See the section on *Contraception in*

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AYA Patients with Cancer for more discussion of appropriate contraception choices for patients with cancer and survivors.

Consequent to treatment and lifestyle-related factors, it is reported that AYAs have a 5- to 15-fold increased risk of cardiovascular morbidity when compared to the general population.[326] Therefore, lifestyle and diet modification are key in AYA cancer survivors to increase survival. AYA patients have nutritional concerns that are different from those of children and adults, especially among younger patients in this population. Adolescents are dependent on their families for food preparation and may experience peer pressure when eating at school or with friends. The INAYA trial, consisting of AYAs aged 18 to 39 years, reported improved nutritional behavior among AYAs following intensified individual nutrition counseling during a 3-month period. Diet/nutrition information has thus been reported as an unmet need among AYA patients.[8] Promotion of healthy lifestyle behaviors and incorporation of physical activity into treatment regimens and post-treatment follow-up may also produce numerous physical and psychological health benefits.[327,328] Regular physical activity integrated into AYA cancer care may also decrease cancer-related fatigue.[329]

NCCN Recommendations for Supportive Care Services/Interventions for Psychosocial and Behavioral Issues

- Discuss with patient/guardian regarding referral of AYA patients for neuropsychological assessment if there are concerns regarding cognitive function (eg, attention, memory, executive function) and/or prior to educational and career transitions, including returning to school/work after treatment.
- If psychological support specialists are present, including psychology, child life specialists, or integrative medicine, have them meet with the patient soon after diagnosis to address any concerns regarding treatment or procedures, provide psychoeducation, and assist with coping mechanisms to reduce anxiety related to procedures.
- Refer AYA patients to a social worker, mental health provider, and community-based resources serving AYA patients to screen for any symptoms of depression, anxiety, suicidal ideation/behaviors, and self-injurious behavior.
- Offer psychosocial support and counseling to help alleviate distress. See the NCCN Guidelines for Distress Management, available at www.NCCN.org.
- Facilitate adherence to treatments by considering flexible treatment dates, consultation times, and procedures, such as weekends and evenings, to enable AYA patients to continue with their treatment without interrupting their school/work or other normal activities.[289]
- Refer patients for educational and career services to address training/education, employment, disability disclosure, vocational adjustment training, and transition services (ie, social services, vocational counseling, occupational therapy, financial counselors).
- For all AYA patients, provide counseling around sexual health conversations and decision-making regarding the risks of treatment-related fertility impairment and discuss the options for fertility preservation prior to initiating treatment.[330]
- For patients with minoritized identities: Practice cultural humility and remail open to the impact of cultural and identity factors in treatment decision-making; offer additional psychosocial support and referrals surrounding stressors, stigma, or rejection related to their sexuality or gender identity; ensure that the record system accurately reflects the patient's pronouns and preferred name; and, for transgender youth, consider referring the patient to a specialized gender clinic for psychosocial support and coordination of gender-affirming medical care at the patient's discretion.
- Refer patients to legal services, if applicable, for estate or legacy planning, child custody concerns, or other legal issues. Refer patients

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to a smoking cessation program if needed (see NCCN Guidelines for Smoking Cessation, available at www.NCCN.org).
- Provide education and/or guidance on cannabinoid use. See the NCCN Guidelines for Antiemesis, available at www.NCCN.org.
- Refer patients with signs, symptoms, or a history of a substance use disorder to a risk reduction or substance use counseling program.
- Provide education about the impact of treatment on sexual health and safe sexual practices, including oral, anal, and vaginal sex, risk for bleeding, prevention of pregnancy, and sexually transmitted infections.
- Provide education about potential diet/nutritional changes associated with cancer treatment and possible interventions. Refer to a Registered Dietitian-Certified Specialist in Oncology (RD-CSO).[331]
- Provide education on physical conditioning and its related health benefits during and following cancer treatment. To address physical impairments and to initiate physical activity interventions, refer AYAs to a rehabilitation specialist (ie, physiatrist, physical therapist, occupational therapist). Note that a medical evaluation and clearance by a physician (such as an oncologist or physiatrist) are recommended before initiating exercise in patients for whom exercise modifications or precautions may be needed.
- Refer patients with sleep disturbance to supportive medicine or behavioral health for interventions. See the NCCN Guidelines for Palliative Care, available at www.NCCN.org.
- Evidence-based integrative therapies/interventions can be considered.
- Refer patients experiencing challenges with their faith or belief in a just or fair world to faith-based resources or activities (eg, church youth groups, mentors).[9,332] Refer to a chaplain or spiritual counselor.
- If cancer and its therapy are expected to be associated with significant morbidity, referral to palliative care is warranted.

Relationships

Social, Peer, and Family Relationships

AYA patients often have to endure lengthy hospital stays under the supervision of health care providers, resulting in significant isolation from their family members, caregiver, and/or peer group.[289,333] Isolation and alienation are common among AYAs diagnosed with cancer, because they often miss out on the life experiences shared by their non-ill peers. Reinforcing relationships with family members, caregivers, peers, and health professionals is an important aspect of life for AYA patients.[283,334]

While some studies have identified family support and cohesiveness as important contributors to a survivor's adjustment, others have identified the important role played by same-aged peers (healthy peers as well as other AYA cancer survivors) in helping AYA patients cope with cancer and overcome feelings of loneliness.[9,283] In one study, AYA patients with cancer (aged 16–22 years) identified social support (including family members, friends, health care providers, and other patients) as their major coping strategy.[335] In another study, some AYA patients and survivors reported that opportunities to meet other young adult survivors were more important than the support they received from family and peers.[330]

Peer support programs assist AYA patients and survivors in establishing and maintaining relationships with their healthy peers as well as with other AYA patients with cancer, offer opportunities to achieve age-related developmental tasks (building interpersonal and problem-solving skills), and promote positive psychosocial growth.[330,336] Peer support also provides AYA patients with an opportunity to address areas of shared concern, such as uncertainty about the future, establishing autonomy while being increasingly dependent on family and friends, sexual identity, and impaired fertility, thereby reducing feelings of social isolation.[336]

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AYA peer support groups have been developed in a variety of formats, including face-to-face meetings, camp style formats, or online support groups.[337,338] Social networking groups focused on supporting AYA patients are particularly helpful for exchanging support, informational and emotional, through providing advice and empathizing with other AYA patients dealing with cancer.[338] Summer camps and adventure programs where participants are physically challenged have resulted in improvements in self-confidence, independence, and social contacts.[9,337] Many of the AYA patients may not be interested in conventional cancer support groups but are willing to participate in social networking events involving other AYA patients, survivors, and family members.[9] Indeed, studies of AYA patients and survivors indicated that 73% of patients currently receiving therapy and 74% of off-treatment survivors reported that their needs for retreats and camp programs were unmet.[339,340]

Communications with Health Care Professionals

Communicating information to AYA patients can be challenging, especially since there are several subgroups within the AYA population with different levels of cognitive and emotional development. It is very important to establish direct communication with the patients on an individual basis, with sufficient sensitivity to each patient's needs and preferences.[27] While some patients prefer not to receive direct communication about their cancer, others may desire a more prominent role in their comprehensive care. For the latter group, information should be provided directly to patients in a developmentally appropriate manner, allowing time to process the information and deliver information in a caring manner.[341] AYA patients prefer that information about their cancer and cancer-related risks be communicated to them in a manner that is positive, respectful, and nonjudgmental.[289] In a pilot project aimed at eliciting the views of AYA patients with cancer, humor, closely followed by expertise and knowledge, was identified as the most important characteristic that patients would like to see in their nurses.[342] Since there is evidence that AYA patients are willing to use the internet to obtain health information and support, it is also helpful to provide them with a list of recommended and reliable age-appropriate online sources to access information about their cancer, particularly with regard to treatment and late effects, fertility preservation, mental health counseling, peer support groups, diet, and nutrition.[337,340,343] See NCCN online resources for patients and survivors, available at www.NCCN.org.

NCCN Recommendations for Supportive Care Services/Interventions for AYA Patient Relationships

- Promote collaborative communication between AYA patients and parents, caregivers, children, spouse/partners, other family members, siblings, friends, and/or social network.[28]
- Encourage early in the treatment process the completion of a medical power of attorney/health care proxy and a living will at age of majority.
- Provide access to AYA-specific advanced care planning guides to determine a health care proxy.
- Provide information to identified family members, caregivers, and partners about psychosocial support and behavioral services to increase awareness of the possible psychosocial issues associated with cancer diagnosis in AYAs.
- Refer patients to resources that particularly offer support for AYA patients who are parents, such as those for respite care.
- Provide AYA-specific education and access to activities to provide psychosocial support and reduce boredom, anxiety, and depression. Such interventions include inpatient recreational activities to foster social connections with their peers as well as outpatient social networking events, camps, and retreat programs.
- Provide education and access to AYA-specific psychosocial support, for example, support groups for AYAs, parents, and caregivers, in addition to psychoeducational programs.

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- Provide education and access to supportive/palliative care and integrative oncology for stress and symptom management, including mind-body therapies, acupuncture, tai chi, yoga, and music therapy.
- Create flexible visiting hours and an environment that will encourage peers to visit AYA patients.[289] Ensure staff are trained to recognize normative expectations for AYA interactions with peers.
- Establish direct communication with the individual patients, providing developmentally appropriate information about their cancer, treatment options, and potential side effects, thus reinforcing the importance of AYA involvement in decision-making.[27,330]
- Some AYA patients prefer not to share information about their cancer with their family in an effort to shield their family members or caregivers from some of the things they themselves worry about. Therefore, obtain their permission to share information with identified family members or other members of their support system, and encourage completion of a health insurance portability and accountability act (HIPAA) release form.
- Provide psychoeducation and assistance exploring and documenting advance directive preferences.
- If the AYA patient is >18 years, provide information on and the necessary forms that legally allow medical information to be shared with caregivers of the patient's choice.
- Always conduct medical and psychosocial care in the language preferred by the patient, family, or caregiver. Use certified interpreters and do not rely on family members, caregivers, friends, or non-certified medical staff for interpretation.

Socioeconomic Issues

AYA patients, ≥26 years, are much more likely to be uninsured or underinsured individuals than adults or children, with many of them in a transition between their parent's insurance and their independent insurance.[17,73] Young adult survivors of childhood cancers are more likely to report health-related unemployment, lower rates of health insurance coverage, and more difficulties obtaining coverage compared to their siblings.[344] An analysis of 9353 AYA patients with HL showed that having either public or no health insurance was associated with poorer HL-specific survival, compared to patients with private or military insurance (HR, 2.08; 95% CI, 1.52–2.84).[345] Furthermore, unemployment and lack of health insurance appear to be significant predictors of psychological distress in the childhood cancer survivor population.[346] Uninsured AYA patients are also less likely to participate in clinical trials.[90] As described above, advanced stage of cancer at diagnosis and lack of health insurance were significantly associated with longer time to cancer diagnosis in AYAs.[74] Greater rates of unemployment and lack of health insurance among AYA patients and survivors are also associated with limited access to long-term follow-up care.[281] Results from the AYA HOPE study, a population-based cohort study of 523 AYA patients with cancer (aged 15–39 years at diagnosis from 2007–2009), suggest that lack of health insurance is also associated with poor health-related quality of life among AYA patients with cancer.[347]

Financial toxicity is a concern, as AYA patients with employment also experience problems in obtaining affordable health and life insurance due to their pre-existing cancer history.[9] Even those with relatively comprehensive insurance may be liable for substantial out-of-pocket expenses related to treatment, such as transportation costs associated with traveling for treatment, accommodations, meals, and child care as well as expenses not related to treatment.[9] AYA patients who are financially independent also have to face an additional burden of loss of income because of their inability to work during treatment. Once treatment is completed, AYA patients with cancer also require long-term follow-up care for monitoring and treatment of late effects.

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NCCN Recommendations for Supportive Care Services/Interventions for Socioeconomic Issues

- Link qualified AYA patients to Medicaid, Social Security, and/or other disability insurance.
- For patients with limited or no insurance coverage, provide information regarding medication assistance programs, hospital pharmacy vouchers, or low-cost medication programs.
- Direct AYA patients to legal resources and/or advocates for assistance with understanding health insurance coverage and the transition to independent health insurance coverage.
- Educate AYA patients about the benefits for which they may qualify (eg, short- or long-term disability, state disability benefits, Social Security benefits, public assistance).
- Provide information on obtaining financial assistance for fertility-based services. Local and institutional grants may be available.
- Provide school support, education services, and work reintegration for patients.
- Refer patients for career counseling and/or education support as indicated. Encourage discussion with guidance counselors and educators about the impact of cancer care on education.
- Refer to mental health experts as needed to assess the psychosocial impact of financial toxicity (eg, loss of employment, withdrawing from school, not being able to socialize with friends due to decreased income).
- Provide a referral for transportation assistance programs (eg, van ride programs, voucher programs) for AYA patients who have to travel to receive treatment.
- Provide information about recommended and reliable online resources and financial support programs to access information related to their cancer.
- Integrate financial assistance for AYA cancer survivors into their survivorship care plans.
- Consider the need for long-term follow-up care for monitoring and treatment of late effects long after treatment has been completed.

Survivorship Issues

Late Effects in AYA Cancer Survivors

AYA cancer survivors are at increased risk for late effects related to cancer treatment, and the risk for long-term effects is dependent on the age at initial diagnosis and the type of treatment.[348-350] In addition, the risk for many late effects may also be influenced by family history, lifestyle behaviors, and comorbid health conditions. Age at treatment exposure modifies the risk of some late effects (eg, breast cancer following chest RT, cardiomyopathy following anthracycline chemotherapy) but not others (eg, ischemic coronary artery disease following chest RT).[351,352] Improvements in RT and surgical techniques may help reduce late effects.[353]

Much of the understanding of the long-term outcomes of AYA cancer survivors comes from the CCSS, which includes long-term survivors of childhood and adolescent cancers who were diagnosed prior to age 21 years.[354,355] No such large cohort studies have addressed the survivorship issues related to cancer diagnosed in young adult patients between the ages of 22 and 39 years. Outcomes from the CCSS among those diagnosed between ages 15 and 20 years are particularly relevant for the NCCN Guidelines for Adolescent and Young Adult Oncology. Among adult survivors of childhood and adolescent cancer, Oeffinger and colleagues reported that by 30 years after the cancer diagnosis, the cumulative incidence of a chronic health condition was 73%, with a cumulative incidence of 42% for severe, disabling, or life-threatening conditions or death. Importantly, the risk for a chronic health condition (ie, long-term or late effect) was similar for those diagnosed with the primary cancer in adolescence or in childhood.[354]

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Other reports have also documented the prevalence of treatment-related adverse health status and the risk of late morbidity leading to hospitalizations among AYA cancer survivors.[356-359] In a retrospective analysis of 5-year survivors of young adult cancer (*n* = 902), the presence of at least one late morbidity leading to hospitalization was higher in survivors than in the control group (50.4% and 37.9%, respectively), and the adjusted risk of this morbidity for survivors was 1.4 times higher than for the control group.[358] Other analyses of survivors of young adult cancers showed that hospitalization rates are highest for survivors of upper gastrointestinal cancer, leukemia, urologic malignancies, brain cancer, and HL.[360,361]

A report that examined the health status of 4054 AYA cancer survivors revealed a significantly higher prevalence of current smoking (26% vs. 18%); obesity (31% vs. 27%); cardiovascular disease (14% vs. 7%); hypertension (35% vs. 29%); asthma (15% vs. 8%); disability (36% vs. 18%); and poor mental health (20% vs. 10%) and physical health (24% vs. 10%) among AYA cancer survivors compared to those who had no history of cancer.[356] In another large cohort study that included adult cancer survivors (245 patients, aged 15–19 years and 12 patients, aged 20–24 years at the time of diagnosis), impaired organ dysfunction (pulmonary, auditory, endocrine, and nervous system) was the most prevalent of all the adverse health outcomes.[357] In a report that evaluated the quality-of-life outcomes in 8375 AYA cancer survivors (diagnosed with cancer between ages 15–39 years) relative to the same aged controls, AYA cancer survivors were two times more likely to report fair or poor general health than the control group. The limitations in quality of life persisted across gender, race, ethnicity, and age.[359]

While several single cancer studies have assessed long-term outcomes among HL and testicular cancer survivors across the AYA age range, the long-term outcomes of survivors of other cancers occurring in young adulthood, such as breast, ovarian, and thyroid cancer or melanoma, remain understudied. Since there is a paucity of literature on survivorship issues related to cancer diagnosed during adolescence and young adulthood, the findings from the CCSS and similar studies focusing on childhood and adolescent cancer survivors could be extrapolated to the survivors of AYA cancers, albeit with caution. Increased adherence to long-term follow-up guidelines may contribute to improvement in health status of AYA cancer survivors.[356]

Some of the more common late effects among AYA cancer survivors are discussed below.

Secondary Cancers

AYA cancer survivors (aged 15–39 years at the time of diagnosis) are at significant risk of developing a variety of secondary cancers compared to the general population,[36,362] as well as cancer survivors diagnosed during either childhood or older adulthood.[37] Although secondary cancers may be caused by a hereditary syndrome, they are thought to be largely caused by treatment exposure.[353] The risk and specific types of secondary cancers are widely dependent on the type of initial cancer diagnosis and treatment exposure,[363-365] although the most common secondary malignancies are breast cancer, gastrointestinal cancer, genital cancers, and melanoma.[37] RT exposure is particularly associated with risk of secondary cancers.[37,362] In a retrospective matched cohort study consisting of 10,574 AYA cancer survivors, the 20-year cumulative incidence of SMN was found to be 12.5%. An elevated risk for SMN in the same site as the primary neoplasm was noted in AYA survivors of breast cancer, melanoma, and testicular cancer. In this study, risk factors for SMN were found to be older age at diagnosis, female sex, white race/ethnicity, advanced stage at first cancer diagnosis, and treatment with RT. Finally, AYA cancer survivors diagnosed with SMNs were found to have a 7-fold increased mortality risk

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compared with those who did not develop SMNs, highlighting the importance of SMN surveillance as appropriate in the AYA population.[38]

AYA survivors of HL diagnosed between ages 21 and 39 years are at an increased risk of developing secondary cancers.[364] The most frequently observed secondary cancers are breast, lung, thyroid, and gastrointestinal cancers.[351] AYA patients with HL treated with chest RT are at significantly increased risk of developing secondary breast cancer, and the risk for secondary breast cancer among HL survivors is strongly associated with age at diagnosis and mediastinal RT dose.[104,366-368] In a cohort of 770 survivors who had been diagnosed with HL before age 41 years, the risk of developing breast cancer increased with increasing RT dose (≥38.5 Gy).[366] In an international, population-based study of 3817 HL survivors diagnosed at ≤30 years of age, Travis and colleagues reported that for those treated at age 25 years with a chest RT dose of at least 40 Gy without alkylating agents, the estimated cumulative absolute risk of developing breast cancer by age 35, 45, and 55 years was 1.4%, 11.1%, and 29.0%, respectively.[367]

Alkylating agent-based chemotherapy for HL has been associated with a modestly increased risk for secondary lung cancers in patients diagnosed at ≤40 years, and the risk increased with both increasing number of cycles of alkylating agents and the cumulative dose.[369] In this study, the risk of secondary lung cancer was substantially higher among survivors who smoked (9.6% due to treatment alone compared to 63.3% due to the combination of treatment and smoking). In a collaborative British cohort study that assessed the risk of developing secondary cancers in 5798 patients diagnosed with HL between 15 and 34 years of age, the 20-year cumulative risk of secondary cancer was 13% and 18%, respectively, for chemotherapy alone and combined modality therapy.[370] Risks for secondary lung cancer, NHL, and leukemia were significantly higher after treatment with chemotherapy alone, whereas combined modality therapy was associated with a higher risk for these and several other cancers.[370]

AYA survivors of testicular cancer are also at a significantly increased risk of developing secondary cancers, including contralateral testicular cancer, leukemia, malignant mesothelioma, and cancers of the lung, colon, esophagus, stomach, and pancreas.[371,372] In a population-based study of 29,515 testicular cancer survivors, the 15-year cumulative risk of developing contralateral testicular cancer was almost 2%, which is 12-fold higher than that of the general population.[373] In an international, population-based study of 40,576 testicular cancer survivors, the cumulative risk of developing solid tumors by age 75 years was slightly higher for patients with seminomas than for patients without seminomas who were diagnosed at age 35 years (36% and 31%, respectively).[374] The combination of chemotherapy and RT was associated with a larger risk of secondary solid tumors than RT alone, although the difference was not statistically significant.[374] Secondary leukemia related to chemotherapy with topoisomerase II inhibitors and alkylating agents has also been reported in testicular cancer survivors. In one study, the cumulative incidence of secondary AML was 0.5% at 2 years after treatment with high-dose chemotherapy (with a median cumulative etoposide dose of 4.9 g/m^2) and autologous HCT.[375] In another study involving 42,722 one-year survivors of testicular cancer, the estimated excess cumulative leukemia risk was 0.23% at 30 years after testicular cancer diagnosis.[376] The risk for secondary AML was higher for patients treated with chemotherapy compared to those treated with radiotherapy alone.

The risk for secondary malignancies among survivors of cervical and breast cancers, NHL, and melanoma has been assessed in only a few cohort studies.[377-379] Among 104,760 one-year survivors of cervical cancer, patients heavily treated with RT were at increased risk for secondary cancers at sites in close proximity to the cervix beyond 40 years of

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follow-up. The 40-year cumulative risk for any secondary cancer was higher among patients diagnosed before age 50 years than among those diagnosed after age 50 years (22.2% and 16.4%, respectively).[377] In a population-based cohort of 376,825 one-year survivors of breast cancer from the Scandinavian cancer registries, patients diagnosed at ≤40 years with localized disease were particularly at risk of developing a secondary cancer at ≥30 years after breast cancer diagnosis.[378] In an analysis of 28,131 patients from the Swedish Cancer Registry, the risk of developing subsequent solid tumors after NHL during the first decade was higher among patients diagnosed between 20 and 39 years of age compared to those who were ≥40 years at the time of diagnosis.[379] In the Surveillance, Epidemiology, and End Results (SEER) database analysis of 89,515 melanoma survivors, patients diagnosed at <30 years had the highest risk of developing secondary cancers (with breast, prostate, and NHL being the most common cancers) at >20 years after initial diagnosis. HCT and RT to head and neck also increased the risk of subsequent cancers in the oral cavity.[380]

Long-term AYA survivors of pediatric-predominant cancers, including ALL, AML, CNS tumors, and bone and soft tissue sarcomas, are also at risk of developing secondary cancers. Among the survivors of ALL and AML, CNS tumors were the most common secondary cancers (24%) followed by thyroid cancer (22%). For patients who survived for at least 5 years after initial diagnosis, the cumulative incidence of secondary cancer at 30 years was 3.9% and 4.3%, respectively, for ALL and AML.[381]

The risk is especially higher among patients diagnosed at a younger age (≤17 years for ALL and CNS tumors; ≤18 years for bone and soft tissue sarcomas).[382] Among long-term survivors of bone cancers at 25 years after diagnosis, the cumulative incidence of subsequent cancers is higher for those diagnosed with Ewing sarcoma compared to those diagnosed with osteosarcoma (9.0% and 5.4%, respectively).[383,384]

Clinicians who provide care for AYA cancer survivors must implement and evaluate methods for improving awareness of SMNs. They must also implement appropriate surveillance strategies for early detection of these malignancies.[385] An annual mammogram and breast MRI for the purpose of secondary breast cancer screening are recommended for survivors treated with a chest RT ≥10 Gy, including those <30 years of age.[386,387] Refer to the NCCN Guidelines for Breast Cancer Screening and Diagnosis for information about breast cancer screening recommendations and the NCCN Guidelines for Genetic/Familial High-Risk Assessment: Breast, Ovarian, Pancreatic, and Prostate for information about individuals who may be at high risk for breast cancer, available at www.NCCN.org.

A colonoscopy, for the purpose of secondary colorectal cancer screening, is recommended (ie, every 5 years, multitarget stool DNA/RNA-based test every 3 years, or blood-based cell-free DNA test every 3 years), based on informed decision-making, starting at age 30 years or 5 years after completion of RT, whichever occurs last, for patients treated with abdominal or pelvic RT or TBI. Screening for secondary AML or myelodysplasia should be done by assessing complete blood count (CBC) and bone marrow examination as clinically indicated based on symptoms. Routine endocrine, ophthalmology, and dental evaluation (dental examination and cleaning every 6 months) are recommended for long-term AYA cancer survivors following any treatment exposure to the head or upper neck regions.[380] Refer to the NCCN Guidelines for Genetic/Familial High-Risk Assessment: Colorectal, Endometrial, and Gastric, available at www.NCCN.org.

Cardiovascular Complications

Cardiovascular complications (including, but not limited to, congestive heart failure [CHF], myocardial infarction [MI], pericardial disease, and valvular abnormalities) are the leading non-malignant cause of death among survivors of AYA cancers, compared to the general population.[388-

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[391] Additionally, lifestyle factors, including poor physical activity and dietary habits, unique to the AYA population increase the risk of cardiovascular mortality.[326] A retrospective cohort study comparing 5673 two-year AYA survivors of cancer to 57,617 controls showed that the cancer survivors were more likely to develop cardiovascular disease (adjusted incidence rate ratio [IRR], 2.37; 95% CI, 1.93–2.93), with risk being highest among survivors of breast cancer (adjusted IRR, 3.63; 95% CI, 2.41–5.47) and leukemia (adjusted IRR, 4.23; 95% CI, 1.73–10.31).[392] A systematic review and meta-analysis of 64 studies of specific cardiovascular late effects in 143,606 survivors of childhood or adolescent cancer reported that the weighted average prevalence was 19.7% for hypertension and 2.3% for stroke in this population.[393]

Mediastinal RT and anthracycline-based chemotherapy are the strongest risk factors for late cardiovascular complications in AYA survivors of HL.[352,394,395] The Children's Oncology Group (COG) Long-Term Follow-Up Guidelines recommend cardiac imaging as appropriate according to the cumulative anthracycline and/or RT dose.[385] In a British cohort study of 7033 patients with HL, the risk of death from MI was highest for patients <35 years at the time of treatment with supradiaphragmatic RT.[395] Patients treated with anthracyclines were at increased risk for MI within 1 year after first treatment, whereas the risk for MI among patients treated with supradiaphragmatic RT and vincristine without anthracyclines increased sharply after the first year of follow-up.[395] In another study of 1474 survivors of HL <41 years at the time of treatment, mediastinal RT increased the risk of MI, CHF, and valvular disorders, whereas the addition of anthracyclines to RT elevated the risks for CHF and valvular disorders.[352] The 25-year cumulative incidence of CHF after mediastinal RT and anthracyclines was 8%. An analysis of 15,815 survivors of childhood cancer, including participants from the CCSS (*n* = 12,407), showed that the anthracycline daunorubicin may be less cardiotoxic than doxorubicin (HR, 0.45; 95% CI, 0.23–0.73).[396] Methods to mitigate anthracycline toxicity include the administration of dexrazoxane, a cardioprotective agent, as well as the use of continuous infusion therapy rather than bolus treatments, the preference for epirubicin or liposomal doxorubicin, and the use of beta-blockers and angiotensin-converting enzyme inhibitors as appropriate.[397-401]

Cisplatin-based chemotherapy is associated with long-term risk for cardiovascular complications in testicular cancer survivors.[402-404,] In a Dutch study of 2512 testicular cancer survivors, non-seminomatous testicular cancer survivors <30 years at diagnosis treated with mediastinal RT and chemotherapy with cisplatin, vinblastine, and bleomycin were at increased risk for MI within 20 years of treatment.[402] Haugnes and colleagues reported that treatment with cisplatin, bleomycin, and etoposide and/or RT was associated with increased risk for cardiovascular disease in testicular cancer survivors; chemotherapy alone or in combination with RT significantly increased the risk for MI.[403]

Survivors of brain tumors, leukemia, NHL, and bone and soft tissue sarcomas treated with anthracyclines and cardiac irradiation are also at significantly higher risk of adverse cardiovascular complications. However, the majority of patients included in these studies were <21 years at the time of diagnosis.[405] Other reports have also documented increased cardiovascular complications among survivors of lymphoma, brain tumor, leukemia, and testicular cancer.[389,406]

Pulmonary Complications

Analysis of data from the CCSS showed that pulmonary complications (eg, asthma, chronic cough, emphysema, lung fibrosis) are more frequent in survivors (*n* = 20,690) than in sibling controls (*n* = 4027).[407] AYAs must therefore be counseled regarding avoidance of tobacco and smoking cessation as appropriate. Chemotherapy (eg, bleomycin and alkylating agents such as busulfan, carmustine, and lomustine), chest RT, and craniospinal irradiation are associated with pulmonary toxicity and can

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compromise pulmonary function in AYA cancer survivors.[385,404,408,409] Age at diagnosis (15–21 years compared with age <15 years) and pulmonary toxic chemotherapy alone or combined with chest RT were associated with a significantly increased relative risk of lung fibrosis and pleurisy.[408] The cumulative incidence increased up to 15 to 20 years after diagnosis. Other complications include recurrent pneumonia, chronic cough, supplemental oxygen use, and shortness of breath.

A large international study reported a significant increase in mortality from respiratory diseases among testicular cancer survivors treated with chemotherapy compared to the general population.[410] Risk factors for pulmonary toxicity include age at diagnosis, cumulative bleomycin dose, reduced glomerular filtration rate, kidney dysfunction, and stage IV disease at presentation.[411] Haugnes and colleagues reported that among 1049 testicular cancer survivors, those treated with chemotherapy combined with pulmonary surgery or large cumulative cisplatin doses had significantly reduced pulmonary function compared with those treated with surgery alone.[412] Bleomycin dose was not associated with restrictive lung disease. Instead, in a multivariate model, cisplatin dose (P = .007) and age at diagnosis (P = .008) were associated with the risk for restrictive lung disease. Therefore, AYA cancer survivors, particularly those who received RT and chemotherapeutic agents known to cause pulmonary toxicity, are advised to undergo baseline and interim pulmonary function screening as recommended by the COG Long-Term Follow-Up Guidelines.[413]

Neurologic Complications

AYA survivors of brain tumors and those treated with intrathecal chemotherapy, high-CNS penetrating chemotherapy, and/or cranial RT are at increased risk for neurologic and/or neuropsychological complications, including hearing impairments, cataracts and other vision problems, seizure disorders, migraines, and coordination and motor control problems.[414-421] However, these findings are relevant to survivors diagnosed at ≤21 years. Neuropsychological screening for difficulties in school or work is recommended for all AYA survivors of cancer so that providers identify the need for referral early in the survivorship phase.[422] In particular, patients with brain tumors, or those treated with cranial/craniospinal radiation, intrathecal chemotherapy, and/or high-dose methotrexate, should be referred for neuropsychological evaluation. In addition, a routine evaluation for symptoms of peripheral neuropathy should be conducted.

Long-term AYA survivors of testicular cancer who were treated with cisplatin-based chemotherapy are at risk for neurologic complications such as sensory neuropathy, tinnitus, hearing impairment, and Raynaud's disease.[404] Among 1814 survivors of testicular cancer included in a Norwegian observational study, Raynaud-like phenomena were the most frequently reported complications (39%), followed by paresthesia of the hands or feet (29%), and tinnitus and hearing impairment (22% and 21%, respectively) by patients treated with chemotherapy compared to those not treated with chemotherapy.[423] The incidences of paresthesia of the feet were also higher among patients treated with RT.

Patients treated with vincristine, docetaxel, or paclitaxel are also at risk for long-term peripheral neuropathy.[424] A cross-sectional study of 80 ALL survivors who had been treated with vincristine found that 33.75% had neuropathy as measured electrophysiologically, although the study reported significant improvement over time.[425] Another study of 37 ALL survivors who had been treated with vincristine reported that 29.7% of patients showed abnormalities in nerve conduction studies.[426] A study of 605 breast cancer survivors at least 2 years out from diagnosis reported that survivors treated with docetaxel or paclitaxel were more likely to experience peripheral neuropathy (31% for docetaxel, 44% for paclitaxel) than those who were not treated with these chemotherapies (17% for no

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chemotherapy, 20% for other chemotherapies).[427] Routine evaluation for symptoms of peripheral neuropathy is recommended.

Stroke, although relatively uncommon, is a devastating neurologic complication in AYA survivors of brain tumors and leukemia treated with cranial RT and survivors of HL treated with mantle field RT.[418,428-430] In a retrospective cohort study of 5-year survivors of HL (*N* = 2201), those treated with RT to the neck and mediastinum were particularly at increased risk for stroke and transient ischemic attack.[431] The incidences were higher among patients diagnosed at <21 years than those diagnosed between 21 and 30 years. The standardized incidence ratio was 3.8 and 3.1, respectively. In an analysis of CCSS data, out of 271 survivors who reported a stroke, 26% reported a second stroke.[430] Predictors of recurrent stroke included history of brain tumor, exposure to cranial RT (total dose ≥50 Gy), older age at first stroke, and hypertension.[430]

Nephrotoxicity

Long-term kidney dysfunction has been reported in survivors of testicular cancer treated with infra-diaphragmatic RT and cisplatin-based chemotherapy. In one study with long-term follow-up, kidney impairment was observed in 8% of patients treated with abdominal RT alone compared to a 14% reduction in patients with chemotherapy with or without RT. Age at treatment and type of treatment were associated with impaired kidney function. Screening for kidney insufficiency, other kidney complications, and electrolytes are recommended for those at high risk, including patients treated with platinum-based therapies and ifosfamide-based chemotherapy.

Bladder and Bowel Symptoms

Cancer survivors treated with pelvic RT, especially those administered concurrent cyclophosphamide, are more likely to experience bladder and bowel symptoms. Hemorrhagic cystitis may occur consequent to RT and/or cyclophosphamide administration (now coadministered with mesna in high-dose protocols) and may progress to long-term bladder fibrosis and contracture in some cases.[432] A study of 104 survivors of cervical or endometrial cancer found that the severity of bladder and bowel symptoms were significantly associated with pelvic RT treatment.[433] Two studies of long-term gynecologic cancer survivors (*N* = 77 and *N* = 519) found that approximately 12% to 17% of survivors treated with pelvic RT develop symptoms of bowel incontinence.[434,435] Bowel incontinence was associated with the mean radiation dose, with the larger study reporting that mean doses >50 Gy carry the greatest risk.[434] Cancer survivors who experienced bladder and/or bowel incontinence as a result of pelvic RT therapy have reported considerable distress and a decreased quality of life as a result of their symptoms.[433,436] Screening of urinary bladder function may be performed as appropriate.

Endocrine Complications

Cranial and/or spinal RT, TBI, and targeted RT to the neck, abdomen, pelvis, and testes are associated with neuroendocrine late effects in survivors of AYA cancers.[173] The most common endocrine complications include growth hormone (GH) deficiency, thyroid gland abnormalities, gonadal dysfunction, obesity, diabetes mellitus, and decreased fertility.[99,163,437] AYA cancer survivors treated with an RT dose of ≥18 Gy to the hypothalamic-pituitary-adrenal (HPA) axis are at high risk for GH deficiency, whereas those treated with an RT dose of ≥40 Gy to the HPA axis are at risk of developing central hypothyroidism, gonadotropin deficiency, and central adrenal insufficiency.

GH deficiency can be observed within 5 to 10 years after treatment with RT doses >15 Gy.[173] Secondary thyroid cancers, hypothyroidism, and, to a lesser extent, hyperthyroidism are more common among AYA survivors of brain tumors, ALL, HL, and those who have undergone HCT.[414,438,439] Testicular cancer survivors treated with chemotherapy and RT are at greater risk for hypogonadism.[404] Low testosterone levels and testosterone

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replacement have been reported in 34% and 4% of testicular cancer survivors, respectively.[440] In a study of semen variables in 214 adult cancer survivors following alkylating agent administration (measured in cyclophosphamide equivalent doses [CED]), a negative correlation was discerned between increasing CED and sperm concentration. Additional findings included the possibility of azoospermia, oligospermia, and normospermia with CED.[441] Risk of endocrine complications may increase over time, indicating a need for lifelong evaluation and screening as appropriate.[437,442] Routine endocrine, ophthalmologic, and dental evaluations are recommended for patients with selected radiation exposure and/or TBI for HCT. Consider multidisciplinary consultation, which includes a radiation oncologist for the optimal method to reduce radiation-induced late effects.

Long-Term Follow-up

As discussed above, AYA cancer survivors have a high risk of developing a wide range of late effects. Development of a "Cancer Treatment Summary and Survivorship Care Plan," including periodic evaluation with focused history, physical examination, and screening based on treatment exposure, and risk for treatment-related late effects, should be an integral part of caring for AYA cancer survivors.[349,443,444] Medical and psychosocial care should be conducted in the language preferred by the patient, their family, or caregiver. Certified interpreters should be used; do not rely on non-certified medical staff or family members, caregivers, or friends for interpretation. Vaccinations are recommended for AYA cancer survivors; please refer to the NCCN Guidelines for Survivorship, available at www.NCCN.org, for additional guidance on vaccines.

As aforementioned, the screening recommendations included in the NCCN Guidelines for Adolescent and Young Adult (AYA) Oncology are adapted from the COG Long-Term Follow-Up Guidelines for Survivors of Childhood, Adolescent, and Young Adult Cancers, available at survivorshipguidelines.org.[385] See *Screening Recommendations for AYA Survivors* in the algorithm for specific recommendations based on the treatment exposure and timing and intensity of screening. These recommendations may be adapted based on additional risk factors.

Issues related to insurance, clinical team composition (presence of a provider knowledgeable in childhood cancer), scheduling (availability of flexible scheduling), and comprehensive nature of the care provided were identified as patient-perceived facilitators for the transition of survivorship care in young adult survivors of childhood cancer.[445] The models for AYA survivorship care include cancer center follow-up (primary treatment team or specialized long-term follow-up clinics), follow-up by the patient's primary care physician (PCP), or a combination of both (shared care model).[444,446] Some studies have suggested that a shared care model involving both the primary oncology team and the PCP is feasible and may facilitate appropriate care in childhood cancer survivors.[447-449] A relationship between the AYA cancer survivor and their primary health care provider for routine health issues is encouraged in addition to support of end-of-treatment transition to a PCP with clear communication of the PCP's role within the oncology team. Due to the transient lifestyle of the AYA population (ie, attending college far from home, traveling to different locations to establish careers), it is imperative for the initial care team to communicate with the new care team (once established). AYAs should be screened for risky behavior such as substance use, tobacco/nicotine use, and binge drinking/excessive alcohol use on a regular basis and counseled on cessation. AYAs should be educated on safe sex practices/contraceptive options if they are not trying to actively conceive regardless of their infertility risk post treatment. AYAs should also be asked about sexual dysfunction on a routine basis and how it may impact interpersonal relationships or self-image.

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Risk stratification of survivors based on the current medical issues and prior treatments may be helpful to determine the different levels of follow-up in the shared care model.[446,450,451] Survivors at low risk for late effects (treated with surgery alone and/or chemotherapy with no RT, not including alkylating agents, anthracyclines, bleomycin, or epipodophyllotoxin) can be transitioned to their PCP soon after completion of therapy. Survivors at moderate risk for late effects (treated with low- or moderate-dose chemotherapy with no RT, chemotherapy-containing alkylating agents, anthracyclines, bleomycin, or epipodophyllotoxin) can be evaluated by their oncology team or PCP on alternating years. Survivors at high risk for late effects, such as those treated for CNS cancers or those treated with HCT, any RT, high-dose alkylating agents, anthracycline, bleomycin, or epipodophyllotoxin, should be followed annually by their oncology team and continue follow-up care with their PCP regularly. Cancer survivors with cardiomyopathy, cardiovascular risk factors, and a history of pelvic radiation who become pregnant should be referred to maternal-fetal medicine with consideration of a high-risk obstetrics referral. The International Late Effects of Childhood Cancer Guideline Harmonization Group has developed additional recommendations on counseling and surveillance of obstetric-related risks in childhood, adolescent, and young adult cancer survivors.[452]

Palliative and End-of-Life Considerations

Palliative care is interdisciplinary care of patients with life-threatening illnesses, and malignant as well as non-malignant cancer. The goal of palliative care in patients with cancer is to control symptoms, relieve emotional and physical suffering from adverse effects of treatment, and improve quality of life for patients and their families, regardless of the stage of the disease or the need for other therapies.[453,454] Clinicians with expertise in end-of-life care should facilitate discussion of difficult medical issues such as nutrition/hydration, sedation, treatment cessation, and place of death.[453] Ongoing psychosocial support is extremely important during the transition to end-of-life care. Clinical teams need to be aware of and work with their palliative care teams with regard to local availability and regulations for home-based palliative care and concurrent care, continuing to receive disease-directed therapy while enrolling in hospice. See the NCCN Guidelines for Palliative Care, available at www.NCCN.org.

The end-of-life phase of therapy involves alleviation of suffering, communication between all parties involved, and supportive care services when needed. Palliative care services for AYA patients should be provided by a team with expertise in understanding the psychosocial, social, spiritual, emotional, developmental, and financial issues that are unique to this age group.[453,455-457] Introduction of palliative care for symptom management and psychosocial support should occur before the patient's condition is considered "palliative" in order to provide the best possible care.[458-460] Palliative care is appropriate even when patients are being treated with curative intent, and there is growing consensus that AYA patients should have access to palliative care services from the time of diagnosis, across the cancer trajectory, until end-of-life care.[458]

Patients, caregivers, and health care professionals should be taught that palliative care is an integral part of their comprehensive cancer care.[456] AYA patients usually do not make decisions in isolation. While some AYA patients have the ability to make life and death decisions independently, many are either not the primary decision maker or they rely intensely on input from parents, spouses, significant others, caregivers, and/or other family members.[453] It is imperative to determine who the AYA wants to involve in decision-making and consider appropriate referrals and the psychosocial needs of the patient's family members (including children and young siblings), friends, and caregivers.[458]

If prognosis is poor, consider initiating a discussion with a palliative care team or hospice services early in treatment to provide continuity of care and support for both the patient and family throughout the cancer

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experience.[458,459] Discussion about end-of-life preferences and the formulation of an advance directive/advance care planning document should begin at the time of initiating treatment, but details should be individualized according to the preferences of the AYA patient and family.[461] Several retrospective studies have shown that >50% of AYA patients with cancer receive medically intensive care at the end of life, indicating a need for better understanding of care preferences in these patients.[462-464] In one of these studies, a review of the health records of 12,938 deceased AYA patients with cancer showed that 59% of patients received at least one intensive intervention at the end of life and 30% received two or more. Patients treated at non-specialty cancer centers were more likely to receive two or more intensive interventions than those at specialty centers.[462]

It is imperative for health care professionals not to assume that AYA patients may be less inclined to discuss death and other end-of-life issues and make sure that the care is consistent with the patients' goals.[465-468] In an exploratory study of 50 adolescent patients (aged 15–21 years) with and without chronic illness, adolescents were willing to discuss end-of-life decision-making by taking part in a one-on-one survey administered by a researcher.[469] AYA patients' opinions about end-of-life care vary across this age group. Exploring individual preferences for end-of-life care and providing interventions specific to the needs of this patient population could significantly improve end-of-life care.[461,470] In one retrospective review, a significant number of adolescents dying of cancer felt that discussions about end of life occurred very close to death, allowing very little time to psychologically prepare for death.[471] Additionally, during palliative and end-of-life care, AYA patients may be able to and wish to continue to engage in their day-to-day activities, even if some activities are in discord with medical advice (eg, participating in strenuous physical activity).

Patients' goals, dreams, and desires to leave a legacy are important considerations to address.[456] Involve child life specialists and/or psychosocial team members to discuss legacy projects and memory work with the patient, their family, or caregivers. Note that the preferred location of death may be influenced by individual, family, and cultural differences.[472] While many adolescents indicate a preference for dying at home, the majority die in hospitals.[473-476] Others may choose to die in the hospital due to regional scarcities of home hospice, caregiver demand at the end of life, or personal preference.[477] It is critical to make every effort to query and support the patient's preferred location of death. Supportive care programs should have a bereavement plan that maintains a list of bereavement resources and at least once makes contact with an AYA's parents, spouses, significant others, caregivers, and/or other family members after the AYA's death.

Summary

AYA patients with cancer should be recognized as a distinct population that has unique medical, developmental, and psychosocial needs. It is important for physicians to identify issues specific to the AYA population and recommend appropriate interventions with the aim of improving clinical outcomes. Most importantly, all AYA patients should have access to age-appropriate supportive care as well as medical subspecialty services appropriate for their cancer diagnosis.

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